Likelihood approximations distort the ion channel kinetic information in macroscopic currents

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Abstract Some ion channel mechanisms leave a mark in the record that does not depend on fitting a model: the flip state of $P2X_2$ showed up as a delay before the macroscopic current rose. The asymmetric activation we reported from the same recordings two decades later leaves no such mark: it rests on evidence ratios under a recursive likelihood, and a published control without the recursion put a non-conformational alternative first. How much likelihood approximations distort what they report, and in which directions, was never measured for macroscopic currents. The diagnostics are classical: for a correctly specified likelihood the score, the log-likelihood's gradient, averages to zero at the true parameters and its covariance equals the reported Fisher information. We tested eight likelihoods, from least squares on the mean current, which leaves the correlated gating fluctuation unmodelled, to a filter conditioning each interval average on both ends. The sweep runs over channel number, noise and interval in a two-state scheme, the smallest that isolates the approximation's own error, ten thousand exact simulations of the process they approximate at every point. Those that leave the correlation unmodelled misreport their information by up to an order of magnitude, threefold on the error bar, wherever the residual retains memory, and no single rescaling repairs it, because the distortion has directions in parameter space. The published control's likelihood is among them. Neither partial correction improves on the likelihood it corrects, and a white residual does not certify calibration. The filter conditioning on both ends is calibrated over almost the whole measured range, with departures growing toward few channels and low noise. The noise at which the least-squares error bar becomes calibrated lies one to three decades above the noise at which the fluctuations stop carrying the unitary current: no measured region gives both.

Introduction

Some mechanisms leave a mark in the record that does not depend on fitting a particular model. The flip state we proposed for the $P2X_2$ receptor is one: it shows up as a delay before the current rises (***Moffatt and Hume, 2007***). As inferences get finer, claims are reached that leave no such mark. Two decades later nine kinetic schemes for the same receptor were ranked by Bayesian evidence over the same recordings, in work of our own (***Moffatt and Pierdominici-Sottile, 2025***). Under MacroIR, which averages the current over each acquisition interval and conditions it on the data already observed, a conformational scheme came first; a published control, MacroINR, the same likelihood without the recursion, put the subunit-specific allosteric alternative first, overturning a margin of a factor of six. The direction was reported, the control systematically underestimating the evidence for schemes with conformational intermediates; but the choice between the arms rested on argument, and nothing in those recordings settles which likelihood tells the truth, the way the delay once settled the flip.

What the recordings hold is the reason. Single-channel durations are read off the record (***Neher and Stevens, 1977***; ***Colquhoun and Hawkes, 1981***) and macroscopic currents follow a deterministic relaxation (***Hodgkin and Huxley, 1952***), but most recordings are in neither place: the averaging is incomplete, so a fluctuation around the mean current survives and carries the memory of the gating. That memory is where the remaining information is, since channel number and unitary current enter the mean as a product and no fit to the mean separates them.

Current practice fits the mean and discards the fluctuation. The four approaches surveyed by ***Clerx et al.*** (***2019***) are deterministic fits with a root-mean-square objective, and the IonBench benchmark states that it does not cover optimisation approaches for stochastic models (***Owen and Mirams, 2025***). Discarding the fluctuation usually leaves the point estimate standing. What it damages is the count: samples closer together than the channel's relaxation time are correlated, so a recording supplies fewer independent observations than it has samples while the fit counts them all, and the interval it reports is narrower than the spread it delivers.

Likelihoods that model that correlation have been available for decades, from stationary power spectra and the exact non-stationary two-time covariance (***Katz and Miledi, 1970***, ***1972***; ***Anderson and Stevens, 1973***; ***Conti et al., 1980***; ***Sigworth, 1981***) to whole-record likelihoods (***Celentano and Hawkes, 2004***; ***Milescu et al., 2005***; ***Moffatt, 2007***; ***Stepanyuk et al., 2011***; ***Münch et al., 2022***), differing in what they condition on, MacroIR the most recent (***Moffatt and Pierdominici-Sottile, 2025***). Where they break was stated when they were published and never located: the recursive member that samples instantaneously was expected to degrade as the channel count falls and as the record is averaged, its non-recursive counterpart to have no useful regime at all, the damage falling on the error rates rather than on the estimates (***Moffatt, 2007***).

No statistic computed on a single recording settles it either: one recording is one draw from the sampling distribution, and with the generating model unknown a departure may belong to the approximation or to the model. The question has to be put from outside, over an ensemble drawn from known parameters, and for this class of model that is possible: the forward process simulates exactly where its likelihood cannot be evaluated exactly.

The instruments are classical. For a correctly specified likelihood the score, the gradient of the log-likelihood, has mean zero at the generating parameters and its covariance across recordings equals the reported Fisher information (***Huber, 1967***; ***White, 1982***); how far an approximation departs from either identity is a measurable quantity rather than a test to be passed. Goodness of fit does not answer the question, since an approximation can reproduce the mean current closely and still misreport the information by more than an order of magnitude, and comparing two approximations against each other does not answer it either, because both may be wrong. Near the maximum that same matrix enters the Bayesian evidence through a volume term and a rescaling of the effective sample size, so misreporting it carries the error into every comparison built on it. The correction is derived elsewhere; this paper measures its input and computes no evidence, and it never benchmarks against non-stationary fluctuation analysis (***Stepanyuk et al., 2014***), which answers a different question, returning the channel number and the unitary current with no likelihood and no kinetics, from many exchangeable sweeps.

We measured both departures over a plane whose coordinates are the number of channels, the acquisition interval in units of the relaxation time, and the instrumental noise in units of the gating noise. The rate constants are held fixed. The scheme is two states at an open probability of one half, driven by a single concentration jump, which isolates the error the approximation itself commits; whether it bounds anything richer is a conjecture, not a result of this paper. Ten thousand recordings were simulated at every cell and scored by the eight members of the cost ladder, from least squares to the filter conditioning each interval average on both ends. An implementation error shows up as a departure like any other, so the same test checks the code.

The plane separates two failures usually named as one: what a method gets wrong, and what it says about how wrong it is. On the first the ladder is nearly flat. What separates the members is what they report about those estimates: the ones that leave the correlation unmodelled misreport their information by an order of magnitude wherever the residual retains enough memory, while the one conditioning on both ends of the interval is calibrated over almost the whole plane. The map includes that member in what it audits, since a method whose limits are unmapped has been endorsed rather than characterised.

Making the measurement takes an ensemble. Using it does not: the integrated autocorrelation of the standardized least-squares residual reads the effective interval count off a fit already being made. It also picks up the analogue filter on a record sampled faster than its own bandwidth, and that is a number the reader chose as they chose the sampling rate. What is new here is narrow, and the likelihood's construction is not part of it: that was published with the $P2X_2$ analysis. Its derivatives are. The gradient of the log-likelihood is analytic and the Fisher information follows from the same pass (Methods). Standard errors from fluctuation data go back to ***Anderson and Stevens*** (***1973***) and analytic gradients to ***Milescu et al.*** (***2005***). What this work adds, for a non-stationary protocol read from a single record, is an interval whose calibration has been verified rather than assumed.

The likelihood family and its diagnostics

A macroscopic current is the summed signal of a population of ion channels, each an independent continuous-time Markov chain over a few conformational states. N_{ch} is the number of channels and K the number of kinetic states of one channel ($K = 2$ throughout this paper, carried symbolically). Each channel has a generator (rate) matrix $\mathbf{Q} \in \mathbb{R}^{K\times K}$, hence the row-stochastic transition matrix $\mathbf{P}(t) = e^{\mathbf{Q}t}$ over an elapsed time t, with entries $P_{i\to j}(t)$ from state i to state j, and a conductance γ_k in state k, collected in $\boldsymbol{\gamma} \in \mathbb{R}^K$.

Writing the likelihood of one recorded sample exactly, given the kinetic parameters and the samples before it, calls for two objects, and neither is available here: a microscopic description keeping each channel discrete, which is explicit and ruled out by cost, and the law of the conductance averaged over the window each sample is recorded over, which has no closed form at any channel count although its low-order moments do (Appendix 1). Every member replaces both with Gaussians, and they differ in which object the second Gaussian describes, what it is conditioned on and how its variance is accounted.

The observable is an interval average

A recorded sample is the current averaged over the acquisition window rather than sampled at an instant, because the current is low-pass filtered in analogue form before it reaches the converter. Writing the sampling period as Δ and taking the weighting to be uniform across it, the quantity a likelihood must describe is the windowed average

$$\bar{y}_t \;=\; \frac{1}{\Delta}\int_{t-\Delta}^{t} y(\tau)\,\mathrm{d}\tau \;+\; \eta_t, \tag{1}$$

where $y(\tau)$ is the instantaneous population current and η_t is the instrumental noise accumulated over the window. This interval average is the physical observable. When Δ is much shorter than the fastest relaxation of $\mathbf{Q}$ the average sits close to the instantaneous current and the distinction is negligible; when Δ is comparable to or longer than that relaxation the two diverge, and a likelihood built on instantaneous sampling describes the wrong random variable.

Uniform weighting is a first approximation to what the electronics do. The simulator and every member share it (Methods), so the data and the likelihood describe the same observable and the window itself does not enter the comparison made here. Appendix 1 bounds what the approximation misses, as the filter's cutoff against the acquisition window, and gives the remedy, which is to block-average a record down to that scale before analysis.

The two Gaussian closures fail in different regimes

To close a distribution is to replace it by the Gaussian with the same mean and covariance. Two closures are available, and every member makes the first; whether it pays for the second is set by what it predicts. The current at an instant is $\boldsymbol{\gamma}$ contracted against the occupancies at one time, and the linear image of a Gaussian is Gaussian; the interval average is a functional of the path inside the window and needs a closure of its own. An instantaneous member drops that evolution instead, which misspecifies the random variable rather than closing it, and in the unhelpful direction, since averaging is what brings the law closer to Gaussian.

The two Gaussian approximations

Occupancy closure. One probability per occupancy vector, a count that grows with N_{ch}, is replaced by a Gaussian in the per-channel pair $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the same size at every channel count and living on the simplex. *Validity:* two conditions, not one. The channel count must be high enough for the counts to behave as a continuum, which fails as N_{ch} falls; and the recorded sample must not resolve the step in predicted current made by moving one channel between states, which fails as the instrumental noise falls. The second is why the boundary of the failing region in the Results runs across the plane of channel count against noise rather than down a line of constant N_{ch}.

Interval-signal closure. The law of the conductance averaged over one window is replaced by a Gaussian carrying its first two moments. The exact law is telegraph-like, a mixed law closed only at $K = 2$ and proliferating with K, which neither the few distinct conductances channel models carry nor the independence of the channels rescues (Appendix 1). *Validity:* the window must span many transitions, or the channels must be many enough for their interval averages to sum to a Gaussian, or the instrumental noise must be large enough to make the departure immaterial. The last of the three is exact rather than heuristic: noise independent of the gating current adds to the variance of the recorded sample and leaves every cumulant above the second unchanged, so the standardized departures shrink as the noise grows.

The two share two of their three controlling quantities. Both are relaxed by the channel count, which drives the counts towards a continuum and the sum of the channels' interval averages towards a Gaussian, and both are relaxed by the instrumental noise. Only the window separates them, and it enters the interval-signal closure alone. That is why the Results sweep a plane of channel count against noise at one window before moving the window along an axis of its own, and why a departure measured at a single window cannot be assigned to one closure rather than the other.

The moments the closure keeps are closed-form single-channel objects, independent of N_{ch} and of the topology (Appendix 1), which is what justifies the closure beyond the two-state case run here.

What they discard does not vanish. It reappears as temporal correlation the likelihood does not predict, which is what the correlation term of the distortion measures (below). Two names follow, for what the two closures predict rather than for regions the measurement can partition. The *microscopic* regime is few channels, where the occupancy closure is expected to fail and only the microscopic description is exact. The *telegraphic* regime is where the interval average is still the switching signal rather than a Gaussian, which a short window, a small channel count and a low instrumental noise each push a recording towards. Both are predictions of the closures, and what the Results measure is their product.

Two design quantities are reported throughout in natural units, marked by a tilde. Time is measured in the channel time constant $\tau = 1/k_{\mathrm{off}}$, the closing time constant, and current in the unitary current i, giving the acquisition interval $\widetilde{\Delta} = \Delta\, k_{\mathrm{off}} = \Delta/\tau$ and the instrumental noise $\widetilde{S} =$

$S\,k_{\text{off}}/i^2$, with S the instrumental noise power spectral density, so $\widetilde{S}$ is the noise power accumulated over one channel time constant measured against the single-channel signal power (Methods). The channel count and the instrumental noise govern the distortion through their ratio, which is the variable that half of the design plane collapses in; the boundaries of what is measurable at all follow a second ratio and do not (Results).

Least squares models no gating, and the family above it varies in window and recursion

Before any of the choices above comes a prior question: does the method model the gating fluctuations at all? Classical nonlinear least squares answers no. It fits the deterministic mean current and assigns one constant variance to every residual. Least squares is therefore not a member of the family. It is the bottom rung of the ladder of computational cost this paper walks, and it is the anchor against which every member above it is measured, because it is the method most readers are using.

The methods differ along two axes. One is the window: how much of the acquisition interval the predicted current is conditioned on. The other is recursion: a recursive (prequential) member (***Dawid, 1984***) factorizes the likelihood as $\prod_t q_t(\bar{y}_t \mid \bar{y}_{1:t-1})$, so the occupancy distribution carried into each window is the posterior left by the previous ones, while a non-recursive (open-loop) member factorizes as $\prod_t m_t(\bar{y}_t)$, each window propagated from the initial condition and never conditioned on the observed trace.

The window decision is not the family's property alone: any method that predicts a recorded value has to make it, at every level of the gating description and least squares included, which is why least squares has two arms here.

The names are read off the two axes. The suffix is the recursion axis, `NR` for a non-recursive member and `R` for a recursive one; the prefix is the window axis and records what the member predicts: none for the current at an instant, `M` for the interval mean given the state at the window's start, `I` for the interval mean with both ends of the window conditioned on. The `M` is the mean conductance and not an abbreviation of macroscopic; every member of the family is macroscopic. Crossing the two axes gives five members rather than six, because without an update the state at the end of a window is never observed and the two interval prefixes describe the same method on the non-recursive side; it carries `I` for the interval it predicts, and Table 1 records that what it conditions on is the start state alone. A sixth member sits beside the lattice, where a third letter records the form of the variance rather than the window: `VR` is `MR` with the residual interval variance in place of the total one, the one member whose variance form is a choice rather than a consequence of what it conditions on (Appendix 2). The two least-squares arms sit off this lattice, `LSE` fitting the instantaneous deterministic mean and `ILSE` the same mean averaged over the interval, where the `I` records the window setting and nothing else. Each is the open-loop member at its own window setting with the gating variance replaced by one scale common to the whole record, marginalised rather than fitted, and what the least-squares arm discards is the variance; the arm is also run in two parameter configurations, which are different statistical models, so every least-squares number below is scoped to one of them (Methods). Six and two are the eight members the Results compare, and Table 1 lists the name each goes by there.

Two things follow. `IR` occupies the top rung below the intractable exact likelihood, so its standing is structural and set before any measurement; and the lower rungs are established likelihoods rather than constructed foils, the non-recursive instantaneous member being the independent-interval likelihood of ***Milescu et al.*** (***2005***) and the recursive instantaneous member the macroscopic filter of ***Moffatt*** (***2007***) and the generalized filter of ***Münch et al.*** (***2022***). Above all of them sits the exact likelihood; the same model can nonetheless be simulated forward exactly, so the simulation realizes the top of the ladder and is the ground truth every rung below it is judged against.

The interval likelihood is the instantaneous one applied to the boundary state

The interval likelihood is the instantaneous one applied to the boundary state, and the instantaneous one is where it starts. The recursive member of ***Moffatt*** (***2007***) carries a Gaussian over the channel occupancies and conditions it on each recorded sample: in the per-channel normalization used throughout, with $\boldsymbol{\gamma}$ the per-state conductance and ϵ^2 the instrumental variance,

$$y^{\text{pred}} = N_{\text{ch}}\, \boldsymbol{\mu}^\top \boldsymbol{\gamma}, \qquad \mathbf{g} = \boldsymbol{\Sigma}\boldsymbol{\gamma}, \qquad \sigma^2 = \epsilon^2 + N_{\text{ch}}\, \boldsymbol{\gamma}^\top \boldsymbol{\Sigma}\boldsymbol{\gamma},$$
$$\boldsymbol{\mu}^{\text{post}} = \boldsymbol{\mu} + \frac{\mathbf{g}}{\sigma^2}\,\delta, \qquad \boldsymbol{\Sigma}^{\text{post}} = \boldsymbol{\Sigma} - \frac{N_{\text{ch}}\, \mathbf{g}\mathbf{g}^\top}{\sigma^2}, \qquad \delta = y^{\text{obs}} - y^{\text{pred}}. \tag{2}$$

This is the Gaussian conditioning of the pair (state, observation) on the value recorded. The gain $\mathbf{g}$ is the covariance between the state a channel is in and the current it carries, and it is the whole of what brings the observation to the state: the predicted variance divides it, and nothing else in the update touches the data. Every moment here is carried on a single channel, and the factors of N_{ch} are where the counts over the ensemble enter (Appendix 1). Per channel is a normalization: conditioning on a shared recorded value correlates the channels, so from the first update onward $\boldsymbol{\Sigma}$ is $\text{Cov}(\mathbf{N})/N_{\text{ch}}$ rather than one channel's own covariance (Methods, Appendix 1).

What it gives up is not exactness but *sufficiency*: the three moments it consumes are exact functions of $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, but $\text{Var}[\bar{y} \mid \mathbf{N}]$ depends on $\mathbf{N}$, so $(\mathbf{N}, \bar{y})$ is not jointly Gaussian and Eq. 2 is not the exact Bayesian update, a heteroscedastic emission treated as if it were homoscedastic. Nor can it be used on the interval average, which is not a function of the occupancy at any one time, so no conductance vector on the K states delivers its mean.

On the K^2 pairs of states a channel occupies at the two ends of the window one does. A *boundary state* is the pair $(i \to j)$ of those two states; an arrow marks an object that spans the window, so $P_{i\to j}(\Delta)$ is a transition across it, $\bar{\gamma}_{i\to j}$ the conductance averaged over it given that pair, and $\bar{v}_{i\to j}$ the variance that average still has once the pair is fixed. A channel that starts in i and ends in j contributes $\bar{\gamma}_{i\to j}$ on average, so the interval average is linear in the boundary counts and Eq. 2 applies to them with nothing added to it,

$$\boldsymbol{\mu} \to \boldsymbol{\mu}_{0\to\Delta}, \qquad \boldsymbol{\Sigma} \to \boldsymbol{\Sigma}_{0\to\Delta}, \qquad \boldsymbol{\gamma} \to \bar{\gamma}_{i\to j}, \qquad \epsilon^2 \to \epsilon^2 + N_{\text{ch}}\, \boldsymbol{\mu}_{0\to\Delta}^\top \bar{\boldsymbol{v}}, \tag{3}$$

where $\boldsymbol{\mu}_{0\to\Delta}$ and $\boldsymbol{\Sigma}_{0\to\Delta}$ are the mean and covariance of the boundary state, $\mu_{i\to j} = \mu_{0,i} P_{i\to j}(\Delta)$, and $\bar{\boldsymbol{v}}$ collects the $\bar{v}_{i\to j}$, so the last substitution charges to the instrumental term what the conductance still varies by once the pair is fixed. The start index is then summed away, the next window having no use for it. That is `IR`, and every other interval member is one of these substitutions made in part, each closed by a Gaussian (Appendix 1).

The pairs collapse before they are built. Eq. 3 puts $\boldsymbol{\Sigma}_{0\to\Delta}$ in two places, the predicted variance and the gain, and each contracts in closed form to K-vectors and $K \times K$ matrices, so no $K^2 \times K^2$ object is formed anywhere and one window costs K^3 however many channels the patch holds (Appendix 1). Chaining intervals leaves a single state variable per junction, since the end state of one interval is the start state of the next, so the record stays Markov across the windows as well as inside them.

In a recursive member the posterior pair becomes the prior for the next window; in a non-recursive member it is not carried forward. Carrying it forward is itself a closure, exact only while the end marginal is a sufficient summary of the boundary posterior, so the recursive members' Gaussian information is model-based rather than the exact information of the process.

Every member above therefore reports an information it has assumed. What follows is how far that assumption is from the information the member delivers against data from the true process, and how it is measured.

The distortion is a matrix, and only the residual check runs on a real recording

Everything below is read off ensembles of exactly simulated recordings at two parameter vectors, the simulation truth θ_{sim} and the pooled optimum θ_{pool}, the fit of the member under test to all the

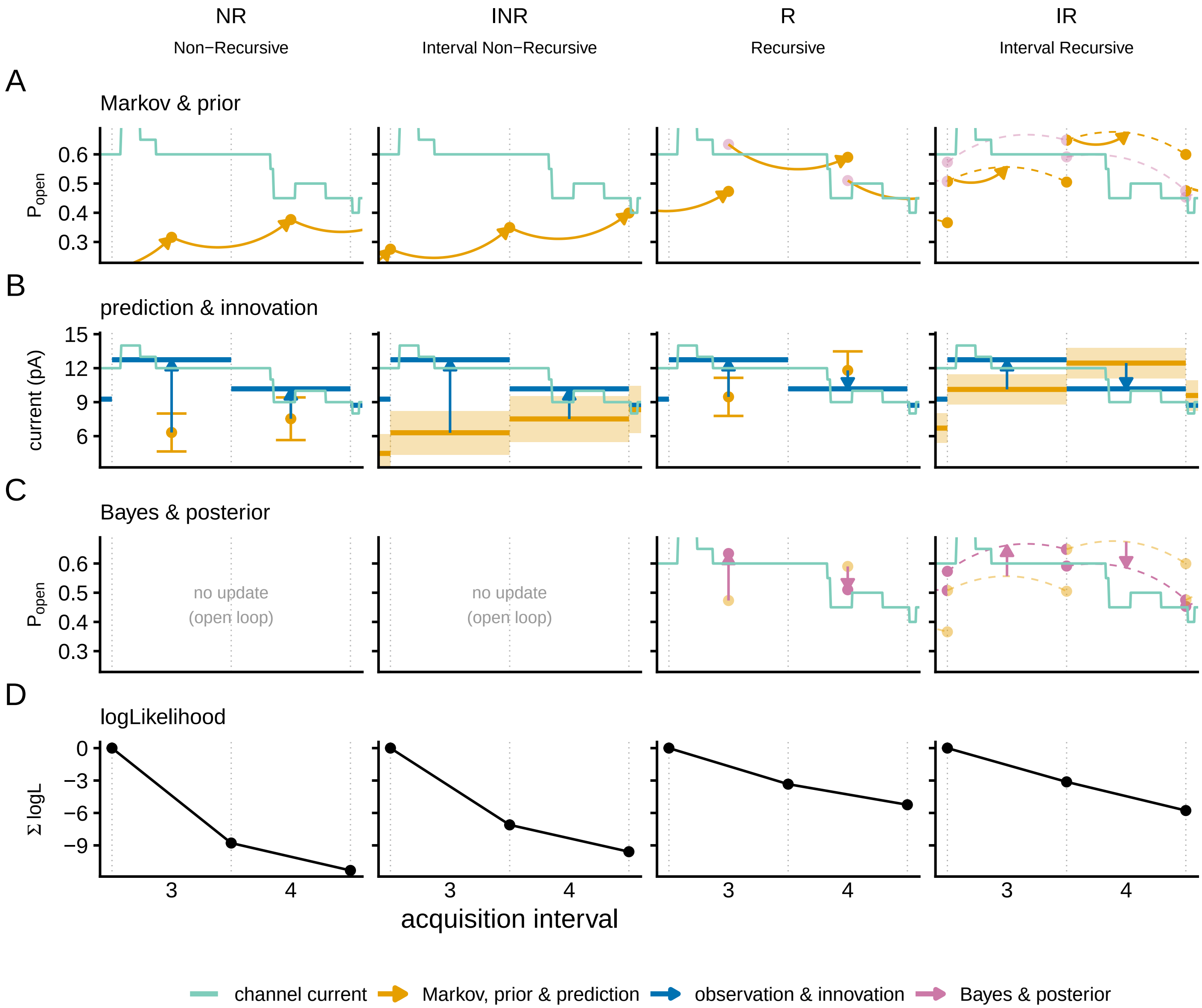

Figure 1. One filter step, and what each of the two axes changes. A simulated patch of $N_{\text{ch}} = 20$ two-state channels. The green trace is the current the channels carry, resolved inside the acquisition window, and the recording keeps one number per window, the average over it; dotted lines are the window boundaries and the axis counts windows, each 2 ms. At the lowest noise of the design plane, $\widetilde{S} = 0.01$, the recorded value sits on the average of that trace. Columns are the two axes crossed, the open-loop gating likelihood (`NR`), its interval-averaged partner (`INR`), the recursive filter (`R`) and the interval-conditioned filter (`IR`); the other four members of Table 1 are settings of the same cycle and are not drawn. Rows follow one turn of it, prior, prediction, posterior and score, with the posterior absent from the open-loop columns. In the two occupancy rows, `IR` alone carries a pair of occupancies, one at each end of the interval, tied by a dashed arc that is their covariance and implies no path through it. No number is quoted from this simulated recording.

Member	The states are conditioned on	Predicted mean	Predicted variance beyond the instrumental term	In words
Open loop: no update; every window is propagated from the initial condition				
`LSE`	nothing: there are no states	the deterministic mean current at the middle of the window	none: one scale over the whole record, marginalised rather than fitted (Methods)	the classical fit, un-averaged arm
`ILSE`	nothing: there are no states	the same mean current averaged over the window	the same constant	the classical fit, or the least-squares arm
`NR`	an instant, the middle of the window	the current at that instant	the spread of the instantaneous conductance across the states the channels occupy	the open-loop member
`INR`	the state at the start of the window	the interval-mean current given that state	the spread of that interval mean across the occupied states, and the variance of one channel's interval mean given its start state	the open-loop interval member
Recursive: the posterior of one window is the prior of the next				
`R`	an instant, the middle of the window	the current at that instant	the spread of the instantaneous conductance across the states the channels occupy	the recursive instantaneous filter
`MR`	the state at the start of the window	the interval-mean current given that state	the spread of that interval mean across the occupied states, and the variance of one channel's interval mean given its start state	the one-endpoint member
`VR`	the start, and both ends for the interval variance alone	the interval-mean current given the start state	the spread of that interval mean across the occupied states, and the variance of one channel's interval mean given both its endpoints	the variance-corrected rung
`IR`	the states at both ends of the window	the interval-mean current, the unobserved end state marginalized	the spread of the interval mean across the occupied pairs of endpoint states, and the variance of one channel's interval mean given both its endpoints	the boundary-conditioned member

Table 1. The eight members as settings of one construction. One conditioning choice runs through each row: the states an interval is conditioned on are the states its conductance is computed from and the states its update is taken on, so the second column governs the gain as well as the two columns beside it. Reading down, the second column is the window axis and the two blocks are the recursion axis. `LSE` and `ILSE` differ in the mean alone, which isolates the acquisition average with nothing else moving, and `NR` and `R`, and `INR` and `MR`, carry identical Gaussians and differ only in whether the recording is fed back into the states. How the remaining pairs differ is set out with the algebra in Appendix 2. In print, `IR` is MacroIR and `INR` is MacroINR. All eight are run over the design plane; the four that cross the two axes are drawn as one turn of the filter cycle in Figure 1, the equation-level specification is Appendix 2, and the dispatched flags and data keys are in Supplementary File 1.

recordings of one cell at once; the two answer different questions and every figure states which one it shows (Methods).

Three quantities check the same specification with increasing strength, and in the same order they ask for more than an experiment can give. The first is the standardized residual $r_t = (y_t - \mu_t)/\sigma_t$, with μ_t and σ_t^2 the likelihood's own predictive mean and variance for interval t: mean zero, unit variance and no autocorrelation across intervals if the one-step predictive distribution is correct *at the parameters it is evaluated at*. It is the only one of the three an experimentalist can run on a real recording, needing no ensemble and no known truth, and it does need a fitted point. A fit with the baseline and the noise level free absorbs most of the first two properties; the temporal structure is what no fit absorbs. That structure is summarised by the integrated autocorrelation (***Sokal, 1997***) of the residual, $\tau_9 = 1 + 2\sum_{k=1}^{9} \rho_k$ with ρ_k its lag-k correlation: it reads one for a white residual, and above one it counts how many intervals a fit effectively has for every interval it believes it has. Nine positive lags is what the bootstrap's memory allows. The window is in the symbol because it bounds the statistic at 19 and blunts it wherever the sampling is fine enough for the residual to stay correlated past it.

The second is the score. With $\ell(\theta)$ the total log-likelihood and $S(\theta) = \nabla_\theta \ell(\theta)$ its gradient, correct specification requires $\mathbb{E}[S(\theta_{\text{sim}})] = 0$. Projected through the parameter covariance, a nonzero mean score becomes a displacement in parameter units, $b = \mathbf{H}^{-1}\,\mathbb{E}[S]$, with $\mathbf{H}$ the Gaussian Fisher information the likelihood itself reports: how far the estimate is pushed. This one cannot be run on an experiment at all, needing a known truth and an ensemble to take the expectation over.

The third and strongest is the information itself. Correct specification requires $\mathrm{Cov}[S] = \mathbf{H}$, the covariance taken over independent recordings of the same cell, which no experiment supplies: one recording gives one score. An approximation can pass the residual and score checks and still violate this one by an order of magnitude. Because the failure is directional, we record it as a matrix,

$$\mathbf{C} = \mathbf{H}^{-1/2}\,\mathbf{J}\,\mathbf{H}^{-1/2},$$

with $\mathbf{J} = \mathrm{Cov}(S)$ the covariance of the total score, cross-interval terms included, so that $\mathbf{C} = \mathbf{I}$ exactly under correct specification. This is the classical information-matrix-equality comparison ***White*** (***1982***), measured here rather than tested, and read as a property of each approximation rather than as a hypothesis to accept or reject. We fix the direction by convention: $C_{ii} > 1$ means the likelihood under-reports the uncertainty in parameter i and is over-confident, $C_{ii} < 1$ means it over-reports and is conservative. The anchor $\mathbf{H}$ is the likelihood's own Gaussian Fisher information, formed from the predictive moments and their parameter derivatives that the score already requires, so it costs no extra likelihood pass and is positive semidefinite by construction (Methods). The comparison is not circular: $\mathbf{H}$ is the information the likelihood claims and $\mathbf{J}$ the information it delivers against data from the true process. All three checks run on the whole cost ladder; the bottom rung needs a qualification, its predictive variance being one constant rather than a gating quantity, and the Results state it where the number is quoted.

The same matrix repairs what it measures. The sandwich covariance ***Huber*** (***1967***); ***Godambe*** (***1960***)

$$\Sigma = \mathbf{H}^{-1}\,\mathbf{J}\,\mathbf{H}^{-1} = \mathbf{H}^{-1/2}\,\mathbf{C}\,\mathbf{H}^{-1/2}$$

substitutes the information the likelihood actually carries for the information it reported. It and the bias vector b are first-order approximations, verified without leaning on the expansion against the empirical covariance of the estimates over the simulated recordings (Figure 2, Appendix 3).

The distortion splits into two physically distinct parts. The sample distortion $\mathbf{C}_s = \mathbf{H}^{-1/2}\,\mathbf{J}_s\,\mathbf{H}^{-1/2}$, built from the within-interval score covariance $\mathbf{J}_s = \sum_t \mathrm{Cov}(s_t)$ with $s_t = \nabla_\theta \ell_t$, measures per-sample departure from Gaussianity, to leading order the third and fourth cumulants of the interval current. The correlation distortion $\mathbf{R} = \mathbf{J}_s^{-1/2}\,\mathbf{J}\,\mathbf{J}_s^{-1/2}$ measures the cross-interval correlation of the score, which $\mathbf{J}_s$ misses and which is the local time correlation of the current that ***Milescu et al.*** (***2005***) named as the dominant error of non-recursive fitting. One parameter at a time, F_t is the Gaussian

Fisher information of interval t alone and $J_t = \mathrm{Var}(s_t)$, and accumulated it is $J_T = \mathrm{Var}(\sum_{t\le T} s_t)$, not the sum of the J_t, that the reported F_T has to match, so per-interval and accumulated calibration are not equivalent. How the two parts compose is Appendix 3.

The correlation part has a reader-facing translation. Where $\mathbf{R}$ is one number times the identity, that number is the factor by which a likelihood over-counts its intervals. Where $\mathbf{R}$ is anisotropic, or $\mathbf{C}_s \neq \mathbf{I}$, no such correction suffices at any factor and the full matrix is needed.

A matrix is not a number, and two scalars summarise it without pretending to replace it. With λ_i the eigenvalues of $\mathbf{C}$ on the retained subspace, of dimension d (Methods), $\bar{e} = d^{-1}\sum_i \log\lambda_i$ and $s^2 = \mathrm{Var}_i(\log\lambda_i)$, the population variance with the same divisor, the two reported throughout are the exponentials

$$m = e^{\bar{e}} = \left(\prod_i \lambda_i\right)^{1/d}, \qquad a = e^{s}, \tag{4}$$

so that m is the typical factor by which the reported information is wrong, the geometric mean of the per-direction factors, and a is their multiplicative spread from direction to direction, both one at correct specification. An estimated a exceeds one even where the truth is isotropic, so a measured spread is evidence of a real one only above the floor that finite sampling sets (Appendix 3). In error-bar units take the square root of any of these, since the distortion is a ratio of informations: 1.15 is a 7% error on a confidence interval, 2 is 41%, and 4 turns a nominal 95% interval into about 68%.

Results

A recursive likelihood cannot be checked against a record by eye: it predicts each sample from the samples before it, so it follows the record whether or not the uncertainty it reports is the uncertainty it delivers. What follows measures it against least squares, on recordings simulated exactly from the process every member approximates.

Throughout, $\widetilde{\Delta}$ and $\widetilde{S}$ are the acquisition interval and the instrumental noise in the natural units defined above, a cell is one combination of channel count, instrumental noise and acquisition interval at which recordings are simulated, and the eight members are those of Table 1. The two least-squares arms are one flag apart, `LSE` fitting the conductance at the sample and `ILSE` fitting it averaged over the acquisition window, and both are run over the whole design plane, on the same cells as each other and as every other member.

The window and the recursion repair different moments, and the correction is checked against the fits themselves

Figure 2 reads the ladder at one design cell, and each panel shows the two quantities the rest of the paper maps. The cloud away from the truth is the bias. The reported ellipse away from the empirical one is the distortion, and it takes two numbers rather than one: its magnitude, the typical factor by which the reported uncertainty is wrong, and its anisotropy, how much that factor changes from one direction to another (m and a of Eq. (4), the two orange numbers of each panel). Both are one when the member is calibrated, and both are read here from a cloud of a thousand maximum-likelihood fits, each taken on ten simulated recordings at once, where every later figure reads them from the score at one anchor and never from the estimates, and the two constructions agree only to first order (Appendix 3).

The truth and the cloud mean coincide only for the two least-squares arms and for `INR` and `IR`; the other four macroscopic members are displaced, within the dispersion of their own cloud on the rates and beyond it on the unitary current and the channel number. The noise level is biased in `NR` alone (Figure 2–source data 1). That the mean current pins the product of the channel number and the unitary current, while only the fluctuations can separate the two, is the identifiability structure known for models of this kind (***Hines et al., 2014***). The open circle of each panel is where the score and the information alone put that displacement, with no fitting, and it lands on the same side as the cloud and close to it: on the unitary current -0.287 against a measured -0.228 for `MR`, and

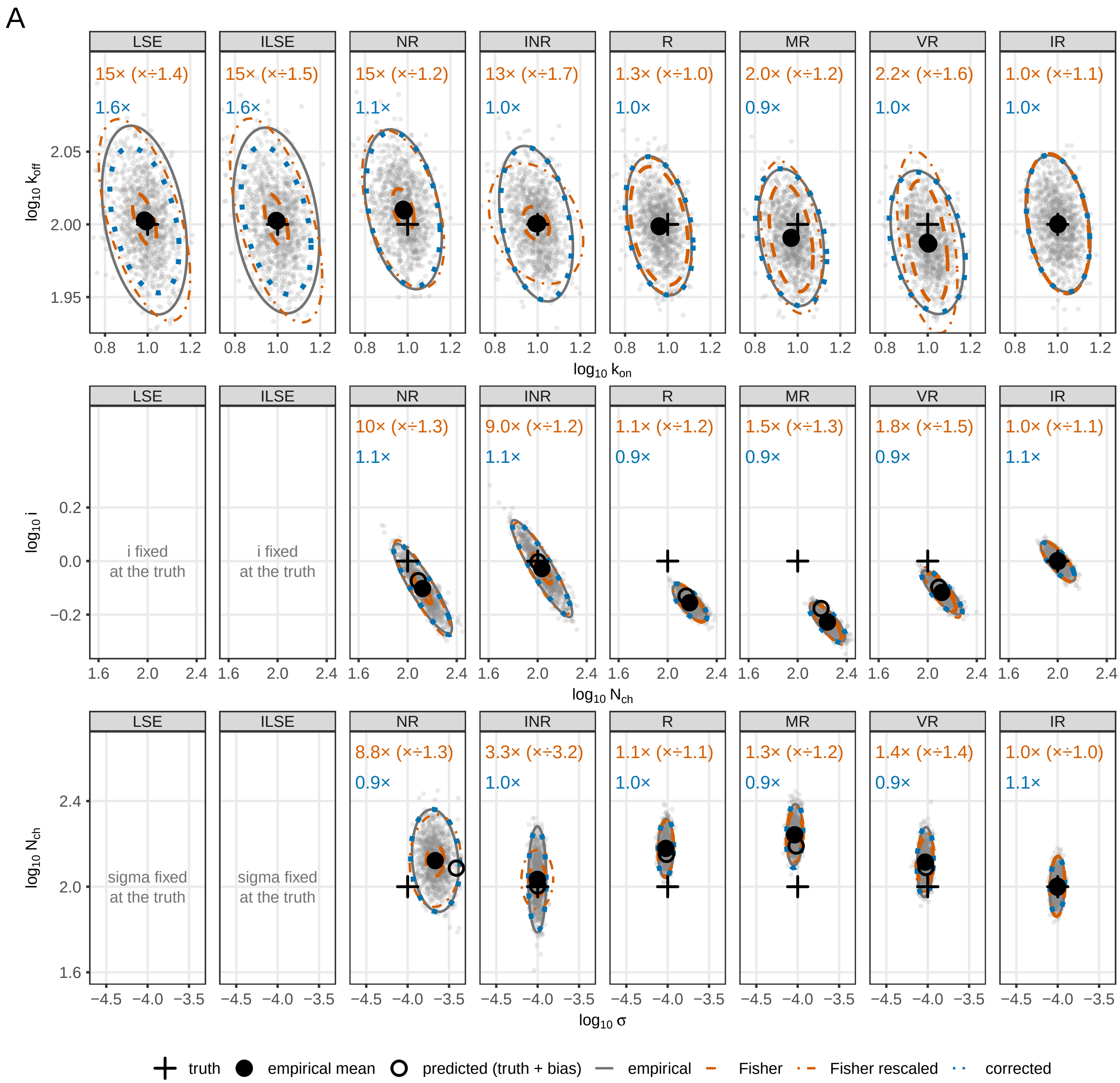

Figure 2. Parameter recovery and the calibration of the reported interval, for the whole ladder at one design cell. $N_{\mathrm{ch}} = 100$, $\widetilde{S} = 0.01$, $\widetilde{\Delta} = 0.1$; $10{,}000$ recordings in groups of ten, so one thousand maximum-likelihood estimates per member. Columns are the eight members in ladder order, rows the kinetic, amplitude and noise pairs, with σ the instrumental noise level. Each panel carries the cloud of estimates and four 95% ellipses at its mean, named in the key. Above the cloud, the orange numbers are the magnitude of the distortion and, in parentheses, its anisotropy; the blue number is the empirical ellipse over the corrected one. All three read 1.0 for a calibrated member. Covariances are the Gaussian-Fisher family at the pooled estimate, divided by the group size.

Figure 2—figure supplement 1. The sandwich prediction against the empirical multivariate distribution of the estimates, as a Mahalanobis quantile-quantile plot.

Figure 2—source data 1. First-moment bias, measured and score-predicted, with 95% bootstrap intervals.

Figure 2—source data 2. Distortion magnitude, anisotropy, corrected ratio and eigendirections, per member and parameter pair.

nothing for `IR`, where the cloud resolves none. Figure 3 shows the score that produces it. On the rates no member is off by more than a tenth and the order there is not the ladder's, `R` sitting last of the eight and the rate bias of `IR` and `VR` not resolving at all; the plane is where the least-squares arms' rate bias is established (Figure 4A). The interval average is what removes the displacement, in all three families that carry both arms, and recursion on its own does not: it leaves `R` the least centred member of the eight in the rates and further out in the unitary current than the open-loop member it is built from.

The distortion orders the ladder the other way, and not monotonically: the orange numbers read one for `IR` on every pair, within a third of it for `R`, behind `R` in every pair for the two one-endpoint corrections built on it, and an order of magnitude higher, evenly, for the members that leave the correlation unmodelled, `INR` excepted in the noise pair, the one direction its interval variance repairs. What recovers the calibration is the second endpoint entering the gain (Appendix 2).

The factor is not even one number per member: `INR`'s noise-pair distortion factors into more than ten along the channel-number direction and one along the noise axis, the member that repairs the noise estimate reporting the noise uncertainty exactly right and the channel number with a standard error more than three times too small, so no single rescaling serves.

What does serve is a matrix. The distortion is measured as one, and the same matrix corrects the covariance it measures (the sandwich, Appendix 3), which the rest of the paper reports. Whether that correction is right is what this figure settles, dividing the empirical ellipse of each panel by the corrected one: at one, the correction has recovered the covariance the estimator actually has. It does for every gating-aware member, and at a second cell, where each fit is taken on a hundred recordings, it lifts the coverage of `IR`'s nominal 95% region from 0.914 to 0.947 in all six directions at once (Figure 2–figure supplement 1). It fails alike for the two least-squares arms, at a factor of 1.6 in both, two members that differ in every first-moment quantity, which puts the failure in what they share; they are also the two arms whose analytic Fisher the differenced check flags (Supplementary File 1), at a size consistent with the residual measured here.

Least squares does estimate the channel number, to within 0.012 in $\log_{10}$, but with the unitary current pinned at its true value that estimate is the mean current under another name, which is why its other two columns are empty.

The error bar fails for two reasons, one within a sample and one across them, and they can cancel

The miscalibration of Figure 2 is a property of the likelihood and not the optimiser: Figure 3 scores one common set of a thousand recordings at the same cell, interval by interval, with no fitting. Row (A) separates the families, the recursive members following the record where the open-loop members and both least-squares arms predict the ensemble, and annotates the probability each gives it: the boundary-conditioned member on top, the two least-squares arms 119 nats below (Figure 3–source data 1). Over a common sample those differences are Kullback-Leibler differences, so the ladder orders by distance to the truth: exactly for the six whose scale is fixed, and as a lower bound for the two that integrate theirs out (Eq. 8), which a proper prior would only lower. The ladder ranks; it cannot test, and the rows below it can: `INR` sits fifteen nats above `NR` and is the worst of the eight in the accumulated ratio, 20.8 against 13.6.

The three diagnostics are kept apart, each with the value a correct likelihood must give: the residual, its variance in (B) and its memory in the black series of (G), one and zero; the score, its mean in (D), its variance against the information in (E) and (F) and its memory in the coloured series of (G), zero, one and zero; the information alone in (C). Only the residual rows need no knowledge of the truth.

The score's mean says whether the estimate will be displaced. Per interval only `IR` sits at the chance level of 0.05, departing from zero at 0.04 to 0.07 of the informative intervals, those whose information is above the numerical floor, against 0.10 for `INR`, 0.19 for both least-squares arms and 0.41 to 0.73 for the displaced four. The count is not the displacement, since opposite signs cancel;

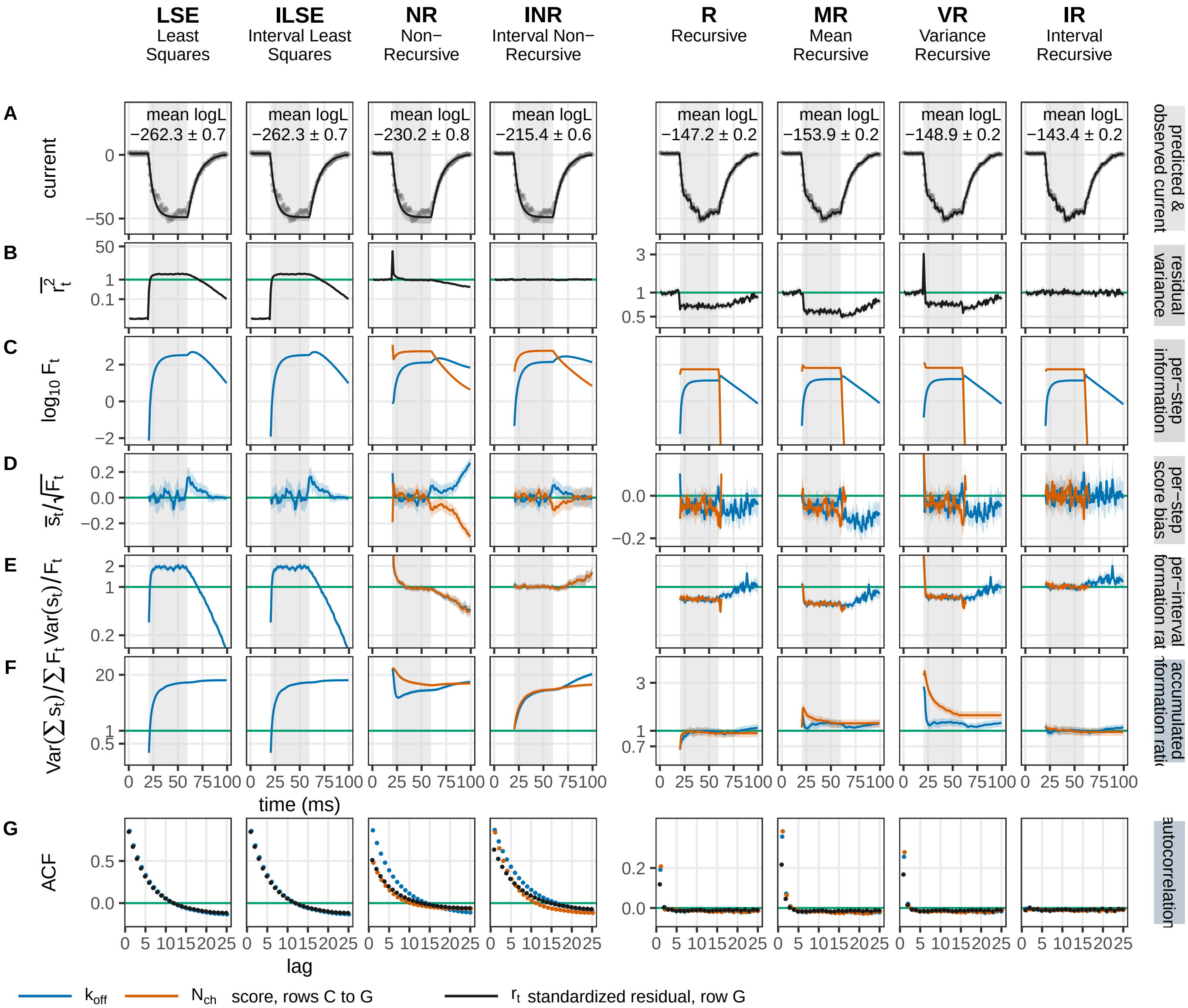

Figure 3. The calibration cascade in time. One set of 1,000 recordings ($N_{\mathrm{ch}} = 100$, $\widetilde{S} = 0.01$, $\widetilde{\Delta} = 0.1$) is scored by each of the eight members, in cost-ladder order and with no fitting. Rows A and B are properties of the data, in grey; C to G are per parameter, the closing rate and the channel number in colour. (A) An example recording, its band the member's own predictive ± 1 s.d., annotated with the ensemble-mean log-likelihood. (B) The mean squared standardized residual. (C) The per-interval Fisher information. (D) The mean per-interval score over $\sqrt{F_t}$, standardized so that the two parameters share an axis. (E) The per-interval information ratio $\mathrm{Var}(s_t)/F_t$. (F) The same ratio accumulated, J_T/F_T. (G) The autocorrelation against lag, of the residual in black and of the score in colour. Bands are 95% bootstrap intervals over the recordings and curves stop where the information reaches the numerical floor, which is also where an interval stops counting as informative in the text and in the source data. Non-negative quantities are on a logarithmic axis and signed ones on a linear axis, with the calibrated value on the ticks; the four open-loop columns share a y-scale block distinct from the four recursive ones.

Figure 3—figure supplement 1. Per-step information against score variance, for the opening rate and the closing rate.

Figure 3—figure supplement 2. The same, for the unitary current and the channel number.

Figure 3—figure supplement 3. The residual and the score in the three parameters not coloured above.

Figure 3—source data 1. The calibration cascade as numbers: per member and coloured parameter, the fraction of informative intervals whose score departs from zero and whose information ratio departs from one, the accumulated ratio J_T/F_T with its interval, the lag-one score autocorrelation with its interval, and the ensemble-mean log-likelihood.

that is the accumulated score against the whole information matrix, $b = F^{-1}\,\mathrm{E}[\sum_t s_t]$, in marginal standard errors, and it reproduces Figure 2's split: the four centred members within a quarter of a standard error of zero, the displaced four out by half to two on the unitary current and the channel number (Figure 4A).

Misreported uncertainty has two sources. The score variance is what the dispersion should equal, and per interval no member misses it by more than 1.6 at the median; accumulated, the four members that model no correlation in time miss it by 12 to 21 and the four recursive ones stay below 1.5 (E against F). The residual gives the per-sample half with no parameter at all: its mean square is one only for the two interval-averaged members, below one for `R`, `MR` and `VR`, and 1.15 for `NR`, most of that excess in the one interval where the agonist steps on, where the occupancy changes inside the window its variance cannot see. The marginalised scale pins both least-squares arms at exactly one over the recording, while they run from 0.010 before the agonist to 1.91 over the pulse.

The second source is correlation in time, which no per-sample check sees: `INR` has the calibrated residual variance of `IR` and reports its accumulated information twenty times wrong. The accumulated ratio is the per-interval one times a correlation factor: `IR` carries both at one, 1.07 and 1.01; `R` reaches a comparable 1.07 from two errors of opposite sign, a score that varies 28% less than the information it claims on a single interval, an error bar 18% too wide, against a correlation that inflates the accumulated variance by 48%. The cancellation belongs to the point of evaluation: at the pooled optimum, where the estimates land, `R`'s factors read 1.09 and 1.40, both above one, for 1.51, while `IR` reads the same at either (Figure 4–figure supplement 1). At lag one the score autocorrelation on k_{off} is -0.004 for `IR`, its 95% interval covering zero, against 0.191 for `R` and 0.864 for `NR`: the local correlation ***Milescu et al.*** (***2005***) named as the dominant error of independent-interval fitting, now at the level of the inference. The residual carries the same memory, 0.85 for either least-squares arm, 0.51 for `NR`, 0.12 for `R` and -0.01 for `IR`, the half a real recording gives. These are the two mechanisms Figure 4B maps.

Row (C) alone says where each parameter is measured. The channel number and the opening rate reach the numerical floor within six intervals of the agonist being removed, while the closing rate survives through the mean and the unitary current through the variance of the channels still open (Figure 3–figure supplements 1 and 2): a member that conditions on the past already carries the number still open, so a parameter reaching the recording only through it has nothing left to give, while the noise level keeps a white score throughout (Figure 3–figure supplement 3). The two open-loop members go on reporting information about the channel count through the washout, 15 to 16% of their pulse total against 0.5 to 0.8%, which their accumulated ratio charges them for as the same information counted twice.

Only the interval members are centred, and only the boundary-conditioned one is calibrated, everywhere on the design plane

Figures 2 and 3 measure one recording condition. Figure 4 asks what changes over the plane: the same two measurements at every cell of a grid in the sampling interval, the instrumental noise and the channel count, scored by the same eight members, six here and `MR` and `VR` in Figure 4–figure supplement 2.

The verdict does not change. Across the plane `INR` misses the channel number by two per cent and `IR` by one, where the other four macroscopic members miss it by a fifth to a third, and `IR` is the only member whose error bar is calibrated, leaving the 15% band on five per cent of the cells every member shares against `R`'s twenty-nine and seventy for the other four. They all miss in the same direction: nine in ten of the cells outside the band report an error bar too narrow rather than too wide. Where they fail, the four that leave the correlation unmodelled are off by a factor of about two to three on the error bar, and `R` and `IR` by about a tenth. Both partial corrections shave the first moment and lose the second (Figure 4–figure supplement 2): `VR` cuts `R`'s channel-number bias by about a third and `MR` by a sixth, while on that parameter their distortion climbs with the channel

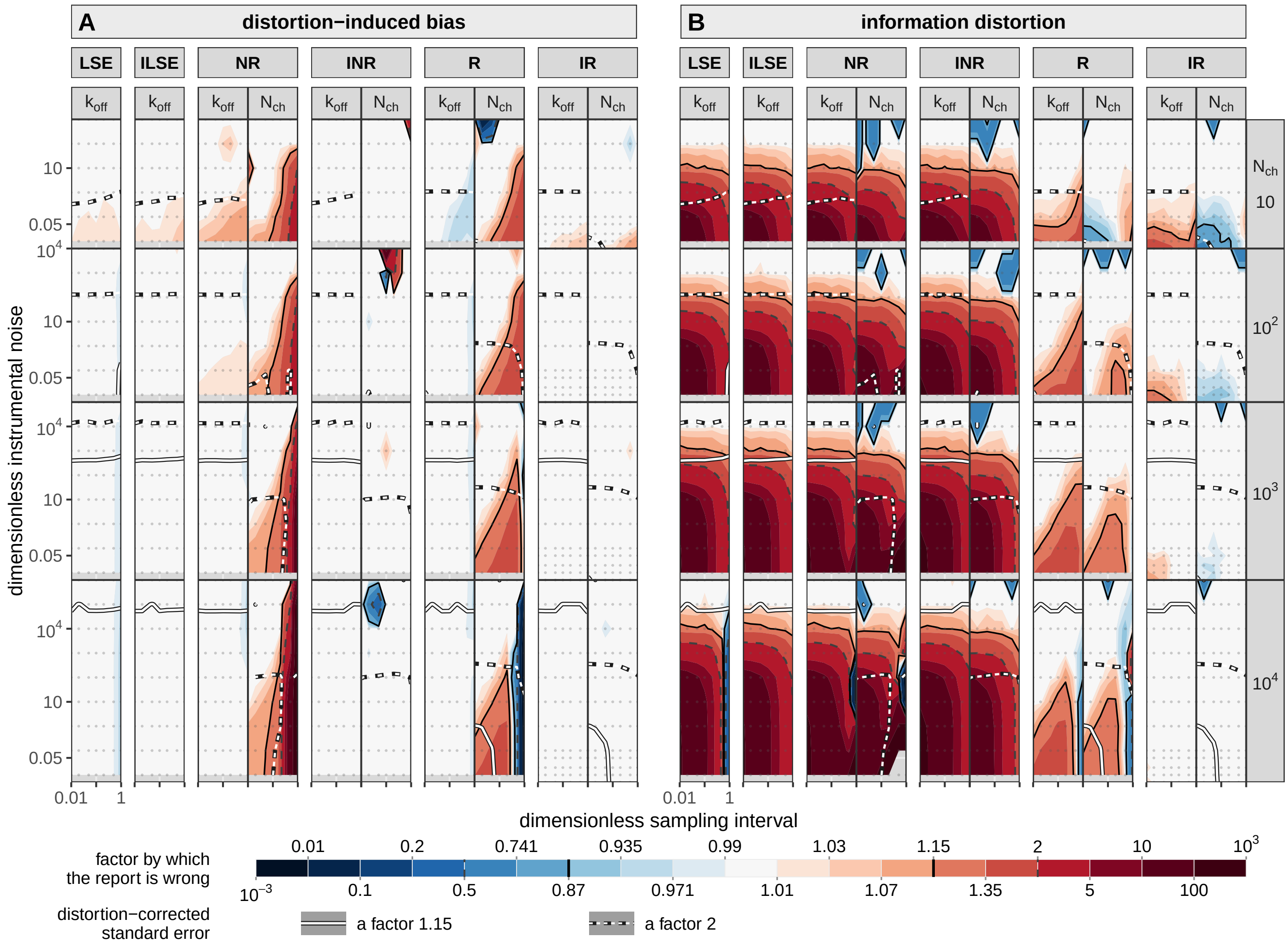

Figure 4. The design plane: how wrong each member's report is, in both moments, on one scale. Ten thousand recordings are simulated at every cell of the grid and scored by each member of the cost ladder. (A) The distortion-induced bias, evaluated at the simulation truth θ_{sim}: how far the estimate is displaced. (B) The information distortion, evaluated at the pooled optimum θ_{pool} (Methods): how far the uncertainty a member reports is from the one it delivers. Columns nest twice, member and then parameter, k_{off} and N_{ch}. Both least-squares arms are on the grid and each holds one column per half rather than two, their four-parameter configuration fixing the unitary current and the noise level at the simulated values (Methods). Rows are the channel count, each given a height proportional to the range of instrumental noise the sweep reaches there. Inside a panel the axes are the sampling interval in units of τ and the dimensionless instrumental noise, one decade the same distance on both. Colour is shared by the two halves: the factor by which the report is wrong, centred on one, pale right and saturated wrong in either direction, blue conservative, clipped at $10^{\pm 3}$. What a factor means differs between them: in (A) it is a factor on the parameter, so 2 means the estimate is out by a factor of two, while in (B) it is a ratio of variances, so 2 means the reported error bar is $\sqrt{2}$, about 40%, too narrow. Lines carry two codes: colour says which quantity, dark hairlines cutting the colour field and white lines on a dark casing cutting the distortion-corrected standard error, which asks whether the parameter can be measured at all rather than whether its error bar is calibrated; dash says which criterion, solid a factor 1.15 and dashed a factor 2, the two thresholds used throughout.

Figure 4—figure supplement 1. Figure 4's second moment factored into a per-sample part and a temporal-correlation part, over the same plane.

Figure 4—figure supplement 2. The same plane for the four rungs of the recursive family, in both moments.

Figure 4—figure supplement 3. Can the distortion be undone by counting fewer intervals? The magnitude of the information distortion against its anisotropy.

Figure 4—figure supplement 4. What each member achieves: the distortion-corrected standard error.

count, to 1.10 and 1.16 at ten thousand channels where `R` stays at 1.01. On the rates both sit on `R`, so the trade is confined to the parameter they were built for. Neither reaches either interval member.

What the plane adds is where each verdict breaks. `IR`'s closing-rate error bar is calibrated everywhere except at the two smallest channel counts with the least noise, most of that corner being ground only `IR` was swept on, below the noise floor the rest share. It is the channel count that repairs it: its closing-rate distortion converges from 1.37 at ten channels to 1.006 at ten thousand, as both the current and the state distribution become Gaussian by the central limit.

Instrumental noise repairs everyone. As it grows the recorded current becomes Gaussian whatever the member assumes: the closing-rate distortion of least squares falls from 13 to 1 as the noise rises by five decades, and `R`, which starts at 1.5, is inside the same band after two. Colour at the noisy end of each row should not be read cell by cell: ten thousand recordings often cannot resolve a bias of an eighth there, and the variation between neighbours is sampling noise.

A short sampling interval attenuates the missing average without removing it. The channel-number bias of `NR` runs from a factor of eleven at the coarsest sampling to nine per cent at the finest, and `R`'s from four fifths of the parameter to seven per cent, while `INR` stays under two and a half per cent and `IR` under one at every interval. The two least-squares arms part only at the coarsest interval, where the arm that does not average picks up a rate bias close to twenty times the other's and understates its own information (Figure 4–figure supplement 3); that single flag is why both are on the grid. That is the whole of what the average buys a fit that does not model the gating variance: it keeps a window comparable to the kinetics from aliasing the deterministic mean, and buys nothing at any finer one. To the two members that do model it the same step is worth a factor of 9 to 50 on the same rate offset at every interval, and 240 on the channel number at the coarsest.

Three qualifications. The deep blue block in `R`'s channel number at ten thousand channels and the coarsest interval is where it cannot tell the unitary current from the channel number: moving the two together costs it almost no likelihood, so both wander, -2.79 in $\log_{10}$ here against $+2.87$ on the current the figure does not draw. Grey is absence and not a result: the strip along the bottom of each row is noise that member was not swept at, and the blank patch in `NR`'s second moment at the same corner is where its distortion is not positive at all. And the anchor behind every factor was checked against a differenced Fisher information and overturns no verdict, least squares being the one it would make more conservative (Supplementary File 1).

How much gating noise a recording carries can be read from the classical fit, and it chooses a likelihood rather than correcting one

A reader holding one recording does not know its channel count or its instrumental noise. But what governs the distortion is their ratio, and the least-squares fit has access to it: the integrated autocorrelation of its standardized residual, τ_9, reads 1 when the residual is white and grows as the gating fluctuation comes to dominate the instrumental noise. Cells that share the ratio at one sampling interval agree in τ_9 to a median of 1% across four decades of channel count (Figure 5–figure supplement 1), so a reader who measures it has measured how much of their record's variance is gating. Figure 5 plots each likelihood's distortion against its inverse, the fraction of intervals the classical fit effectively holds, with the dashed line marking what a deflation by that factor would predict.

At the right edge, where the instrumental noise dominates, the residual is white and the cheap fit is honest: below $\tau_9 = 1.20$ `ILSE` reports its information within a fifth of the truth in every cell of the plane, and the two least-squares arms are interchangeable there at every interval but the coarsest. Away from it `ILSE` and `INR` run out to factors of 57 and 33 while `R` stays within a factor of 1.7 and `IR` within 1.08. What the axis cannot carry is direction. The equality fails through the correlation of the score across intervals (Figure 3) and the score is a vector, so what is misreported is a matrix, and any estimate of its size, however good, leaves the anisotropy still to be estimated. That is the bar, spanning a median factor of 2.4 across directions for `ILSE`; the dashed line, where

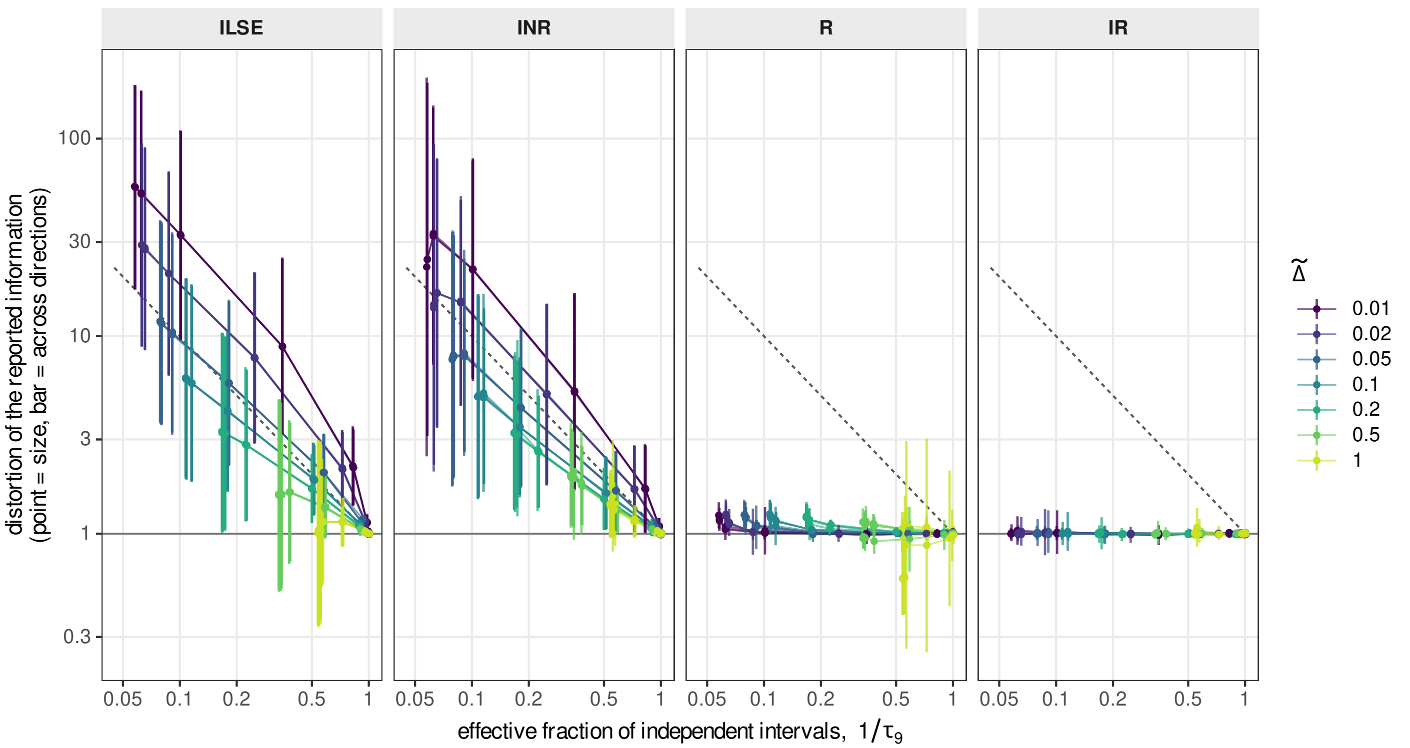

Figure 5. The repair a reader would apply, against the number they would apply it with. Every mark is one cell of the design grid, scored over the same $10{,}000$ recordings as Figure 4, with no binning. The horizontal axis, shared by the four panels, is $1/\tau_9$ of the standardized residual of the interval-averaged least-squares arm `ILSE`: the fraction of its intervals the classical fit effectively holds, and the multiplier a reader would apply to the interval count. It reads 1 for a white residual, and its left edge at $1/19$ is set by the nine-lag window rather than by the data. The un-averaged arm gives the same value to two digits at every interval but the coarsest. The vertical axis, also shared, is the distortion of the reported information: the point is its magnitude and the bar runs from that magnitude divided by its anisotropy to the magnitude multiplied by it, one standard deviation of $\log \lambda$ either side of the geometric mean, so it shows the typical spread across directions and not its extremes (Eq. 4). The error-bar scale is its square root. The dashed line is the deflation itself, $\text{distortion} = \tau_9$: a mark on it is a report the reader could repair with the number they measured, and the length of the bar is what no single factor removes. No threshold is drawn, the criterion belonging to the reader. Colour is the acquisition interval $\widetilde{\Delta}$; each line joins one channel count at one interval swept over the instrumental noise, so twenty-eight are drawn per panel and seven can be seen, the four channel counts landing on one another (Figure 5–figure supplement 1). The four members are the two axes of the family crossed (Table 1). In the `ILSE` panel both axes come from one fit; scaling a residual by a constant leaves its autocorrelation unchanged, so the horizontal axis is not self-referential. Each panel reports what its own fit says about its own parameter set (Methods); nothing is pooled.

Figure 5—figure supplement 1. The instrumental noise and the channel count act only through their ratio.

Figure 5—figure supplement 2. Each member against the memory left in its own residual, rather than in the least-squares residual.

a reader applying that number would land, falls inside those bars rather than through the points.

Two qualifications. The subscript is the window: the sum runs over nine positive lags, so τ_9 is bounded at 19 and understates the memory wherever fine sampling keeps the residual correlated past it, which is why the same $\tau_9 \approx 2.9$ carries a distortion of 8.9 at the shortest sampling interval and 1.6 at $\widetilde{\Delta} = 0.5$, and why the interval sits beside it in colour. And the axis is the classical fit's in all four panels, so a mark of `IR` does not say that `IR` left that memory anywhere. Each member against the memory in its own residual is Figure 5–figure supplement 2.

Sampling a hundred times faster buys the noise level and nine per cent of a rate

Everything so far asks whether a member's report is calibrated. The complementary question is what the report is worth, and the standard error each member delivers answers it: the square root of the diagonal of its distortion-corrected covariance (Figure 4–figure supplement 4). Read along the acquisition interval it says something the design axes do not otherwise make visible: sampling faster is not more data, it is more *independent* data.

Sampling a hundred times faster, from $\widetilde{\Delta} = 1$ to $\widetilde{\Delta} = 0.01$ at $N_{\mathrm{ch}} = 10^4$, divides the error on the instrumental noise level by 11.8 to 20.5 across the noise sweep, at or above the factor of ten a hundredfold increase in independent samples would give. It divides the error on the channel number and the unitary current by 6.7 and 7.7 at the lowest noise every member reaches, a gain that decays monotonically with instrumental noise, to 1.7 at $\widetilde{S} = 10$ and to nothing beyond. And it divides the error on either rate by at most 1.09, anywhere on the plane.

The three answers are one statement, ordered by the correlation time of whatever each parameter is measured from. The instrumental noise level is the scale of a white process, so every additional sample is independent and its relative error follows the sample count: measured against the interval, the log-log slope is 0.60 for `IR` and 0.52 to 0.60 for the other three that carry it, against the 0.5 of an inverse square root. It is the one parameter whose error the sampling rate sets and the noise level does not, varying by a quarter over the whole sweep. The amplitude pair is measured from the gating fluctuation, correlated on the scale of the kinetics, so faster sampling buys the part of its spectrum above that scale, which is real and bounded, and nothing once the instrumental noise has covered the fluctuation. The rates are measured from the shape of the deterministic transient, which any interval shorter than the relaxation already resolves, so the extra samples are redundant. Averaging n samples of a correlated process divides the variance by n/τ_9 and not by n, and τ_9 is the quantity Figure 5 puts on its horizontal axis.

One consequence belongs to the reduction rather than to the acquisition. This axis is not the sampling rate of the rig: a record acquired fast and uniformly is reduced before it is fitted by grouping consecutive samples and averaging them, and the average of window averages is the average over the window they span, so the reduced record is the same observable at a larger Δ. The numbers above are what that grouping costs: almost nothing for a kinetic question, the whole gain for an amplitude one. The interval members take the window as an argument rather than as a constant, so the groups need not be equal and the schedule can follow the transient; the members written for an instant describe a different random variable once the record is grouped at all.

In the few-channel corner the boundary-conditioned member misreports two parameters in opposite directions

Figure 6 resolves the corner where `IR` departs. Finding that corner was half the object of the measurement, since a diagnostic that never fires on the member it is applied to says nothing about what it can resolve.

The departure is two-sided within a single cell. At ten channels and $\widetilde{S} = 0.005$ the reported error bar of the closing rate is too small by a factor of 1.62 at the shortest interval and that of the channel number too large by a factor of 0.65, so one likelihood is over-confident about a rate and conservative about a count in the same recording. The two parts of the decomposition answer to the acquisition interval in opposite directions: the correlation part of the closing rate falls as

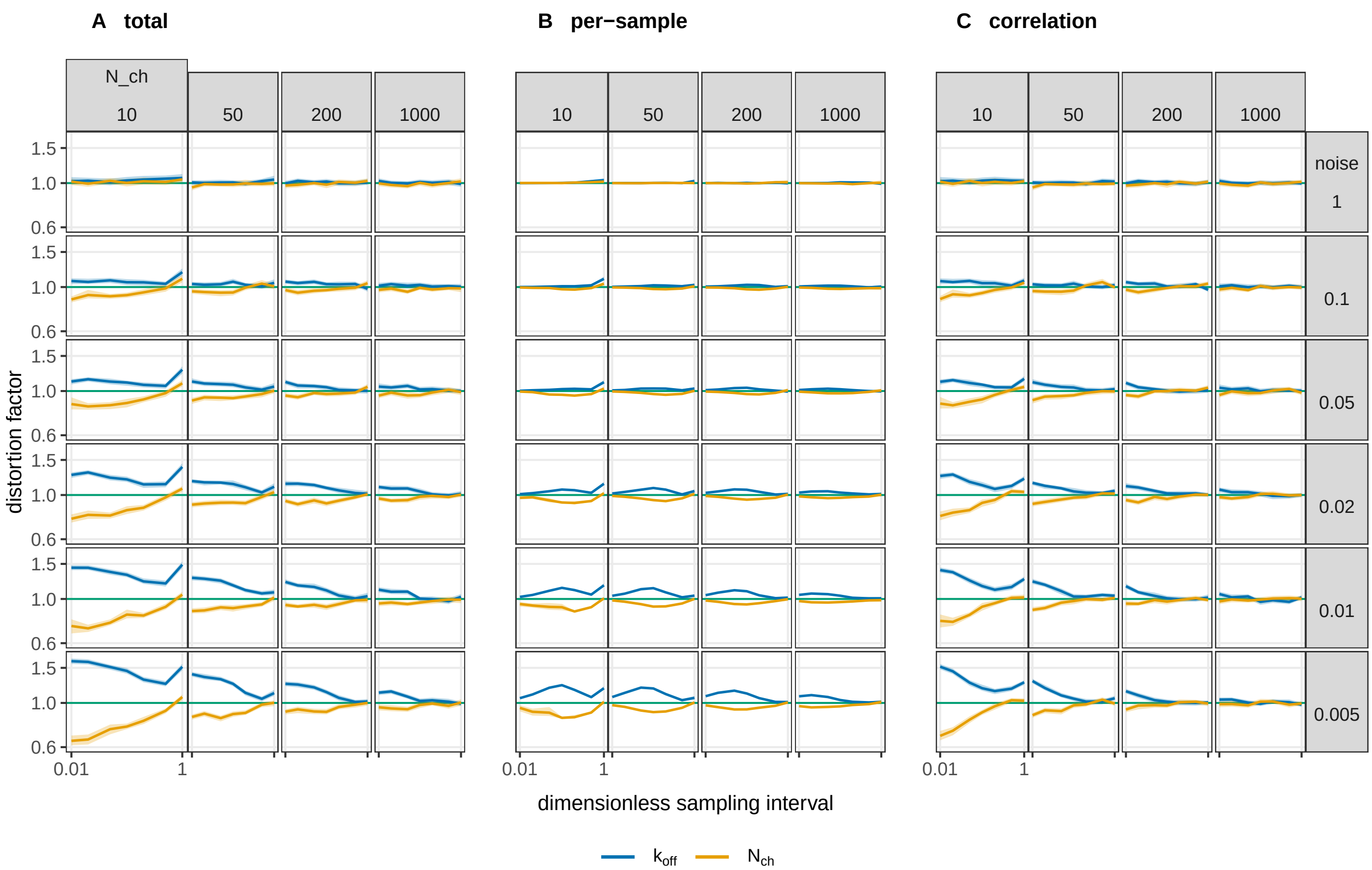

Figure 6. Where the boundary-conditioned filter stops reporting a calibrated error bar, and which of the two discarded moments is responsible. The same measurement as Figure 4, the Gaussian information distortion at the pooled optimum on the same runs, resolved on the corner where IR departs: on Figure 4's scale a factor of 1.5 is a faint tint and here it needs an axis. The three blocks are the distortion and the two parts it decomposes into: (A) the total, (B) the part carried by the per-sample non-Gaussianity of the score, and (C) the part carried by its correlation across intervals; their product reproduces the total to within 1% over the whole grid. Each block is a grid of the design plane in the orientation of Figure 7: the channel count increases to the right, the dimensionless noise $\widetilde{S}$ increases upward, and inside every cell the horizontal axis is $\widetilde{\Delta}$ from 0.01 to 1, shared by every panel and labelled once per block. The vertical axis is the distortion factor on a logarithmic scale, shared by all twelve columns so that they can be compared, with the green line at 1 marking a report that is neither over- nor under-confident. Colour is the parameter. Bands are 95% percentile bootstrap intervals over recordings, of median half-width 0.03. The two readings the figure supports, and the limits on them, are in the text.

the interval coarsens while the per-sample part rises, so sampling more coarsely buys back the memory and pays for it in the per-sample non-Gaussianity. Both parts vanish by $\widetilde{S} = 1$, where the instrumental noise has made the emission Gaussian enough that neither closure is strained and the recording is outside both regimes.

Two limits travel with the figure. The family applies both of its Gaussian approximations at every step and the score sees only their product, so a departure measured here cannot be assigned to one of them; separating them needs the microscopic recursion, a companion paper's subject. And the grid is truncated where the effect grows: the closing rate's departure is still accelerating at the lowest simulated noise, reading 1.22, 1.37 and 1.52 over the last three noise steps at ten channels, so the figure locates the onset and not the far side.

A recording loses its parameters in a fixed order, and the amplitude goes before least squares becomes calibrated

What a recording can be asked for is set by two properties of the preparation: how many channels the patch holds, and how much noise the rig adds. Those are the axes of Figure 7, which they cut into five regions, each with its own answer to which parameters can be recovered and with which method. Of the eight methods on the ladder only two are ever the right answer here, `IR`, the boundary-conditioned member, and the classical fit; the rest are biased or miscalibrated wherever they are cheap (Figure 4). Read upward at a fixed channel count, which is a noisier rig or a smaller unitary current, the regions come in a fixed order. While the gating fluctuation dominates, use `IR`: it recovers the amplitude and the rates, all four parameters in five cells of every six, with a calibrated interval. The unitary current is the first thing to go, because the fluctuation stops separating it from the channel number and the mean current pins only their product. Its loss opens the band this map exists to show, 1.7 decades of noise tall: there the fluctuation is already too small to give the current, and still large enough to make the least-squares interval several times too narrow, so the rates survive but `IR` is still the only one that can report them. Above the band the classical fit is calibrated too and is then the one to use, being the cheaper; higher still the rates go as well and nothing is measurable. A recording with few channels begins below all of this, in the hatched corner, where `IR` recovers the parameters but its own reported interval has to be corrected first (Figure 6). No measured cell offers the amplitude and a calibrated classical interval at once.

Two ratios put those boundaries where they are, because the record carries two observables that scale differently with the channel count. Whether the gating fluctuation stands above the instrument is a ratio of variances, constant along $\widetilde{S} \propto N_{\mathrm{ch}}$ (Figure 5–figure supplement 1); whether the deterministic transient stands above it is a ratio of amplitudes, constant along $\widetilde{S} \propto N_{\mathrm{ch}}^2$. Every line in the figure is one, the other, or a combination, and its slope says which: 1.03 where the classical fit becomes calibrated and 1.11 where the unitary current stops being measurable, both fluctuation-limited; 2.09 and 2.20 for the two rates, which live in the transient; and 1.47 for the channel number, which is read from both. The one line that falls, -0.50, is not a signal-to-noise condition at all: it is the edge of the telegraphic regime, where the interval average stops being the switching signal and becomes Gaussian enough for the closure `IR` assumes, and both more channels and more instrumental noise help it.

The three configurations are placed by the channel counts each can hold (Methods). An excised outside-out patch and a whole-cell recording fall in `IR`'s territory over most of their range, each crossing one boundary at one corner, so these lines run through the conditions people record in and not around them. Two floors matter at the bench: under about 17 channels the channel number cannot be recovered at any noise, and under 52 the opening rate. The oocyte, above 10^5 channels, sits past the simulated grid; what carries out there is the ordering and not the positions, since the two lines converge by only 0.09 decades per decade of channels, leaving 1.4 at ten million.

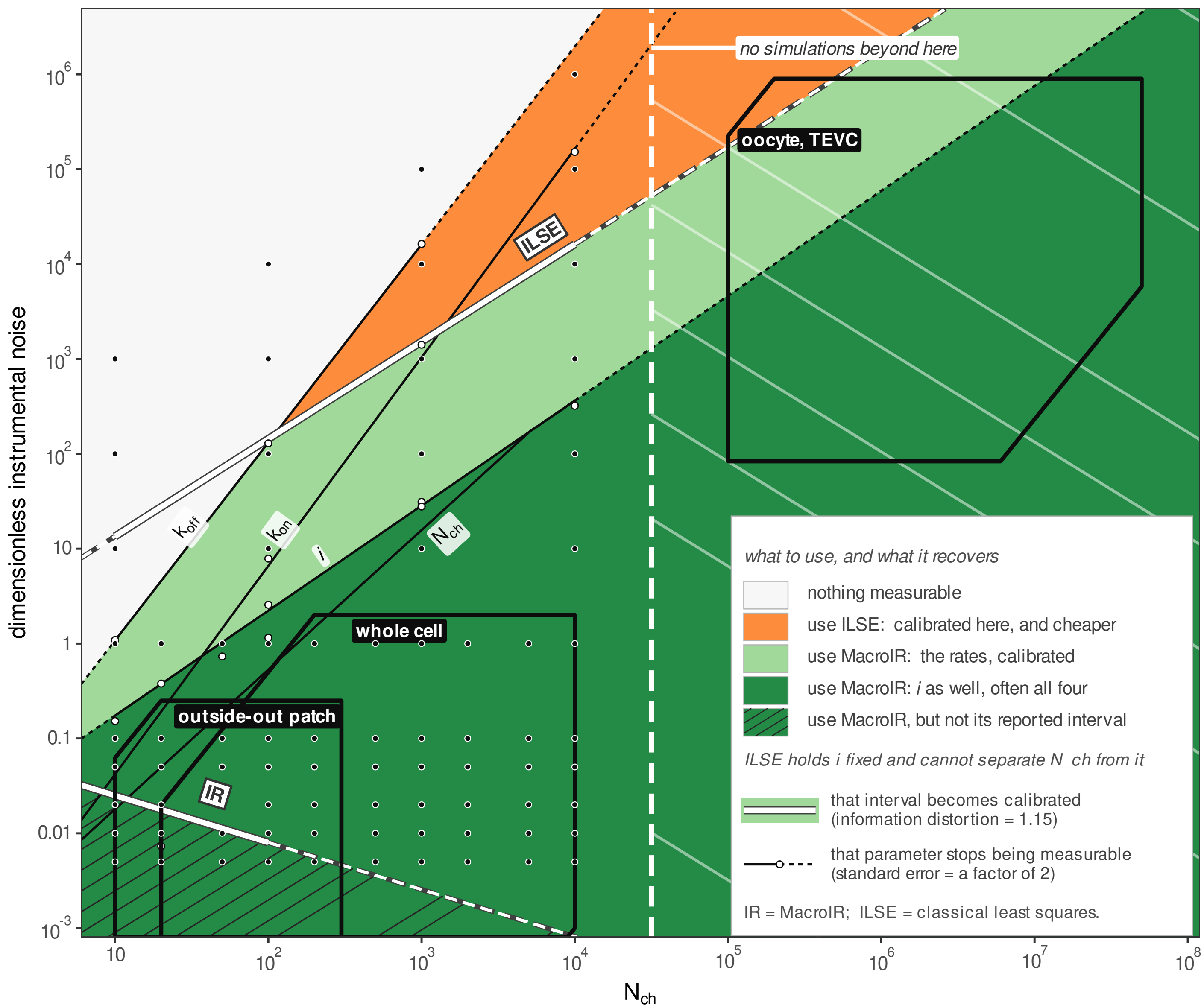

Figure 7. What a macroscopic recording can be asked for, and which method can deliver it. The design plane, channel number N_{ch} against dimensionless instrumental noise, at one acquisition interval ($\Delta \cdot k_{\mathrm{off}} = 0.1$) and no other. Every cell is filled with which parameters survive there, counting one as recovered when its distortion-corrected standard error is inside a factor of two. Five fills run up the panel, near-white, amber, pale green and dark green, with the corner in which the recursive likelihood's own interval (`IR`) is not calibrated hatched over the last of them rather than replacing it; the key names what each is worth, in their vertical order on the panel so that it reads as a section through the map, and the text reads them. Two classes of line cross the plane and are drawn differently because they answer different questions: a white line cased in dark marks where a method's reported error bar becomes calibrated (information distortion = 1.15, a 7% error on the reported standard deviation), one per method, and a thin black line marks where a parameter stops being measurable, one per parameter. Two of the four black lines carry the partition, those for i and k_{off}; the other two subdivide the dark green region without partitioning it. Each line is a power law fitted through its own measured crossings, drawn as open dots, solid over the range of channel numbers in which those crossings were found and dashed outside it; the six fitted slopes, and what each of them says about the observable its line answers to, are in the text. Filled dots are the 80 simulated cells, 5 to 10^4 channels and 0.005 to 10^6 in dimensionless noise; past the last simulated column the fills are veiled and the veil is labelled. The noise axis is dimensionless by construction (Methods), so that a whole-cell rig at 1 pA RMS in a 1 kHz band recording 1 pA channels with a 10 ms relaxation sits at $\widetilde{S} = 0.1$. Three recording configurations are placed on the same plane as heavy outlines with reversed tabs, an excised outside-out patch, whole cell, and an oocyte under two-electrode voltage clamp (TEVC): their channel ranges, given in the text, are derived from the smallest and largest current each configuration can hold, while their vertical extent is read from published traces and is indicative. The map is a concept map and not a phase diagram: a looser criterion moves each boundary by up to a decade in noise without changing the layout. All of it is the two-state scheme at a single open probability and a single concentration-jump protocol.

Discussion

A usable calibrated likelihood for most experimental conditions on macrocurrents is demonstrated here for two states: the boundary-conditioned member we published with the $P2X_2$ analysis (***Moffatt and Pierdominici-Sottile, 2025***). Over most of the design space it reports an uncertainty close to the one it delivers, and where it does not, the departure is measured rather than assumed. The reported interval is mildly the wrong width in the few-channel, low-noise corner, and there the repair is the sandwich the diagnostic already supplies (Figure 4–figure supplement 4).

A simple statistic tells an experimenter which regime a recording is in: the integrated autocorrelation time of its standardized least-squares residual, read off a fit already being made, with no ensemble and no known truth. As the instrumental noise grows the recorded current becomes Gaussian whatever the member assumes, and the window-averaged least-squares fit recovers a calibrated error bar there, its closing-rate distortion falling from thirteen to one across five decades of noise (Figure 4). The refuge has a price: the noise at which that error bar becomes calibrated lies one to three decades above the noise at which the fluctuations stop carrying the unitary current, so no measured region gives both (Figure 7). The reading is necessary and not sufficient: memory implies distortion, and whiteness on its own certifies nothing (Figure 5).

The acquisition average is the macroscopic form of the missed-event problem. A single-channel record loses the transitions that fall inside its dead time (***Colquhoun and Sigworth, 1995***); a macroscopic record loses the ones that fall inside its sampling window, where the excursions of every channel are averaged together before anything is recorded. What is recorded, in both places, is not the state of the process at any one time.

At first we expected the one-endpoint member to be a fair approximation: a mean conductance plus a noise of its own, the emission form ***Münch et al.*** (***2022***) give to open-channel noise, with that noise computed from the averaging. It is not. It shaves the amplitude bias and loses the second moment, ending worse on the parameter it was built for than the member that ignores the averaging altogether, 1.16 against 1.01 at ten thousand channels. We tried the variance alone next, keeping the one-endpoint update and taking the interval variance from the two-endpoint construction, in case a mis-stated variance was the whole of it. It buys back more of the amplitude bias and loses the second moment in the same way, and neither partial correction reaches either interval member (Figure 4–figure supplement 2).

The Markov property came to the rescue. Conditioning on both endpoints is what makes the interval average a function of the states again, and it is also what leaves one state variable per junction, so the same choice recovers the average and keeps the chain Markov across the record (Appendix 1). The integration and the recursion are bought together, and what comes out is a white score, the innovation whiteness that has certified an optimal filter since ***Mehra*** (***1970***).

What is left over factors into two parts that say different things, a per-sample failure and a failure across intervals, and the recursion reaches the second: the local time correlation ***Milescu et al.*** (***2005***) named, now measured (Figure 4–figure supplement 1).

The calibrated member and the control select different mechanisms

A calibrated likelihood finds immediate application. The results indirectly add support to the hypothesis of an asymmetric activation mechanism in P2X2, as IR favored it. The alternative synchronous mechanism was favored by INR, an approximation already distorted in the simplest two state model. Whether IR remains valid on highly complex conformational models is the subject of future research.

The channel number stays in the model and stops informing the record

Two parameters go quiet once the agonist is removed, and for different reasons. The opening rate leaves the generator with the ligand, and reaches the record afterwards only through the occupancy the pulse left behind, which is why it takes a few samples to fall silent rather than none. The channel number stays in the generator and is cancelled anyway: a member conditioned on the past already carries the number of channels still open, so a parameter that reaches the record only through that number has nothing left to give, and its information falls to the numerical floor within six intervals (Figure 3–figure supplements 1 and 2). The members that do not condition go on claiming a sixth of their pulse total about the channel count across the washout. A short pulse therefore buys the closing rate and the unitary current, and not the other two. The numbers are from one design cell.

Grouping samples is the same measurement at a larger window

A second result speaks to how a record should be reduced before it is fitted. Grouping consecutive samples is the same measurement at a larger Δ, with a proportionately smaller instrumental variance and no sample discarded (Results), so the temporal structure of the record becomes something to design: a record can be reduced by orders of magnitude for a kinetic question and hardly at all for an amplitude one; after a concentration change the groups can grow geometrically with the transient they follow, which is legitimate rather than an approximation because the likelihood takes the window as an argument and not as a constant; and the cost of the fit follows the design, since the recursion is linear in the groups it is given.

MacroIR is an integrated-measurement augmented Kalman filter (***Zadrozny, 1988***), and its predictions coincide with that construction to about 10^{-8}. The equations that make that agreement checkable are in Appendix 1; what the augmentation supplies, what has to be computed for the chain, and the prior art on both sides are in Appendix 5.

Münch et al. (***2022***) did not overlook the averaging: they treat a finite integration time as a misspecification, measure the bias it causes and report that the algorithms they compare incur a similar one. Their observation is instantaneous and their states are propagated by the matrix exponential of the interval, which places their filter at the recursive instantaneous rung of the ladder measured here, the one this study finds least centred of the eight in the rates. Its information is calibrated at the parameters that generated the data and not where the estimates land, and a coverage count is taken at the estimates. What a robustness check on estimates does not show is what the same choice costs the reported uncertainty.

Cost has been the standing objection to filtering (***Del Core and Mirams, 2025***), and it is a real one. What this study adds is the other side of the exchange: the price is now quantified in units of misreported uncertainty rather than argued (Appendix 5). The anchor behind those factors is not an artefact of the analytic form either: re-anchoring every distortion on a Fisher information differenced from the score overturns no verdict, and the one direction it moves is to enlarge both least-squares arms by about a third, so the least-squares verdict printed here is the conservative one (Supplementary File 1).

Where exchangeable sweeps are plentiful, the classical route is strong and should be said to be. Non-stationary fluctuation analysis (***Sigworth, 1980***; ***Heinemann and Conti, 1992***) returns the channel number and the unitary current with no likelihood at all and a sweep-level bootstrap returns honest intervals with no model, and whether that suffices depends on how many sweeps a preparation tolerates and whether they are exchangeable; how it composes into a complete workflow is not addressed here. The case for a filter is strongest where that route is weakest: few sweeps, rundown, conditions that are not exchangeable, a single precious recording, and an experiment that wants kinetics and amplitudes jointly from the same record. What the alternatives that repair an interval without changing likelihood answer instead is in Appendix 4.

The measurement has boundaries of its own, and the first is the method it certifies

The first boundary is the method's own. The boundary-conditioned member is an approximation and is not exempt from the failure this paper is about. Over the same grid its distortion runs from 0.645 on the channel-number direction at $N_{\mathrm{ch}} = 10$, where it over-states its own uncertainty, to 1.701 on the closing rate at $N_{\mathrm{ch}} = 5$, where it under-states it, so the departure is two-sided and does not become safe by erring conservatively on average. Both excursions sit where the open population is a small integer count, which is the microscopic regime the occupancy closure predicts, and non-normality on its own cannot produce a factor-two distortion anywhere on the grid. What produces the departure is not known. The decomposition puts it in the term that accumulates across intervals, and the five candidate criteria measured against the observed boundary were all criteria on the per-sample cumulants of the occupancy, so all five failed to locate it; the measured channel-count exponent of -0.25 matches neither the -0.5 a skewness mechanism would give nor the -1.0 of a kurtosis one, and it is not a clean power law. Stated as a conjecture and in those words: the limits behave like a balance that has to come out just right, several quantities moving together in a way that leaves the departure small over four decades, which is the signature of a conserved quantity or a comparable invariant that has not been identified here.

A second boundary is the measurement model. The instrumental contribution is taken to be Gaussian and white, as the simulator emits it (Methods); where a recording departs from that, through a trapped charge in the headstage or amplifier flicker, the diagnostic would report a distortion belonging to the noise model and not to the gating closure, and we have not run that; unmodelled discrepancy of this kind is documented to produce exactly this overconfidence in whole-cell current fitting (***Lei et al., 2020***).

A third boundary is the diagnostic's own. The comparison is between two independent implementations, the likelihood and the exact simulator, so everything downstream of the point where they separate is tested: the propagation, the update, the emission arithmetic, the accumulation. Upstream they share one specification, the rate matrix built from the parameters, the conductance vector, the protocol and the sampling times, and a diagnostic built on their agreement cannot see an error there. The general remedy is different in kind, verifying each object against the theorems it must satisfy, and that is a piece of work with its own design, not undertaken here.

The perimeter is a model, and Methods declares it as one: two states at an open probability of one half, a single concentration jump, simulated data, likelihood only. The fixed open probability means a channel count and an opening count differ throughout by a factor of two that no measurement here can settle. What the channel-number axis carries at $p = 0.5$ is the gating variance, so a reader whose receptor sits elsewhere can enter the map at $4Np(1-p)$ in place of N; the substitution is exact for the variance and says nothing about the skewness, which vanishes at $p = 0.5$ and which no cell here probes. Richer schemes, the stationary regime, the few-channel boundary, the filtered kernel and experimental data are named companions rather than omissions, as is the rung the family does not carry, the full-covariance likelihood route, which is set out with the rest of the ladder (Appendix 5), and the kernel is one for a stated reason: the condition that makes its residue negligible is hardest to meet in the telegraphic regime, at windows short against the relaxation, which is where the boundary-conditioned member's advantage is largest. Two states is the easiest case a Gaussian approximation can be given, since the occupancy of an ensemble of independent two-state channels is exactly binomial and its propagation is exact, so every departure reported here is a lower bound on what a richer scheme would show. The interval axis is confounded with the number of observations: every recording is ten channel time constants long, so it carries a thousand observations at the finest sampling and ten at the coarsest, for six parameters, and nothing in the design separates a coarser view of the transient from fewer views of it. No quantity of recorded data identifies the generating model, which is why this diagnostic needs a simulator whose model is known.

One hierarchy holds the four claims apart. That the tested family misreports its information on this scheme, where, by how much and in which directions, is demonstrated; that the mechanism, temporal correlation the likelihood does not carry, operates in richer schemes is suggested by the mechanism's generality and not shown; that the two-state case bounds anything richer is conjectured; and what a corrected covariance would change in a recomputed evidence is not tested.

The reader supplies their sweep count, their rundown, their model size and their tolerance, and decides; what they cannot measure for themselves, and what is supplied here, is whether a reported interval is the interval that is delivered. A mechanism once left a mark anyone could point to; choosing between mechanisms now means navigating by likelihood, and the likelihood is the instrument, which has to be calibrated before what it selects can be believed.

Materials and methods

Minimal model and simulation

We work with a two-state channel scheme, closed (C) and open (O), the smallest scheme in which the acquisition-averaging effect we study is present. The closed-to-open transition is agonist dependent with association constant k_{on} multiplying the agonist concentration; the open-to-closed transition is agonist independent with dissociation rate k_{off}. Only the open state conducts, and the code applies a sign flip so the single-channel current is inward. At the start of every recording all channels are closed, so the occupancy is $(1, 0)$.

The scheme carries six parameters, all estimated in base-ten logarithmic coordinates. The values used to generate the data reported here are $k_{\mathrm{on}} = 10\ \mu\mathrm{M}^{-1}\,\mathrm{s}^{-1}$, $k_{\mathrm{off}} = 100\ \mathrm{s}^{-1}$, a unitary conductance g set to 1 (the engine parameter `unitary_current`, written i where the noise axis is defined below), and a current baseline b set to 1 (both in units of g), while the white-noise level and the mean channel number were swept (below). Current is expressed throughout in units of the unitary conductance g, which corresponds physically to a single-channel current of roughly $1\ \mathrm{pA}$. The dissociation time $\tau = 1/k_{\mathrm{off}} = 10\ \mathrm{ms}$ sets the natural time scale that normalises the acquisition-interval axis. The relaxation at the agonist condition is faster, with rate $k_{\mathrm{on}}[A] + k_{\mathrm{off}} = 200\ \mathrm{s}^{-1}$; the axis uses $\tau = 1/k_{\mathrm{off}}$ by convention. During the agonist step the two rates are equal, $k_{\mathrm{on}}[A] = k_{\mathrm{off}} = 100\ \mathrm{s}^{-1}$, so the open probability the sweep works at is $p = 0.5$. That is where the gating variance of the open population, $N p(1-p)$, is largest and where its skewness, which goes as $1 - 2p$, vanishes.

The emission model is white instrumental noise only. Both the pink (one-over-f) and the proportional noise channels are set to zero, so the term the reader may expect from flicker noise is genuinely absent. For a measurement interval t of duration Δ_t, the predicted current has mean $\bar{y}_t = N_{\mathrm{ch}}\,\mu_t \cdot \gamma_t + b$ and variance

$$\sigma_t^2 \;=\; e_t \;+\; N_{\mathrm{ch}}\, v_t, \qquad e_t \;=\; \mathrm{Current\_Noise}\,\frac{f_s}{n_{\mathrm{samp},t}} \;=\; \frac{\mathrm{Current\_Noise}}{\Delta_t},$$

where N_{ch} is the channel number, μ_t the occupancy mean, γ_t the interval-averaged conductance, f_s the raw sampling rate, $n_{\mathrm{samp},t}$ the number of raw samples averaged into the interval, e_t the white instrumental variance over the interval, and $N_{\mathrm{ch}}\, v_t$ the channel-gating variance. The white variance e_t falls as the interval lengthens, because more raw samples are averaged into each recorded value. The gating variance v_t is the single-interval conductance variance, whose form differs across the algorithm family and is derived in Appendix 1.

The stimulus is a single concentration jump in three segments, a pre-pulse at $0\ \mu\mathrm{M}$, an agonist step at $10\ \mu\mathrm{M}$, and a post-pulse at $0\ \mu\mathrm{M}$, with raw sampling at $f_s = 50\ \mathrm{kHz}$ throughout. Each recorded measurement is the mean of $n_{\mathrm{samp},t}$ consecutive raw samples, so the averaging window is fixed at acquisition. The averaging is uniform over the window and no analogue filter stage is simulated, so a recorded value is a boxcar average of the trajectory, which is the observable every likelihood in the family is written for (Eq. 1). Recordings from a real amplifier are Bessel filtered instead. That kernel is not simulated here, and how much of it the uniform window misses is bounded in Appendix 1, as a condition on the acquisition window against the filter's response time rather than as an open size.

Ground truth is generated by exact simulation of the continuous-time Markov chain (CTMC) by uniformization, so the interval average of the conductance is realised from an actual sampled trajectory rather than from any moment-closure approximation. Uniformization draws the transition times themselves, so there is no within-interval discretisation to choose and none to justify. The data analysed here are simulated by design: a diagnostic that compares what a likelihood delivers against what it reports needs a known truth, and the inference throughout is likelihood only, with no prior and no evidence computation. The likelihood and the simulator are independent implementations sharing one upstream specification, the rate matrix built from the parameters, the conductance vector, the protocol and the sampling times, written out symbol by symbol in Supplementary File 1; what that sharing costs the diagnostic is stated with the other limitations in the Discussion. Every maximum-likelihood grid cell reported here, the least-squares arm included, used 10^4 recordings, with one exception: the cell (the grid is defined below) that carries the coverage result ($N_{\rm ch} = 10$, $\widetilde{S} = 0.005$, `IR`) was run at 10^5, because coverage of a nominal 95% region in six parameters needs a thousand independent fits and those fits are taken at group size 100, so 10^5 recordings are required to produce them. The time-resolved run used $1{,}000$ recordings; the single-recording illustration (Figure 1) was drawn by the substep sampler instead, one 20-channel trace at 250 substeps per interval, and does not enter the quantitative comparison.

The simulation seed was set to 0, which in this engine is the sentinel for a random seed drawn from `std::random_device`, and the resolved seed was never logged. The deposited ensembles are therefore statistically equivalent to, but not bit-for-bit reproducible from, a rerun of the same script. Because each ensemble is large ($1{,}000$ to $10{,}000$ recordings), the reported statistics are stable; the Monte-Carlo standard here is statistical equivalence, not bit identity.

The likelihood family and its members

The members cross the two axes the body sets out: whether the gating fluctuations are modelled at all, and how much of the acquisition interval the interval-averaged conductance is conditioned on. This subsection gives the flags each setting is dispatched under and nothing else.

Least squares is selected by the family flag *family approximation* $= 2$ and runs non-recursive; every macroscopic member takes 0. Three flags set the rest: *recursive*, whether the occupancy covariance is propagated between intervals; the endpoint count $\mathrm{av} \in \{0, 1, 2\}$, which counts how many interval endpoints the hidden state is conditioned on and is the program's name for the window axis of the family; and the *variance form*, which separates `MR` from `VR`. Each least-squares arm shares its mean model exactly with the macroscopic member carrying the same averaging flag, `LSE` with `NR` and `ILSE` with `INR`, and differs from it only in the predictive variance: on the recording of Figure 1 those predicted means agree to 9×10^{-16} interval by interval. The dispatched flags, the data keys, the conditioning each member applies and the cells each was run over are tabulated in Supplementary File 1. `INR` is the member published as MacroINR in ***Moffatt and Pierdominici-Sottile*** (***2025***), where it served as the control against which MacroIR, the member called `IR` here, was compared.

`IR` is the top implemented rung: it conditions on the pair of channel states at the two ends of the interval, which we call the boundary state, and marginalises the interior analytically, as specified by the construction of Appendix 1. An interval-recursive likelihood of this family was first applied to macroscopic recordings, of the P2X$_2$ receptor (***Moffatt and Pierdominici-Sottile, 2025***), and it coincides with a time-integrated Kalman filter, a correspondence stated and quantified in the Discussion. Every member scores with Eq. 5 and the regularizer of Eq. 7 below; the correspondence between the symbols of Appendix 1 and the names the implementation carries is in Supplementary File 1.

The likelihood.

Every member of the family delivers, per window, a predicted mean $\bar{y}_t^{\rm pred}$ and a predicted variance $\sigma^2_{{\rm pred},t}$, and the recorded value is scored under the Gaussian they define,

$$\log q_t \;=\; -\tfrac{1}{2}\left[\log\!\left(2\pi\sigma^2_{{\rm pred},t}\right) + \frac{\delta_t^2}{\sigma^2_{{\rm pred},t}}\right], \qquad \delta_t = \bar{y}_t^{\rm obs} - \bar{y}_t^{\rm pred}, \tag{5}$$

summed over windows. The recursion enters only through what $(\bar{y}_t^{\rm pred}, \sigma^2_{{\rm pred},t})$ are conditioned on: a recursive member carries $(\boldsymbol{\mu}^{\rm post}, \boldsymbol{\Sigma}^{\rm post})$ of Eqs. A20 and A21 into the next window, so Eq. 5 sums to $\log \prod_t q_t(\bar{y}_t \mid \bar{y}_{1:t-1})$; a non-recursive member propagates from the initial condition by Eqs. A14 and A15 alone and the same sum is $\log \prod_t m_t(\bar{y}_t)$. Nothing else in this section is used by the fit.

The initial condition.

The recursion opens with the occupancy the model supplies and with the covariance that occupancy implies for one channel,

$$\boldsymbol{\mu}_0 = \boldsymbol{\pi}, \qquad \boldsymbol{\Sigma}_0 = \mathrm{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top. \tag{6}$$

For the schemes run here $\boldsymbol{\pi}$ is the first basis vector: the record opens at zero agonist with every channel shut, so $\boldsymbol{\Sigma}_0 = \mathbf{0}$ and the spread the filter carries is generated entirely by the propagation of Eq. A15. Eq. 6 is also the one moment at which $\boldsymbol{\Sigma}$ genuinely is a single channel's covariance; from the first update onward the recursion carries $\mathrm{Cov}(\mathbf{N})/N_{\mathrm{ch}}$ instead (Appendix 1), and every factor of N_{ch} here comes from that distinction.

The equations of Appendix 1 are written with the objects the implementation forms, and the correspondence is one to one; it is tabulated symbol by symbol in Supplementary File 1, because a specification whose symbols cannot be found in the code is not a specification.

Two reductions are worth stating because they anchor the family to prior work. Setting the boundary-conditioned conductance variance to zero in the recursive, no-endpoint member (R) recovers the macroscopic-fluctuation update of ***Moffatt*** (***2007***), so R is the recursive ancestor already in print. The single structural difference between the M and I families is that the one-endpoint members drop the boundary cross-covariance term $N_{\mathrm{ch}}\,\widetilde{\boldsymbol{\gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma}}$ that IR retains, the tilde being the boundary-pair contraction defined at Eq. A18. Every member is written out on one template in Appendix 2.

The remaining build flags were held fixed in every reported run; their settings, and the naming they disambiguate, are in Supplementary File 1.

The filter carries three numerical safeguards. The first fixes what the boundary-conditioned moments are where a transition cannot occur; without the other two the recursive members return no value at all at the smaller channel counts.

The first sits upstream of the filter cycle, in the boundary-conditioned moments of Eq. A3, built from $\mathbf{F}_1(\Delta)$ and $\mathbf{F}_2(\Delta)$, the two derivatives of the conductance-tilted semigroup carrying the first and second moments of the integrated conductance over paths from i to j (Appendix 1). Both are quotients by the transition probability $P_{i\to j}(\Delta)$, which vanishes wherever the transition cannot happen within one window and does so over the whole pre-pulse segment of this protocol (Appendix 1), leaving the quotient $0/0$ in exact arithmetic and, in floating point, a ratio of two quantities no larger than the rounding error that produced them. The implementation does not test for that case; it reads each quotient as a posterior mean and adds a conjugate pseudo-count ε to numerator and denominator alike,

$$\bar{\gamma}_{i\to j} = \frac{[\mathbf{F}_1(\Delta)]_{i\to j}/\Delta \;+\; \varepsilon\,\frac{1}{2}(\gamma_i+\gamma_j)}{P_{i\to j}(\Delta)\;+\;\varepsilon}, \qquad \overline{m}_{i\to j} = \frac{2[\mathbf{F}_2(\Delta)]_{i\to j}/\Delta^2 \;+\; \varepsilon\,\frac{1}{2}(\gamma_i^2+\gamma_j^2)}{P_{i\to j}(\Delta)\;+\;\varepsilon}, \tag{7}$$

where $\overline{m}_{i\to j}$ is the second moment of the interval-averaged conductance over the same boundary pair, and the variance follows by subtraction, $\bar{v}_{i\to j} = \overline{m}_{i\to j} - \bar{\gamma}^2_{i\to j}$. The two prior means are the endpoint averages of Eq. A8, so the prior the variance inherits is $\big((\gamma_i-\gamma_j)/2\big)^2$ and is never imposed separately. The weight is $\varepsilon = \varepsilon_{\mathrm{mach}}\,\kappa_F(\mathbf{V})$, machine epsilon times the Frobenius condition number of the matrix of right eigenvectors of $\mathbf{Q}$, which bounds the floating-point noise the spectral reconstruction can leave in $\mathbf{P}(\Delta)$, so the pseudo-count is inert wherever the transition probability stands above that noise, takes over where the quotient carries no information, and introduces no fitted constant. Why ε is keyed to $\kappa_F(\mathbf{V})$, and why the prior is taken from the conductance vector rather than from the marginals of $\mathbf{F}_1$, is in Supplementary File 1.

The second holds the occupancy mean on the probability simplex. Nothing in the Gaussian conditioning of Eq. A20 knows that $\boldsymbol{\mu}$ is a probability vector, and an undamped step can drive its entries negative or above one, so the mean step is scaled by a trust coefficient $\alpha_\mu \in (0,1]$, the largest scaling that keeps the posterior on the simplex. Damping the step is what replaced correcting the state after it. The form of the damping matters here in a way it would not in an ordinary filter, because this paper differentiates the log-likelihood with respect to $\boldsymbol{\theta}$: a hard $\alpha = \min(1,\ c\,\alpha_p)$ leaves a step in $\partial\alpha/\partial\boldsymbol{\theta}$ at the boundary and pumps variance into the score whenever the system sits near it. The coefficient is therefore taken in the smooth form $\alpha = \frac{1}{2}\big(1+c\,\alpha_p-\sqrt{(1-c\,\alpha_p)^2+\epsilon^2}\big)$, a softened minimum of 1 and $c\,\alpha_p$ that is infinitely differentiable in $\boldsymbol{\theta}$ everywhere, with α_p the largest scaling that keeps the posterior mean on the simplex, $c = 0.9$ and $\epsilon = 10^{-4}$. Reimplementing this with a hard minimum reproduces the mean current and corrupts every quantity this paper measures. The covariance takes its full step, and the truncate-then-renormalise history, the two untied trusts and the $\epsilon/2$ bias arithmetic are in Supplementary File 1.

The third is a pair of constants: a binomial-count floor of 5, below which the expected count in a state is too small for the Gaussian treatment of that state and the member falls back to its non-recursive form, and a minimum occupancy probability of 10^{-12}.

The two least-squares parameter configurations

Least squares runs in two parameter configurations, and they are different statistical models. Both select *family approximation* $= 2$, both fit data from the same simulator, and both minimise the same sum of squares. They differ in which parameters are free, so no quantity computed under one may be compared against the same quantity computed under the other. Which parameters are free is independent of the window setting of the previous section, so the configurations and the arms cross: the six-parameter configuration is run on both arms and supplies the least-squares pair of Figure 3, and the four-parameter configuration is run on `ILSE` alone and supplies every grid cell.

The residual scale is not a free parameter of either and is not given a value. It is integrated out under a scale-invariant prior, so with $\mathrm{SSE} = \sum_i (y_i - \mu_i(\boldsymbol{\theta}))^2$ over the n finite recorded values the objective is the marginal

$$\log L(\boldsymbol{\theta}) \;=\; \log\Gamma\big(\tfrac{n}{2}\big) \;-\; \tfrac{n}{2}\log\pi \;-\; \tfrac{n}{2}\log\mathrm{SSE}(\boldsymbol{\theta}), \tag{8}$$

whose maximiser is the ordinary least-squares one, since only the last term depends on $\boldsymbol{\theta}$. What the marginalisation changes is everything downstream of the point estimate. The score is the derivative of Eq. 8 and carries the factor n/SSE rather than a $1/\sigma^2$ that would have to be supplied, and the information the fit reports is

$$\mathbf{H}_{\mathrm{LSE}} \;=\; \frac{n}{\mathrm{SSE}}\sum_i \frac{\partial\mu_i}{\partial\boldsymbol{\theta}}\,\frac{\partial\mu_i}{\partial\boldsymbol{\theta}}^{\top}, \tag{9}$$

the Gauss-Newton curvature with $\hat{\sigma}^2 = \mathrm{SSE}/n$ entering as a plug-in prefactor and not as a coordinate. There is therefore no $\partial\sigma^2/\partial\boldsymbol{\theta}$ term to keep or drop: the scale direction is not in the parameter vector at all, which is also why the classical arm carries no score in the noise direction and why its mean squared standardized residual is pinned at one by construction.

The six-parameter configuration feeds the time-resolved figure (Figure 3). Both arms of it are produced by the same script that simulates the ensemble and scores the macroscopic members, so all eight members of that figure score one common set of recordings. All six parameters are free in base-ten logarithmic coordinates: k_{on}, k_{off}, the unitary current, `Current_Noise`, the current baseline and the mean channel number. The time-resolved figures accumulate the per-interval Fisher information and never invert it, so the flat directions of the least-squares fit are harmless there, and two of them are visible rather than hidden. `Current_Noise` has an identically zero score row and an identically zero Fisher row, because the least-squares mean carries no noise term; and the unitary current and the mean channel number form a rank-one degenerate block along the amplitude ridge $N_{\mathrm{ch}}\,i$, so the unitary current is free in this configuration without being separately identified.

The four-parameter configuration produces every grid cell and, where a cell needs one, the per-group estimate cloud. It fixes the unitary current at 1 and `Current_Noise` at the cell value, leaving four free parameters: k_{on}, k_{off}, the current baseline and the mean channel number. Fixing those two is forced rather than chosen: they are the flat directions of the previous paragraph, and with them free the parameter Fisher information is singular, so no maximum-likelihood covariance exists. Both fixed values are the values the recordings were simulated from. The script builds one parameter vector and passes it both to the simulator and to the fit as its reference point, so the fixed value and the truth are the same number by construction. Least squares is therefore not handicapped in this configuration but helped: two of the six parameters are supplied to it exactly right, while the likelihood members estimate all six. Where it is nonetheless the arm whose reported uncertainty is furthest from the delivered one, that failure is not a lack of information about those two. This is the configuration behind every `ILSE` cell of Figures 4 and 6.

The consequence is carried into the Results. A score, a Fisher information, a distortion matrix or a log-likelihood computed under six free parameters is a different object from the one computed under four, and the two distortion matrices differ in dimension, 6×6 against 4×4, so any scalar that accumulates over eigenvalues is not comparable between them. Every statement about least squares in the Results is scoped to one configuration and names it. The statement that least squares cannot deliver the unitary current is a statement about the four-parameter configuration.

Parameter estimation and the two anchors

Inference in each grid cell is two-stage. First, the ensemble is partitioned into groups of consecutive recordings and a separate maximum-likelihood estimate (MLE) is computed per group, producing a cloud of estimates whose scatter is the empirical parameter covariance. The partition is by position and any remainder is dropped, which costs nothing here because the recordings are independent and identically distributed and 10^4 divides by both group sizes used. The optimiser is Gauss–Newton with Levenberg damping, warm-started at the simulation truth. Its damping schedule and its three convergence limits are in Supplementary File 1. What stops most fits, however, is a fourth criterion the script cannot reach: the optimiser tests the Newton decrement $\frac{1}{2}\, g^{\top} H^{-1} g$ first, requiring it below 10^{-8} for the first ten iterations and below 10^{-4} after that, with the three script-set limits acting as fallbacks behind it. Two group sizes were used, 10 and 100; with $10{,}000$ recordings these give $1{,}000$ groups of 10 and 100 groups of 100. Second, a single joint MLE is computed over all replicates at once (one group of size n_{sim}), giving the pooled estimate θ_{pool}; its near-zero gradient certifies that it is a stationary point of the pooled likelihood.

A group whose refit fails outright is dropped from the cloud, so a cloud can hold fewer estimates than the partition has groups, and its size is conditional on convergence. Every fit that did complete carries its convergence status, decided by the criteria above, into the per-group run log. The figures in the body do not filter on that status; the supplements that restrict themselves to converged fits say so.

Warm-starting the per-group fits at the truth is deliberate. It isolates the error of the likelihood from the error of the optimiser, which is what this paper studies; it would be the wrong choice for a study of an estimator, which this is not.

The two anchors, the simulation truth θ_{sim} and the pooled fit θ_{pool}, are distinct points because the Gaussian macroscopic likelihood is misspecified with respect to the exact simulator. Their difference $\theta_{\mathrm{pool}} - \theta_{\mathrm{sim}}$ is the misspecification bias. Every diagnostic below is computed at both anchors, and the two answer different questions: at θ_{pool} the score is zero by construction so bias washes out, whereas at θ_{sim} the bias is visible. Each figure states which anchor it uses.

Diagnostics and the Gaussian-Fisher anchor

Faithfulness is measured with the three diagnostics the body introduces: standardized residuals, the score, and the information-distortion matrix comparing the covariance J of the score across replicates with the model's own Fisher information. Their definitions, sign conventions and thresholds are given there; this section gives only the computation.

Every diagnostic reported in this paper is anchored on the analytic Gaussian Fisher information, summed over the intervals of a recording and then averaged over the recordings of the ensemble, $\bar{G} = \mathbb{E}\big[\sum_t \mathrm{GFI}_t\big]$. This is the anchor written $\mathbf{H}$ above, and the expectation is not decoration: for the recursive members the predictive moments of interval t depend on the data already seen, so $\sum_t \mathrm{GFI}_t$ is itself a random matrix and the object the score covariance is compared against is its ensemble mean, taken over the same recordings that supply that covariance. Its per-interval term is the summand of

$$\mathbf{H} = \sum_t \mathbb{E}\left[\frac{(\nabla\mu_t)(\nabla\mu_t)^\top}{\sigma_t^2} + \frac{(\nabla\sigma_t^2)(\nabla\sigma_t^2)^\top}{2\sigma_t^4}\right], \tag{10}$$

which follows from the analytic derivative alone and needs no extra likelihood passes. It is positive semidefinite by construction, so no anchor used here can be indefinite.

A second Fisher construction exists in the codebase, a central difference of the analytic score. It is not interchangeable with the analytic anchor: at θ_{sim} it is indefinite over a region of the design plane, and stays so after averaging over the ensemble, which is why the body is anchored on the analytic construction; it is nonetheless the check that anchor is measured against, over the full roster, in Supplementary File 1, which also holds the region, its settings and the direction convention of the comparison. An earlier battery anchored on it (data directory `433ed13`) is retained in the deposit; no number reported in this paper is taken from it.

One edge case governs how a cell is summarised, and it is by design rather than by accident. The anchor can be near-singular where a parameter is unidentifiable, which happens once the open population relaxes (Figure 3–figure supplements 1 and 2); there we diagonalize $\mathbf{H}$ and retain the eigenmodes whose eigenvalue exceeds a tolerance set relative to the largest, $\lambda_i > \lambda_{\max} \cdot 10^{-10}$, evaluate every inverse power on that retained subspace alone, and leave the quantity undefined, not finite, when the retained subspace is empty. The dimension d of Eq. 4 is the dimension of that subspace, so a cell where a direction has been dropped is summarised over fewer directions than one where none has. Undefined is not zero, and the maps render those cells in grey to keep the two apart.

The comparison of J against $\bar{G}$ is defined, with its sign convention and its two scalar summaries (Eq. 4), in the body; on the least-squares branch the anchor is Eq. 9 rather than Eq. 10 and it carries a further premise, that the residuals are homoscedastic, and the Results state that premise where the ratio is quoted.

Uncertainty on every reported statistic comes from a nonparametric bootstrap over groups, reported as percentile intervals at the 2.5, 16, 50, 84 and 97.5% quantiles; the intervals are ordinary empirical percentiles. The diagnostic battery used 100 bootstrap replicates and a maximum lag of 10, which is the truncation of every autocorrelation it reports, of the score and of the standardized residual alike, and the empirical-versus-theoretical distortion capstone used 200. The per-group estimate cloud is deposited raw, and every interval quoted from it in this paper is computed downstream in the figure code, from the raw estimates.

Regime grid and reduction

The acquisition interval was swept over seven levels of $\widetilde{\Delta}$, namely 1, 0.5, 0.2, 0.1, 0.05, 0.02 and 0.01, realised as 500, 250, 100, 50, 25, 10 and 5 raw samples averaged per interval, that is step durations of 10, 5, 2, 1, 0.5, 0.2 and 0.1 ms, giving 10, 20, 50, 100, 200, 500 and $1{,}000$ steps in the record. The total record duration is held fixed at 100 ms across the sweep, so the interval is varied without varying the amount of data. Within each record the pre-pulse, agonist and post-pulse segments hold a fixed ratio of $20{:}40{:}40$ percent of the steps.

The noise axis needs its conversion stated once, because the quantity the figures plot and the quantity the engine consumes differ by a fixed factor. The engine parameter is `Current_Noise`, a white-noise power spectral density in units of g^2 s, and it reaches the likelihood only through the variance of one recorded point, $e_t = \mathrm{Current\_Noise}/\Delta_t$, with no correlation time of its own. The sweep holds that density fixed and shortens the window, so the variance of one recorded point rises as $1/\Delta_t$ and the noise per unit bandwidth is unchanged. In the natural units defined above, $\widetilde{\Delta} = \Delta\, k_{\mathrm{off}}$ and $\widetilde{S} = S\, k_{\mathrm{off}}/i^2$; at the run values $k_{\mathrm{off}} = 100\ \mathrm{s}^{-1}$ and $i = 1$ the reference cell used throughout is $\widetilde{S} = 0.01$. Both $\widetilde{S}$ and the gating fluctuation are per-sample variances, which is what makes them comparable. The instrumental variance of one recorded sample is $i^2\widetilde{S}/\widetilde{\Delta}$, so $\widetilde{S}=1$ is a sample one channel time constant long whose noise carries a standard deviation of exactly one single-channel current, and a decade of that axis is half a decade of current. The gating variance of the same sample rises as the window shortens only until the window is shorter than the channel's own correlation time, after which it saturates, while the instrumental variance keeps rising as $1/\widetilde{\Delta}$; at the open probability of one half the two are equal along

$$\frac{\widetilde{S}}{N_{\mathrm{ch}}} = \frac{2\widetilde{\Delta}-1+e^{-2\widetilde{\Delta}}}{8\,\widetilde{\Delta}},$$

which is $\widetilde{\Delta}/4$ for windows short against that correlation time and a quarter for long ones.

The mean channel number N_{ch} and the noise level are the other two map axes. Every cell entering a body figure is at $n_{\mathrm{sim}}=10^4$, enforced by the figure code, so a cell run at a smaller ensemble cannot enter a map silently. The noise axis is a fixed ladder of twelve levels, and every cell of every algorithm sits on one of them:

$$\widetilde{S} \in \{0.005,\ 0.01,\ 0.02,\ 0.05,\ 0.1,\ 1,\ 10,\ 10^2,\ 10^3,\ 10^4,\ 10^5,\ 10^6\}, \tag{11}$$

denser below one, where the crossover the map measures sits, and by decades above it. The level $\widetilde{S}=1$ is where the instrumental noise over one channel time constant equals the single-channel signal power, so the ladder is dense on the side where the gating fluctuations still carry information and coarse on the side where they do not. Coverage is deliberately not a rectangle and it differs by member: `IR` spans eleven channel counts from 5 to 10^4 across all twelve noise levels, and every other member covers less. The per-member cell manifest is in Supplementary File 1.

Each grid cell is reduced through five stages, from the per-group estimate cloud to the empirical-against-theoretical distortion capstone; the stages and their outputs are listed in Supplementary File 1.

Reproducibility and computational environment

The engine is `macro_dr`, built as C++20, and each analysis is a script in a small typed domain-specific language run in a single process (***Moffatt, 2025***). Every analysis output file stamps the build's short git commit hash as its first row, and the data directories are named by that hash, so results from different code versions cannot overwrite one another.

The engine that produced these results is not the build behind the $P2X_2$ analysis of ***Moffatt and Pierdominici-Sottile*** (***2025***). Every score reported here comes from automatic differentiation of the likelihood with respect to $\boldsymbol{\theta}$, propagated through the recursion itself: value and derivative travel through the same code, carried together by one templated C++ type, so the derivative of the spectral reconstruction is taken where the reconstruction is, with its divided differences and its coincidence limits (Appendix 1). The Fisher information does not arrive with it. It is an expectation under the model rather than a derivative of the recorded log-likelihood, so it is wired by hand, as the Gaussian form of Eq. 10 over the predictive moments and the first derivatives the score has already produced. The distortion quantities follow from the two. The earlier study's fits were obtained by Markov chain Monte Carlo and needed no gradient, so the score-based diagnostics are new here rather than carried over from that work, and the dependence is what forces the trust coefficient above to be infinitely differentiable.

The production batteries ran under SLURM with the linear-algebra back end pinned to a single thread so that OpenMP parallelises the per-simulation and bootstrap loops; the resource settings and the one environment variable that is load-bearing are in Supplementary File 1.

All resampling that operates on a fixed dataset, the bootstraps and the group partitioning, is deterministic and reproducible given the data.

Use of generative artificial intelligence

Generative artificial intelligence was used in the preparation of this manuscript. Anthropic's Claude models, accessed through the Claude Code command line interface, assisted with drafting and editing the manuscript text, with checking numerical claims in the text against the code and the output files they come from, and with writing analysis and plotting scripts. They were not used to generate, alter or select data, to derive the likelihood family or the diagnostics, or to form the interpretation of the results. No figure or figure panel is AI generated. The author reviewed and verified all content, and takes full responsibility for the work, including any part of it that was drafted with such assistance.

Data and code availability

All code needed to reproduce the study is openly available. The analysis pipeline that generates the figures, together with the simulation configurations that define every run, is deposited in a public repository (`macroir-validity`) and archived at Zenodo, with the DOI issued on acceptance. The simulation and inference engine is the `macro_dr` software ***Moffatt*** (***2025***), referenced at the pinned version that produced the deposited data; each output file records the engine commit hash that generated it. No experimental data were used: every dataset in the paper is simulated, and is regenerable from the deposited configurations up to the simulation seed, which was not recorded for the original freeze (Materials and methods); the later differenced-Fisher battery injects an explicit seed through its deposited configuration (Supplementary File 1).

A separate library, `macroir`, carries the boundary-conditioned likelihood, its score, its Gaussian Fisher information and the distortion diagnostic of this paper out of the engine and into a small self-contained C++ core, with an R package and a Python package over it; the diagnostic returns the distortion matrix, its eigenvalue spectrum and the first-order bias from one call in either language. The two language packages are bindings over that one core, and they were checked against it on a single reference cell, which is the evidence for the statement that the three interfaces return the same numbers.

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Author contributions

L.M. conceived the study, developed the theory, implemented the algorithms, designed and ran the simulations, analysed the data, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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Appendix 1

The interval likelihood, derived

The body states what each member of the family conditions on and what that buys. This appendix carries the derivation behind those statements, in the order it is built: the law of the interval average and why it has no closed form, the boundary-conditioned moments and how they are evaluated, the three steps that turn the instantaneous update into the interval one, the start-conditioned members stated from their own assumptions, the two algebraic differences that separate them from the boundary-conditioned one, and both held against the exact law. Three subsections close it and depend on none of the others: the correspondence with the linear-filtering literature, what the uniform acquisition window misses against a real amplifier, and what was checked numerically and to what tolerance. It is written to be checked rather than read in sequence, and every equation the body or Methods refers to is numbered here.

Three marks, and each one says one thing

An arrow means the object spans the acquisition window, of duration Δ: $P_{i\to j}(\Delta)$ is a transition across it, $\bar{\gamma}_{i\to j}$ an average over it, $\bar{y}_{0\to\Delta}$ a recorded value formed from it. **No arrow means one instant**: $\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0$ are the occupancy mean and covariance where the window opens, $\boldsymbol{\mu}_\Delta, \boldsymbol{\Sigma}_\Delta$ where it closes. **A dot marks a collapsed end**: $a_{i\to\cdot} := \sum_j P_{i\to j}(\Delta)\, a_{i\to j} = \mathbb{E}[a \mid X_0 = i]$, as in $\bar{\gamma}_{i\to\cdot}$; for a probability the weight is already inside and the collapse is the plain margin. Across a recording a subscript t indexes the window, as in $\bar{y}_t$; inside a window the two ends are 0 and Δ, and a collapsed object is defined the moment its pair-resolved version is.

The interval average, and how the members are built

A single channel switching among its conductance levels traces a piecewise-constant path, the random telegraph signal when $K = 2$, and the window delivers its integral over one acquisition interval,

$$A_{0\to\Delta} = \int_0^\Delta \gamma(X_s)\, \mathrm{d}s, \qquad \text{(A1)}$$

where X_s is the state of the channel at time s. Its law is mixed: an atom at $\Delta\gamma_i$ carrying the probability of never leaving state i, and a continuous part from the paths that switched. Its density is a series over the number of transitions in the window, exact at every K, closed only at $K = 2$, and proliferating with K. Two things that might rescue it do not. Channel models carry few distinct conductances, every shut state at $\gamma = 0$ and the open states often sharing one γ, so $A_{0\to\Delta}$ is that conductance times the time spent in the open aggregate however large K is; but the aggregate is Markov only if $\mathbf{Q}$ is lumpable, which for the schemes that motivate this work it is not (***Colquhoun and Hawkes, 1981***). And the population multiplies the difficulty rather than adding to it, since the laws of independent channels convolve and a recorded sample constrains only the sum, so the posterior over channels does not factorize (***Moffatt, 2007***).

Every moment in the construction is carried on a single channel. Writing $\mathbf{X}$ for the indicator vector of the state one channel occupies, which is one on that state and zero elsewhere, $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$ is its probability vector and $\boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{X})$ its covariance, which is $\mathrm{diag}(\boldsymbol{\mu}) - \boldsymbol{\mu}\boldsymbol{\mu}^\top$ for a channel nothing has been observed about and vanishes when its state is known. Per channel is a normalization the whole construction is carried in, and not a claim that the channels stay independent under it: conditioning on a shared recorded value correlates them, so from the first update onward $\boldsymbol{\Sigma}$ is $\mathrm{Cov}(\mathbf{N})/N_{\mathrm{ch}}$ and not one channel's own covariance. Before any observation the channels are independent of each other, so the counts $\mathbf{N} = \sum_c \mathbf{X}^{(c)}$ over the ensemble have $\mathbb{E}[\mathbf{N}] = N_{\mathrm{ch}}\boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{N}) = N_{\mathrm{ch}}\boldsymbol{\Sigma}$, which is where every factor of N_{ch} in Eq. 2 comes from.

The boundary-conditioned moments, and how they are evaluated

The ladder needs two conductance objects, both built from the boundary-conditioned single-channel moments $\bar{\gamma}_{i\to j}$ (the mean interval-averaged conductance of a channel that starts in i and ends in j) and $\bar{v}_{i\to j}$ (its variance). Both are moments of the integrated conductance $A_{0\to\Delta}$ of Eq. A1, taken conditional on the chain starting in i and ending in j, and both are available in closed form, with no object larger than $K \times K$ formed anywhere. They are derivatives at $\alpha = 0$ of the conductance-tilted semigroup $e^{(\mathbf{Q}+\alpha\boldsymbol{\Gamma})\Delta}$, the generator biased by the conductance. Writing

$$\mathbf{F}_1(\Delta) = \frac{\partial}{\partial\alpha} e^{(\mathbf{Q}+\alpha\boldsymbol{\Gamma})\Delta}\bigg|_{\alpha=0}, \qquad 2\,\mathbf{F}_2(\Delta) = \frac{\partial^2}{\partial\alpha^2} e^{(\mathbf{Q}+\alpha\boldsymbol{\Gamma})\Delta}\bigg|_{\alpha=0}, \tag{A2}$$

with $\boldsymbol{\Gamma} = \mathrm{diag}(\boldsymbol{\gamma})$, the $(i \to j)$ entry of the first is $\mathbb{E}[A_{0\to\Delta}\,\mathbf{1}\{X_\Delta = j\} \mid X_0 = i]$ and of the second the same expectation with $A^2_{0\to\Delta}$ in place of $A_{0\to\Delta}$; the factor of two is the two triangular halves of the double integral the second derivative unfolds into. Dividing by the transition probability conditions on the endpoints and dividing by Δ turns the integral into an average, so

$$\bar{\gamma}_{i\to j} = \frac{[\mathbf{F}_1(\Delta)]_{i\to j}}{\Delta\, P_{i\to j}(\Delta)}, \qquad \bar{v}_{i\to j} = \frac{2[\mathbf{F}_2(\Delta)]_{i\to j}}{\Delta^2\, P_{i\to j}(\Delta)} - \bar{\gamma}^2_{i\to j}. \tag{A3}$$

Both derivatives are evaluated from an eigendecomposition of the generator, which is the route the implementation takes and the one the P2X2 analysis of ***Moffatt and Pierdominici-Sottile*** (***2025***) was run on. Writing $\mathbf{Q} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{W}$, with $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_K)$ the eigenvalues, $\mathbf{V}$ the matrix of right eigenvectors and $\mathbf{W} = \mathbf{V}^{-1}$, the transition matrix is $\mathbf{P}(\Delta) = \mathbf{V}e^{\boldsymbol{\Lambda}\Delta}\mathbf{W}$ and the two derivatives are the same sandwich with the conductance carried in the eigenbasis. With $\widehat{\boldsymbol{\Gamma}} = \mathbf{W}\boldsymbol{\Gamma}\mathbf{V}$ and $\circ$ the entrywise product,

$$\frac{\mathbf{F}_1(\Delta)}{\Delta} = \mathbf{V}\big(\widehat{\boldsymbol{\Gamma}}\circ\mathbf{E}_2\big)\mathbf{W}, \qquad (\mathbf{E}_2)_{ab} = E_2(\lambda_a\Delta, \lambda_b\Delta), \tag{A4}$$

$$\frac{2\,\mathbf{F}_2(\Delta)}{\Delta^2} = 2\,\mathbf{V}\mathbf{M}\mathbf{W}, \qquad M_{ab} = \sum_c \widehat{\Gamma}_{ac}\,\widehat{\Gamma}_{cb}\,E_3(\lambda_a\Delta, \lambda_c\Delta, \lambda_b\Delta), \tag{A5}$$

where E_2 and E_3 are the first and second divided differences of the exponential,

$$E_2(x,y) = \frac{e^x - e^y}{x - y}, \qquad E_3(x,y,z) = \frac{e^x}{(x-y)(x-z)} + \frac{e^y}{(y-x)(y-z)} + \frac{e^z}{(z-x)(z-y)}. \tag{A6}$$

Both are symmetric in their arguments and both are smooth where the denominators vanish, so repeated and near-repeated eigenvalues are read off the coincidence limits $E_2(x,x) = e^x$ and $E_3(x,x,x) = e^x/2$, with the one-pair-coincident form between them (Supplementary Information). Eq. A4 is the form in which the first moment is given in the supplement of ***Moffatt and Pierdominici-Sottile*** (***2025***); the second moment was not stated there. The eigendecomposition is computed once per agonist concentration and reused by every window at that concentration, after which each window costs the $O(K^3)$ contraction of Eq. A5.

An equivalent evaluation avoids diagonalizing $\mathbf{Q}$ altogether. $\mathbf{F}_1$ and $\mathbf{F}_2$ are the strictly upper blocks of the exponentials of the augmented generators

$$\mathcal{Q}_2 = \begin{pmatrix} \mathbf{Q} & \boldsymbol{\Gamma} \\ \mathbf{0} & \mathbf{Q} \end{pmatrix}, \qquad \mathcal{Q}_3 = \begin{pmatrix} \mathbf{Q} & \boldsymbol{\Gamma} & \mathbf{0} \\ \mathbf{0} & \mathbf{Q} & \boldsymbol{\Gamma} \\ \mathbf{0} & \mathbf{0} & \mathbf{Q} \end{pmatrix}, \tag{A7}$$

of sizes $2K\times2K$ and $3K\times3K$, whose exponentials are block triangular with $e^{\mathbf{Q}\Delta}$ on the diagonal (***Van Loan, 1978***), so that $e^{\mathcal{Q}_3\Delta}$ on its own returns the transition matrix and both moments with no spectral machinery, and where no reliable eigensolver is at hand it is the construction to use. The route taken here is the spectral one because this paper propagates the derivative of the log-likelihood with respect to the kinetic parameters through every object the filter builds, and the eigenvalues and eigenvectors of $\mathbf{Q}$ move with those parameters in closed form, by a first-order perturbation of the same decomposition. Every number reported in this paper comes from the spectral route.

Both expressions are $0/0$ where a transition is unreachable, which is not a numerical edge case here: at zero agonist $k_{\text{on}}[A]$ vanishes identically, so $P_{C\to O}(\Delta) = 0$ exactly over the whole pre-pulse segment. Conditioning on a single transit whose time, in the short-window limit, is uniform over the window makes the average $\gamma_j + (\gamma_i - \gamma_j)\,U$ with U uniform on $(0, 1)$, so the limit is the midpoint in the mean and the uniform-split variance,

$$\bar{\gamma}_{i\to j} \to \tfrac{1}{2}(\gamma_i + \gamma_j), \qquad \bar{v}_{i\to j} \to \tfrac{(\gamma_i-\gamma_j)^2}{12}, \tag{A8}$$

which depends on the conductance vector alone. The implementation does not branch on the singular case: it reads the two quotients of Eq. A3 as posterior means and shrinks them towards an endpoint prior with the same midpoint mean and the second moment $\frac{1}{2}(\gamma_i^2 + \gamma_j^2)$ (Methods), whose implied variance $((\gamma_i - \gamma_j)/2)^2$ is three times the limit above, a deliberately conservative floor, with a pseudo-count whose weight is the floating-point noise of the spectral reconstruction, so the limit is approached smoothly and no fitted constant enters (Methods). The start-conditioned mean conductance is the K-vector

$$\bar{\gamma}_{i\to\cdot} \;=\; \sum_j P_{i\to j}(\Delta)\,\bar{\gamma}_{i\to j}, \tag{A9}$$

which conditions on the state at the interval's start alone and is the object `INR`, `MR` and `VR` use. The boundary-conditioned mean conductance is the same average retained on both endpoints, so it stays a $K \times K$ object rather than collapsing to a vector, and is the object `IR` uses.

One generalization, and the whole family is inside it.

Eq. 2 takes a channel's conductance to be a number once its state is known. That is true of the current at an instant and of nothing else. One step of generality covers every member below: alongside the conductance, supply a *per-state variance* $\mathbf{w}$, the variance the recorded value still has once the state is fixed. Channels being independent given their states, those variances add over the ensemble, so

$$\mathbb{E}\big[\bar{y}\,\big|\,\mathbf{N}\big] = \boldsymbol{\gamma}^\top\mathbf{N}, \qquad \mathrm{Var}\big[\bar{y}\,\big|\,\mathbf{N}\big] = \epsilon^2_{0\to\Delta} + \textstyle\sum_s N_s\,w_s, \tag{A10}$$

and the extra variance is linear in the counts, exactly as the mean is. Taking the expectation over the state distribution turns the predictive variance of Eq. 2 into

$$\sigma^2 \;=\; \epsilon^2_{0\to\Delta} \;+\; N_{\text{ch}}\,\boldsymbol{\gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma} \;+\; N_{\text{ch}}\,\boldsymbol{\mu}^\top\mathbf{w}, \tag{A11}$$

with every other line of Eq. 2 unchanged. Setting $\mathbf{w} = \mathbf{0}$ returns the published filter. This is the same device ***Münch et al.*** (***2022***) add to the same filter to model open-channel noise, from a different motivation; here the state-dependent term will carry the gating variance the chosen state leaves unexplained rather than a property of the pore.

Four contractions, and the members differ in three of them.
Written this way, Eq. 2 contracts a member's objects against its own state distribution and covariance, and it does so four times:

$$\underbrace{\boldsymbol{\mu}^\top\boldsymbol{\gamma}}_{\text{the mean}} \qquad \underbrace{\boldsymbol{\mu}^\top\mathbf{w}}_{\text{the state-dependent noise}} \qquad \underbrace{\boldsymbol{\gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma}}_{\text{what the state contributes}} \qquad \underbrace{\boldsymbol{\Sigma}\boldsymbol{\gamma}}_{\text{state against current, }\mathbf{g}}$$

two against $\boldsymbol{\mu}$ and two against $\boldsymbol{\Sigma}$, and everything else in Eq. 2 is a scalar. Extending the filter to a recorded value that spans the window therefore cannot require a new algorithm. It requires choosing a state variable and supplying $(\boldsymbol{\gamma}, \mathbf{w})$ on it, and the members below are those choices. The per-state variance is not a free third choice beside the other two: Eq. A10 defines it as whatever variance the chosen state leaves unexplained, so naming the state names it.

The three steps that build the interval update

No conductance vector on the K states delivers the mean of an interval average, and on the K^2 boundary pairs one does. That substitution is Eq. 3 of the body, and three steps carry it out here. The two counts are one: the first three substitutions of Eq. 3 are step one, its fourth is step two, and the start index the body sums away is step three. Run Eq. 2 on the pairs, with $\bar{\gamma}_{i\to j}$ in place of $\boldsymbol{\gamma}$ and the boundary covariance in place of $\boldsymbol{\Sigma}$. Charge what the conductance still varies by once the pair is fixed to ϵ^2, as though the recording carried an extra noise of variance $\bar{v}_{i\to j}$ per channel, of a size that depends on the pair that channel took. Marginalize over the start index, which the next window has no use for. Every equation in this subsection is one of those three steps carried out, and the two objects the interval likelihood is usually written with are those two contractions read on the larger space.

Conditioning on the pair rather than on one end is the same device as static condensation, or as the spatial Markov property, applied on the time axis: the filter holds the states at the interval boundaries and marginalizes the trajectory between them analytically. It is not a transition state in the mechanistic sense, and the word *transition* is not used for it anywhere in this paper.

The boundary state's own Gaussian follows from the prior and the propagator. Let $\mathbf{X}_{0\to\Delta}$ be the indicator of the pair a single channel takes, one on the (i, j) it starts and ends in and zero on the other $K^2 - 1$, and let $(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ be the channel's state at the window's start. It leaves i and lands in j with probability $P_{i\to j}(\Delta)$, so

$$\mu_{i\to j} = \mu_{0,i}\, P_{i\to j}(\Delta), \tag{A12}$$

$$(\boldsymbol{\Sigma}_{0\to\Delta})_{(i\to j),(k\to l)} = P_{i\to j}(\Delta)\,\big(\boldsymbol{\Sigma}_0 - \mathrm{diag}(\boldsymbol{\mu}_0)\big)_{ik}\, P_{k\to l}(\Delta) \;+\; \big[\mathrm{diag}(\boldsymbol{\mu}_{0\to\Delta})\big]_{(i,j),(k,l)}. \tag{A13}$$

The two terms of Eq. A13 are the two sources of spread. The first carries the uncertainty already in the prior across the window, once on each index. The second is the uncertainty a channel known to be leaving i still has about where it lands, the covariance of a single draw from row i of $\mathbf{P}(\Delta)$, which is diagonal on the pairs because a channel takes exactly one of them. Subtracting $\mathrm{diag}(\boldsymbol{\mu}_0)$ in the first term is what keeps the prior's own single-channel part from being counted twice, since the second term regenerates it at the resolution of the pairs.

Summing the start index out of Eqs. A12 and A13 gives the occupancy Gaussian at the end of the window,

$$\boldsymbol{\mu}_\Delta = \mathbf{P}(\Delta)^\top \boldsymbol{\mu}_0, \tag{A14}$$

$$\boldsymbol{\Sigma}_\Delta = \mathbf{P}(\Delta)^\top \big(\boldsymbol{\Sigma}_0 - \mathrm{diag}(\boldsymbol{\mu}_0)\big)\mathbf{P}(\Delta) + \mathrm{diag}(\boldsymbol{\mu}_\Delta), \tag{A15}$$

term by term, the diagonal of the one becoming the diagonal of the other. This is the propagation an unconditioned member performs, and it is Eq. A13 seen from the end of the window alone. The first line of Eq. 2, on the boundary state, is the predictive. Its mean is

$$\bar{y}^{\mathrm{pred}}_{0\to\Delta} \;=\; N_{\mathrm{ch}}\,\boldsymbol{\mu}_0^{\top}\bar{\boldsymbol{\gamma}}_0, \tag{A16}$$

since $\boldsymbol{\mu}_{0\to\Delta}$ contracted against $\bar{\gamma}$ is Eq. A9 summed against the prior, and its variance has three terms,

$$\sigma^2_{0\to\Delta} \;=\; \epsilon^2_{0\to\Delta} \;+\; N_{\mathrm{ch}}\,\widetilde{\boldsymbol{\gamma}^{\top}\boldsymbol{\Sigma}\boldsymbol{\gamma}} \;+\; N_{\mathrm{ch}}\,\boldsymbol{\mu}_0^{\top}\mathbf{w}. \tag{A17}$$

The first is the ϵ^2 of Eq. 2, the instrumental noise now taken over the window, which carries the acquisition averaging and decreases with Δ; its parametric form is given in Methods. The third is the noise charged in step two, $N_{\mathrm{ch}}\sum_{i,j}\mu_{i\to j}\bar{v}_{i\to j}$, which is what Eq. A17 writes with $\mathbf{w}$, the start-conditioned interval variance, once the end index is summed. The middle term is $\boldsymbol{\gamma}^{\top}\boldsymbol{\Sigma}\boldsymbol{\gamma}$ of Eq. 2 evaluated on the pairs, and the tilde is what marks that: under it the two $\boldsymbol{\gamma}$ stand for $\bar{\gamma}_{i\to j}$ and $\boldsymbol{\Sigma}$ for the $K^2\times K^2$ boundary covariance, not for the K-vector and the $K\times K$ matrix they name everywhere else. Written out with $\boldsymbol{\Sigma}_{0\to\Delta}$ of Eq. A13 and then collapsed by substituting it,

$$\begin{aligned}\widetilde{\boldsymbol{\gamma}^{\top}\boldsymbol{\Sigma}\boldsymbol{\gamma}} \;&=\; \sum_{i,j}\sum_{k,l}\bar{\gamma}_{i\to j}\left(\boldsymbol{\Sigma}_{0\to\Delta}\right)_{(i,j),(k,l)}\bar{\gamma}_{k\to l}\\ &=\; \bar{\boldsymbol{\gamma}}_0^{\top}\big(\boldsymbol{\Sigma}_0-\mathrm{diag}(\boldsymbol{\mu}_0)\big)\bar{\boldsymbol{\gamma}}_0 \;+\; \boldsymbol{\mu}_0^{\top}\big(\mathbf{H}\mathbf{1}\big),\end{aligned} \tag{A18}$$

where $H_{i\to j}=\bar{\gamma}^2_{i\to j}\,P_{i\to j}(\Delta)$ and $\mathbf{1}$ is the vector of ones. The two summands are the two terms of Eq. A13 in order, and only K-vectors and $K\times K$ matrices survive, so the $K^2\times K^2$ covariance is never built. Reading the tilde as $\bar{\boldsymbol{\gamma}}_0^{\top}\boldsymbol{\Sigma}_0\bar{\boldsymbol{\gamma}}_0$ keeps the first summand and drops the second, which is the single-channel spread of the interval mean across boundary pairs and does not vanish when the occupancy covariance does.

The gain is the same contraction taken once. Built on the pairs it is the K^2-vector $\sum_{k,l}(\boldsymbol{\Sigma}_{0\to\Delta})_{(i,j),(k,l)}\bar{\gamma}_{k\to l}$, and step three sums its start index away, leaving the $\mathbf{g}_{0\to\Delta}$ the interval update uses,

$$\mathbf{g}_{0\to\Delta} \;=\; \widetilde{\boldsymbol{\gamma}^{\top}\boldsymbol{\Sigma}} \;=\; \mathbf{P}(\Delta)^{\top}\big(\boldsymbol{\Sigma}_0-\mathrm{diag}(\boldsymbol{\mu}_0)\big)\bar{\boldsymbol{\gamma}}_0 \;+\; \mathbf{G}^{\top}\boldsymbol{\mu}_0, \qquad G_{i\to j} \;=\; \bar{\gamma}_{i\to j}\,P_{i\to j}(\Delta), \tag{A19}$$

the covariance between the state $\mathbf{X}_\Delta$ a channel ends the window in and the current that same channel contributes to the window average. Over the ensemble it is $\mathrm{Cov}(\mathbf{N}_\Delta,\bar{y}) = N_{\mathrm{ch}}\,\mathbf{g}_{0\to\Delta}$, with $\mathbf{N}_\Delta$ the counts by state at the end of the window, and that is the form the update uses. Again the two summands are the two terms of Eq. A13: $\mathbf{G}$ carries one power of $\bar{\gamma}$ where $\mathbf{H}$ carried two, and the end index is left open instead of being summed against $\mathbf{1}$.

The second line of Eq. 2, marginalized the same way, is the update, with $\delta=\bar{y}^{\mathrm{obs}}_{0\to\Delta}-\bar{y}^{\mathrm{pred}}_{0\to\Delta}$ the innovation,

$$\boldsymbol{\mu}^{\mathrm{post}} = \boldsymbol{\mu}_\Delta + \frac{\mathbf{g}_{0\to\Delta}}{\sigma^2_{0\to\Delta}}\,\delta, \tag{A20}$$

$$\boldsymbol{\Sigma}^{\mathrm{post}} = \boldsymbol{\Sigma}_\Delta - \frac{N_{\mathrm{ch}}\,\mathbf{g}_{0\to\Delta}\mathbf{g}_{0\to\Delta}^{\top}}{\sigma^2_{0\to\Delta}}. \tag{A21}$$

Conditioning and summing the start index commute here, because the sum is linear and the correction is rank one on the index it acts on, so the downdate crosses it unchanged, with $\mathbf{g}_{0\to\Delta}$ standing where the boundary gain stood. What the sum discards is the posterior correlation between the interval current and the state the window *started* in, which costs nothing inside the window and is what the recursion pays for below.

Two of the three steps are exact and one is the approximation. Eqs. A12 and A13 are the true moments of $\mathbf{X}_{0\to\Delta}$, and so are the three the predictive consumes. Writing $\mathbf{N}_{0\to\Delta}$ =

$\sum_c \mathbf{X}^{(c)}_{0\to\Delta}$ for the boundary counts over the ensemble, channels are independent given the pairs each of them took, so given $\mathbf{N}_{0\to\Delta}$ the recorded average has mean $\sum_{i,j} \bar{\gamma}_{i\to j} N_{i\to j}$ and variance $\epsilon^2_{0\to\Delta} + \sum_{i,j} \bar{v}_{i\to j} N_{i\to j}$ exactly, and the state-dependent noise of step two, having no covariance with the counts that carry it, enters $\sigma^2_{0\to\Delta}$ and changes neither $\bar{y}^{\text{pred}}_{0\to\Delta}$ nor $\mathbf{g}_{0\to\Delta}$. That last point is what makes charging it to the noise term legitimate rather than convenient. What is approximate is the Gaussian, in the two places the box above names: the counts are not Gaussian, which is the occupancy closure, and neither is the window average of one channel given its pair, so replacing its law by the two numbers $\bar{\gamma}_{i\to j}$ and $\bar{v}_{i\to j}$ is the interval-signal closure. The two together still do not make the pair $(\mathbf{N}_{0\to\Delta}, \bar{y})$ jointly Gaussian, since its conditional variance depends on the state, so the update conditions the Gaussian that carries the exact moments. The factor N_{ch} in the downdate, and its absence from the mean correction, come from the same place. All three of $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$ and $\mathbf{g}_{0\to\Delta}$ belong to one channel, while what is observed belongs to the ensemble, so the conditioning is done on $\text{Cov}(\mathbf{N}_\Delta, \bar{y}) = N_{\text{ch}}\mathbf{g}_{0\to\Delta}$ against $\text{Cov}(\mathbf{N}_\Delta) = N_{\text{ch}}\boldsymbol{\Sigma}_\Delta$. The rank-one term picks up the channel count twice and the return to one channel gives one back, leaving a single factor; the mean correction picks it up once and gives it back, leaving none.

The start-conditioned members, stated forward

The construction above reaches `IR` by choosing the boundary pair as the state. The members that stop at the interval's start are stated here from their own assumptions rather than as that construction with the collapse brought forward.

`MR` takes the recorded value to depend on the occupancy where the window opens. Formally this asserts

$$\bar{y}_{0\to\Delta} \;\perp\; X_\Delta \;\Big|\; X_0, \tag{A22}$$

that once the state a channel started in is known, the recorded average says nothing further about where that channel ended. Everything `MR` does follows from Eq. A22, and the member is complete without any object larger than the K states. What it must supply is a conductance and a per-state variance on that state, and both are conditional moments of $A_{0\to\Delta}$ given the start alone:

$$(\bar{\gamma}_0)_i = \mathbb{E}\left[\tfrac{1}{\Delta} A_{0\to\Delta} \;\Big|\; X_0 = i\right], \qquad (\sigma^2_{\bar{\gamma}_0})_i = \text{Var}\left[\tfrac{1}{\Delta} A_{0\to\Delta} \;\Big|\; X_0 = i\right], \tag{A23}$$

the first being Eq. A9 and the second the per-state variance Eq. A11 then asks for. Both come from the same tilted semigroup of Eq. A2 that any interval member needs, but from its derivatives contracted against the vector of ones rather than read entry by entry: what `MR` requires are the row sums, and the row sums are integrals of K-vectors, so `MR` is reached without ever forming a pair.

Eq. A22 is an assumption about the process and can be examined on its own terms. It is false for any window of nonzero length: a channel that starts shut and delivers current during the window is more likely to have ended open than one that delivers none, so the recorded average does carry information about the end state beyond what the start state gives. `IR` is the member that declines to assume it, and the boundary pair is the state on which the recorded average is linear with no conditional expectation taken at all. `MR` is therefore a correctly derived filter for a measurement model that is false, rather than an approximate version of `IR`; how much the falsehood costs is Eq. A28 below.

What separates the start-conditioned members from the boundary one

Run as the same three steps on the K states instead of the pairs, with $\bar{\boldsymbol{\gamma}}_0$ in place of $\boldsymbol{\gamma}$, two algebraic differences separate them from the one that conditions on both ends. The first is the form of the third term of Eq. A17. `MR` carries the total per-start-state interval variance, the spread of the interval-averaged conductance of a channel known only to have started in i, while `VR` and `IR` carry the residual variance left once the end state is also conditioned on, $\sum_j P_{i\to j}(\Delta)\,\bar{v}_{i\to j}$. By the law of total variance the two forms differ by the spread of the interval mean across end states,

$$\underbrace{\sum_j P_{i\to j}(\Delta)\big[\bar{v}_{i\to j} + \big(\bar{\gamma}_{i\to j} - \bar{\gamma}_{i\to\cdot}\big)^2\big]}_{\text{total, MR}} = \underbrace{\sum_j P_{i\to j}(\Delta)\,\bar{v}_{i\to j}}_{\text{residual, VR, IR}} + \mathrm{Var}_j\big[\bar{\gamma}_{i\to j} \mid i\big], \tag{A24}$$

so `MR` charges to the observation variance a spread that the end state, once conditioned on, would resolve. The second difference is the boundary cross-covariance term

$$N_{\mathrm{ch}}\,\widetilde{\boldsymbol{\gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma}} \tag{A25}$$

which `IR` retains and both start-conditioned members drop. It enters twice, in the second term of Eq. A17 and in the gain $\mathbf{g}_{0\to\Delta}$ of Eqs. A20 and A21.

The two differences are not independent, and the tempting reading of them, one step that moves the variance followed by one that moves the gain, is wrong in both directions. Taken from a common prior they cancel in the predicted variance: what the boundary term adds to the second term of Eq. A17 is $\boldsymbol{\mu}_0^\top \mathrm{Var}_j[\bar{\gamma}_{i\to j} \mid i]$, which is what the residual form removes from the third, so from the same state `MR` and `IR` assign the interval average the same variance by two different decompositions and the whole of what separates them is the gain, an identity written out at Eq. A37. In the other direction, the predicted variance divides the gain, so changing the variance form changes the update as well: `VR` removes that spread from the third term without restoring it in the second, predicts less variance than either from the same state, and its posterior departs from `MR`'s at once. The equality is therefore between two maps from a prior to a predicted variance, and not between two recordings. Once the gains differ the priors differ, and the variances the two members report along one recording differ with them, which is what the Results measure across the regime grid.

Both members against the exact law

Neither member is measured against the other. Both are approximations to one joint law, that of $(\mathbf{N}_\Delta, \bar{y})$ under the true process, and given $(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ that law has exact first two moments which either member can be held against. Two elementary decompositions supply them, and they say different things about the two blocks.

The variance: both members are exact, and their slots differ.

By the law of total variance, conditioning on the counts where the window opens,

$$\mathrm{Var}\big[\bar{y}\big] = \epsilon^2_{0\to\Delta} + \underbrace{\mathbb{E}\big[\mathrm{Var}(\bar{y} \mid \mathbf{N}_0)\big]}_{N_{\mathrm{ch}}\,\boldsymbol{\mu}_0^\top \mathbf{w}} + \underbrace{\mathrm{Var}\big[\mathbb{E}(\bar{y} \mid \mathbf{N}_0)\big]}_{N_{\mathrm{ch}}\,\bar{\boldsymbol{\gamma}}_0^\top \boldsymbol{\Sigma}_0 \bar{\boldsymbol{\gamma}}_0}, \tag{A26}$$

which is Eq. A17 read with the start-conditioned objects: the middle term is $\bar{\boldsymbol{\gamma}}_0^\top\boldsymbol{\Sigma}_0\bar{\boldsymbol{\gamma}}_0$, the tilde dropped, and $\mathbf{w}$ is the total slot of Eq. A24. Read with the boundary objects instead, the tilde of Eq. A18 in the middle and the residual slot in $\mathbf{w}$, the same quantity is Eq. A17 again. **Both members report the exact variance of the recorded value** given $(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$, so they agree with each other for the reason that each agrees with the truth, and nothing is measured by that agreement. The variable conditioned on in Eq. A26 is the count vector and not one channel's state: conditioning on a single channel's X_0 returns the same expressions only while $\boldsymbol{\Sigma}_0$ is still that channel's own covariance, which it stops being at the first update.

The covariance: one member is exact and the other is short by a named amount. By the law of total covariance, conditioning on the same counts and carried on the per-channel scale,

$$\frac{1}{N_{\rm ch}}\mathrm{Cov}\left[\mathbf{N}_\Delta,\ \bar{y}\right] = \underbrace{\frac{1}{N_{\rm ch}}\mathrm{Cov}\left[\mathbb{E}(\mathbf{N}_\Delta \mid \mathbf{N}_0),\ \bar{y}\right]}_{\mathbf{g}^{\rm MR}_{0\to\Delta}} + \frac{1}{N_{\rm ch}}\mathbb{E}\left[\mathrm{Cov}(\mathbf{N}_\Delta,\ \bar{y} \mid \mathbf{N}_0)\right]. \tag{A27}$$

The first term is the start-conditioned gain $\mathbf{g}^{\rm MR}_{0\to\Delta} = \mathbf{P}(\Delta)^\top\boldsymbol{\Sigma}_0\bar{\boldsymbol{\gamma}}_0$ exactly. The second is what Eq. A22 sets to zero: given the counts the channels are conditionally independent, so it is the per-channel conditional covariance summed over the population, and a conditional independence is precisely the statement that it vanishes. It does not vanish. `IR` carries the whole of Eq. A27, so the difference between the two members in this block is that second summand and nothing else,

$$\mathbf{g}^{\rm IR}_{0\to\Delta} - \mathbf{g}^{\rm MR}_{0\to\Delta} = \left(\sum_i \mu_{i\to j}\, D_{i\to j}\right)_j, \qquad D_{i\to j} = \bar{\gamma}_{i\to j} - (\bar{\gamma}_0)_i, \tag{A28}$$

which is Eq. A19 minus $\mathbf{P}(\Delta)^\top\boldsymbol{\Sigma}_0\bar{\boldsymbol{\gamma}}_0$ written out. The deviation $D_{i\to j}$ is a property of the process and not of either algorithm: it is how far the interval average of a channel that ended in j sits from the average over all the places it might have ended. It has zero row sums, $\sum_j P_{i\to j}(\Delta)D_{i\to j} = 0$, since $(\bar{\gamma}_0)_i$ is that very average, and it therefore has zero mean under $\mu_{i\to j}$ as well, so the gap between the two forms of Eq. A24 is exactly its variance under the law of the pairs. Eq. A28 does not depend on $\boldsymbol{\Sigma}_0$ at all. The same deviation is invisible in one block and decisive in the other because it enters the variance squared and the covariance once: a second moment can sit in either of the two variance terms without the total noticing, which is Eq. A24, while a collapsed first moment has nowhere to go, a noise term having no first moment. That asymmetry between D^2 and D is the whole of the difference between the two members, and it is a consequence of Eq. A22 rather than a defect of the arithmetic that implements it.

The same construction in the linear-filtering frame

Everything above was Gaussian conditioning: a joint law over (state, recorded value), its three blocks, and Bayes. Nothing was borrowed from linear filtering, and the derivation does not need it. The correspondence with that literature is nevertheless exact; the relation is conceded in the Discussion and placed against the prior art in Appendix 5, and the equations that make it checkable are here.

For a state $\mathbf{x}$ and a scalar observation y jointly Gaussian, the filter's update is

$$\mathbf{x}^{\rm post} = \mathbf{x}^{\rm prior} + \frac{\mathrm{Cov}(\mathbf{x}, y)}{\mathrm{Var}(y)}\big(y - \mathbb{E}[y]\big), \qquad \boldsymbol{\Sigma}^{\rm post} = \boldsymbol{\Sigma}^{\rm prior} - \frac{\mathrm{Cov}(\mathbf{x}, y)\,\mathrm{Cov}(\mathbf{x}, y)^\top}{\mathrm{Var}(y)}, \tag{A29}$$

written in the symbols used here, the state covariance that literature writes $\mathbf{P}$ for being $\boldsymbol{\Sigma}$ throughout this paper. It is Eq. 2 up to the per-channel convention, the source of the factors of $N_{\rm ch}$ there and not here. The two frames do not differ in the update at all: both are the conditioning of a joint Gaussian, and what is usually called the gain is the ratio of the cross block to the observation variance. The entire content of either derivation is the computation of the three blocks $\mathbb{E}[y]$, $\mathrm{Var}(y)$ and $\mathrm{Cov}(\mathbf{x}, y)$ for a recorded value that is an average over the window.

The standard difficulty is that $\bar{y}$ is not a function of the state at any single time, so it is not a fixed linear functional of $\mathbf{x}_\Delta$ plus noise. The device is state augmentation: carry the running integral $z_t = \int_0^t \boldsymbol{\gamma}^\top\mathbf{N}(s)\,\mathrm{d}s$ alongside the counts, so that $(\mathbf{N}, z)$ propagates jointly and the recorded value is z_Δ/Δ plus instrumental noise, which *is* linear in the augmented state. This is the treatment of an observation that is a flow rather than a stock, and it is a known construction rather than a new one (***Zadrozny, 1988***). Augmentation moves the problem rather than solving it: the filter now needs the variance of the accumulator and its covariance with the occupancy at the end of the window, and in general those are not free. For a continuous-time Markov chain they are, and by the two derivatives already written as Eq. A2: the $(i \to j)$ entry of $\mathbf{F}_1$ is $\mathbb{E}[A_{0\to\Delta}\mathbf{1}\{X_\Delta = j\} \mid X_0 = i]$ and of $2\mathbf{F}_2$ the same with $A^2_{0\to\Delta}$, which is exactly what the augmented filter asks for.

Condition on the counts where the window opens; given those the channels are independent, and the three blocks are

$$\mathbb{E}\left[\bar{y}\right] = N_{\mathrm{ch}}\,\boldsymbol{\mu}_0^\top \bar{\boldsymbol{\gamma}}_0, \tag{A30}$$

$$\mathrm{Var}\left[\bar{y}\right] = \epsilon^2_{0\to\Delta} + \frac{\mathrm{Var}(z_\Delta)}{\Delta^2}, \tag{A31}$$

$$\frac{\mathrm{Cov}\left[\mathbf{N}_\Delta, \bar{y}\right]}{N_{\mathrm{ch}}} = \frac{\mathrm{Cov}\left[\mathbf{N}_\Delta, z_\Delta\right]}{N_{\mathrm{ch}}\Delta} = \mathbf{P}(\Delta)^\top \boldsymbol{\Sigma}_0 \bar{\boldsymbol{\gamma}}_0 + \Big(\textstyle\sum_i \mu_{i\to j} D_{i\to j}\Big)_j. \tag{A32}$$

Eq. A30 is Eq. A16, Eq. A31 is Eq. A17, and Eq. A32 is $\mathbf{g}^{\mathtt{MR}}_{0\to\Delta}$ plus Eq. A28, which is to say the $\mathbf{g}_{0\to\Delta}$ of Eq. A19. The second term of Eq. A32 needs nothing from the boundary construction: for one channel, $\mathbf{F}_1$ divided by Δ and by the transition probability gives $\mathbb{E}[\mathbf{1}\{X_\Delta = j\}\,\frac{1}{\Delta}A_{0\to\Delta} \mid X_0 = i] = P_{i\to j}(\Delta)\bar{\gamma}_{i\to j}$, the product of the two conditional expectations is $P_{i\to j}(\Delta)(\bar{\gamma}_0)_i$, and their difference is $P_{i\to j}(\Delta)D_{i\to j}$. The deviation of Eq. A28 is therefore what the first derivative of the tilted semigroup leaves once the product of the two first moments is taken out, and the agreement between the two routes is a result rather than an input.

The augmented-state filter and the boundary-state filter compute the same three blocks, so they perform the same update and return the same posterior over the occupancy.

The augmented state has $K+1$ entries against the boundary state's K^2, so *the pair is a device of the derivation and not information the filter must carry*, which is the same conclusion Eqs. A14 and A15 reached from the other side. Given $(\mathbf{N}_\Delta, z_\Delta)$ the recorded value is deterministic, so what Eq. A17 splits between its second and third terms sits entirely inside $\mathrm{Var}(z_\Delta)$, and that is why the cross block needs only $\mathbf{F}_1$ while the variance block needs $\mathbf{F}_2$ as well. The equivalence holds per window, with a reset: the augmented filter must discard the posterior on z and open the next window with $z = 0$, because the next window's integral is a fresh one and is independent of the old given the occupancy.

`MR` is not the augmented filter under any choice of blocks. It is the filter for a misspecified measurement model,

$$\bar{y} = \bar{\boldsymbol{\gamma}}_0^\top \mathbf{N}_0 + v(\mathbf{N}_0), \qquad \mathrm{Var}\left[v \mid \mathbf{N}_0\right] = \epsilon^2_{0\to\Delta} + \mathbf{N}_0^\top \mathbf{w}, \tag{A33}$$

whose expectation over $\mathbf{N}_0$ returns Eq. A17. The measurement noise depends on the state, which is outside the textbook filter: with that extension the two frames give the same equations for `MR` as well, and without it `MR` has no linear-filtering form at all.

Where the frames genuinely differ is the model rather than the arithmetic. A Kalman derivation presumes a linear-Gaussian state process, and the occupancy counts of a channel population are not one: they are multinomial, their transition noise depends on the state they leave, and no linear-Gaussian system generates them. The derivation above presumes none of that: it computes the exact first two moments of the true process, over the pairs where the recorded average is genuinely linear, and replaces the law by a Gaussian only at the two places the box above names, so the two routes arrive at the same equations because the conditioning of a joint Gaussian consumes nothing but the first two moments. The pair survives as the presentation even though the augmented state reaches the same blocks with fewer entries: on the pair the recorded average is linear in the state with no conditional expectation taken, the collapse hands over the next window's prior at no cost, and the per-state variance slot is what makes `R`, `MR` and `IR` one construction with three readings.

The acquisition kernel, and what the uniform window misses

The likelihood implements the boxcar of Eq. 1 exactly, while a real acquisition kernel is that boxcar composed with the amplifier's analogue low-pass, so what is unmodelled is governed by one dimensionless group, the cutoff frequency against the acquisition window, $f_c\Delta$. For any component flat on the scale of Δ, the effect on the two objects the likelihood uses, the variance of a recorded value and the covariance between consecutive ones, is

$f_c\Delta$	0.2	0.5	1	2	5	10	20
variance deficit	61.5%	29.6%	14.7%	7.4%	2.9%	1.5%	0.74%
lag-one correlation ρ_1	64.1%	21.1%	8.6%	4.0%	1.5%	0.75%	0.37%
error on a confidence interval	61%	19%	8.3%	3.9%	1.5%	0.74%	0.37%

From $f_c\Delta = 1$ upward the deficit is $0.147/(f_c\Delta)$ and ρ_1 is $0.075/(f_c\Delta)$, while ρ_2 and every longer lag are zero to four decimals: the residue is a nearest-neighbour effect of known size. The total is conserved, $\mathrm{Var} + 2\sum_k \mathrm{Cov}_k = S/\Delta$ to six decimals at every setting, forced by the filter having unit gain at zero frequency, so what a likelihood allowing no such covariance sees is a misallocation of variance between the diagonal and its neighbours rather than a loss of it. The pole count does not enter: four poles, eight poles and a Gaussian of the same $-3\ \mathrm{dB}$ frequency agree to the third decimal from $f_c\Delta = 0.5$ upward, so the width can be quoted as the Gaussian equivalent $\sigma_f = \sqrt{\ln 2}/(2\pi f_c) = 0.1325/f_c$, a ten-to-ninety rise time of $0.34/f_c$, which is the convention electrophysiology already uses for filter width (***Colquhoun and Sigworth, 1995***). And the kinetics are the protected part: the deficit is first order in $1/(f_c\Delta)$ for the instrumental noise and second order for a component carrying a correlation time of its own, because unit gain at zero frequency preserves the integral of an autocovariance and the block-averaged variance of a short-correlated component depends on that integral alone to leading order. At $f_c\Delta = 5$ the white term loses 2.94% and a gating term with correlation time Δ loses 0.11%, a factor of 27, so the residue lands in the instrumental-noise partition, which is where the Discussion already argues an unmeasured departure of the noise model lives.

What does not scale with the group is the interval holding the concentration jump: group delay plus rise displace and smear the transient by about $0.69/f_c$ at four poles, a fixed time rather than a fraction of Δ, so that one interval carries an error the expansion says nothing about, and the honest recipe discards or flags the first interval after a step.

One property of this family follows from the same algebra and a point-sampling method does not have it. Because the observable *is* the integral, block-averaging a record down to a coarser Δ produces exactly the observable of Eq. 1 at any decimation factor, with no aliasing question to answer; after decimation the only unmodelled piece left is the residual kernel, which is the table above. That is what makes "average the record first" a scoping statement rather than a workaround.

What was verified, and to what tolerance

The identities of the boundary-state construction were checked against a direct contraction over the K^2 pairs rather than asserted. Eqs. A18 and A19 reproduce the explicit contraction of the boundary covariance in 108 random cases to 7×10^{-18} and 3×10^{-16}; the commutation of the start-index marginalization with the update, which is what licenses Eqs. A20 and A21 being written on the K states, holds in 36 cases to 2×10^{-16}; Eq. A28 holds to 1×10^{-17} in 36

cases, independently of $\mathbf{\Sigma}_0$, and against a Monte Carlo of the continuous-time path; Eqs. A12 and A13 were checked against a Monte Carlo of 4×10^5 replicates of 200 channels, worst entry 2.9 standard errors over 256 entries; and the two forms of Eq. A24 agree with the forms that go through the pair-resolved variance to 1×10^{-15} in 16 cases. The whole construction, run on the pairs and marginalized, reproduces Eqs. A14, A15, A16, A17, A18, A19, A20 and A21 to 9.3×10^{-16} relative on two priors, a multinomial start and a perturbed post-update one.

Appendix 2

The interval update, member by member

Appendix 1 gives the cycle for the boundary-conditioned member and names the two places where the others differ. This appendix writes every member out in one template, so that each is fixed by three switches on a single set of equations.

Objects computed once per interval

All members share the quantities built from the rate matrix $\mathbf{Q}$ and the recording filter for a window of length Δ: the transition matrix $\mathbf{P}(\Delta)$; the boundary-conditioned single-channel moments $\bar{\gamma}_{i\to j}$ and $\bar{v}_{i\to j}$; the start-conditioned mean conductance $\bar{\boldsymbol{\gamma}}_0$ of Eq. A9; the start-conditioned interval variance $\mathbf{w}$; and, for the members that ignore the acquisition averaging, the instantaneous per-state conductance vector $\boldsymbol{\gamma}$. Two matrices collect the boundary weights,

$$G_{i\to j} \;=\; \bar{\gamma}_{i\to j}\, P_{i\to j}(\Delta), \qquad H_{i\to j} \;=\; \bar{\gamma}^2_{i\to j}\, P_{i\to j}(\Delta). \tag{A34}$$

The tilde operator

Over a bold symbol, the tilde marks a contraction taken on the K^2 boundary pairs $(i \to j)$ rather than on the K states. (Over a scalar it means something else, a quantity in the natural units of the model, as in $\widetilde{\Delta}$ and $\widetilde{S}$; the two uses are disjoint and never meet in one expression.) It is not a new operator: its two cases are the two contractions the update of Eq. 2 already has, $\boldsymbol{\gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma}$ and $\boldsymbol{\Sigma}\boldsymbol{\gamma}$, evaluated there with the boundary objects substituted. Both are written out in Appendix 1, the bilinear one as the scalar of Eq. A18 and the vector one as the gain $\mathbf{g}_{0\to\Delta} = \widetilde{\boldsymbol{\gamma}^\top\boldsymbol{\Sigma}}$ of Eq. A19, and both collapse to $K \times K$ arithmetic once the boundary covariance Eq. A13 is substituted.

One cycle, three switches

Every one of the six runs the same four steps on each window: propagate the occupancy Gaussian (Eqs. A14 and A15), predict the observation (Eqs. A16 and A17), condition on the recorded average (Eqs. A20 and A21), and either carry the posterior to the next window or discard it. Three switches fix which member is being run.

Switch 1: the window.

It has three settings, an instant, the state at the window's start, and the states at both of its ends, and it selects the conductance object, and with it the second term of the predictive variance and the gain. At the instantaneous setting the observation is treated as a sample taken at the centre of the window: the occupancy is propagated there first, by $\mathbf{P}(\Delta/2)$, the conductance is $\boldsymbol{\gamma}$, and, with $\boldsymbol{\Sigma}$ read in that mid-window frame,

$$\widetilde{\boldsymbol{\gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma}} \;\to\; \boldsymbol{\gamma}^\top\boldsymbol{\Sigma}\boldsymbol{\gamma}, \qquad \mathbf{g} \;\to\; \mathbf{P}(\Delta/2)^\top\boldsymbol{\Sigma}\,\boldsymbol{\gamma}. \tag{A35}$$

At the start-conditioned setting the same two expressions hold with $\bar{\boldsymbol{\gamma}}_0$ in place of $\boldsymbol{\gamma}$ and the full $\mathbf{P}(\Delta)$ in the gain, since the conductance is conditioned on the state at the window's start and the whole window remains to be crossed; the contraction is still an ordinary quadratic form on the K states. The trailing propagator in the gain is not decoration. It carries the update from the frame in which the observation was formed to the frame the next window starts in, which is the end of the current window, so it is the remaining half at the instantaneous setting and the whole window at the start-conditioned one. Omitting it, or using the full window where half is called for, leaves the rank-one down-date in the wrong frame, and the symptom is specific and was observed: a spike in the distortion-induced bias of the recursive instantaneous member at long intervals and large channel counts. At the boundary setting the conductance is conditioned on both endpoints and the two expressions become the tildes, Eq. A18 and Eq. A19. The step from the start state to the boundary therefore does two things at once: it replaces $\boldsymbol{\Sigma}_0$ by $\boldsymbol{\Sigma}_0 - \mathrm{diag}(\boldsymbol{\mu}_0)$ and adds the boundary term ($\boldsymbol{\mu}_0^\top \mathbf{H1}$ in the variance, $\mathbf{G}^\top \boldsymbol{\mu}_0$ in the gain). The subtracted diagonal is not a second approximation: the within-channel part it removes is already inside the boundary term, and dropping the subtraction would count it twice.

Switch 2: recursion.

A recursive member carries $(\boldsymbol{\mu}^{\mathrm{post}}, \boldsymbol{\Sigma}^{\mathrm{post}})$ into the next window. A non-recursive one discards it and propagates the next window's predictive from the initial condition, so its log-likelihood factorizes over windows. The switch changes no formula above; it changes only what is fed to the next one.

Switch 3: the interval variance, total or residual.

It selects the third term of Eq. A17, which is present only when the acquisition averaging is modelled at all, that is at either of the two interval settings of switch 1:

$$(\mathbf{w})_i = \begin{cases} \sum_j P_{i\to j}(\Delta)\left[\bar{v}_{i\to j} + \left(\bar{\gamma}_{i\to j} - \bar{\gamma}_{i\to\cdot}\right)^2\right] & \text{total,} \\ \sum_j P_{i\to j}(\Delta)\,\bar{v}_{i\to j} & \text{residual.} \end{cases} \tag{A36}$$

The two differ by the spread of the interval mean across end states, which is Eq. A24. The switch is free only where the window is conditioned on the start state alone. With both endpoints conditioned on, the residual form is forced whatever the flag says, for the same reason the diagonal is subtracted in switch 1: the between-endpoint spread is already carried by the boundary term, so the total form would count it twice.

The members

Table 1 sets the six members against the three switches, and it is the equation-level counterpart of Table 1 in the body: where that table says in words what each member predicts, this one says which setting of Eqs. A35, A36 and A20 produces it.

Member	window	recursive	interval variance	what it is
`NR`	an instant	no	—	instantaneous samples, windows independent
`R`	an instant	yes	—	instantaneous samples, posterior carried forward
`INR`	the start	no	total	`NR` with the window average restored
`MR`	the start	yes	total	`R` with the window average restored
`VR`	the start	yes	residual	`MR` with switch 3 flipped, gain unchanged
`IR`	both ends	yes	residual (forced)	`VR` with the boundary terms restored

Appendix 2—table 1. The six members as settings of the three switches. Reading down the last column, each row differs from an earlier one in exactly one switch, which is what makes the ladder measurable one step at a time. `IR` is MacroIR; `INR` is the MacroINR of ***Moffatt and Pierdominici-Sottile*** (***2025***). The data keys under which each is dispatched are in Supplementary File 1.

One consequence of the two switches acting together is worth stating, because it is easy to read the table as saying otherwise. Evaluated at the *same* prior, `MR` and `IR` assign the interval average the same total variance. Expanding the total form of Eq. A36 and using $\sum_j P_{i\to j}(\Delta) = 1$ and $\sum_j P_{i\to j}(\Delta)\bar{\gamma}_{i\to j} = \bar{\gamma}_{i\to\cdot}$,

$$\boldsymbol{\mu}_0^\top \mathbf{w}\Big|_{\text{total}} = \boldsymbol{\mu}_0^\top \mathbf{w}\Big|_{\text{residual}} + \boldsymbol{\mu}_0^\top \mathbf{H1} - \bar{\boldsymbol{\gamma}}_0^\top \mathrm{diag}(\boldsymbol{\mu}_0)\, \bar{\boldsymbol{\gamma}}_0, \tag{A37}$$

and the two extra pieces are exactly what switch 1 moves from the third term of Eq. A17 into the second when the window goes from the start state to both ends. The decompositions differ, the sum does not; what that leaves as the whole of the difference, and why the equality is between two maps from a prior and not between two recordings, is Appendix 1.

Two of the six pairs isolate one algebraic change at the point where it is made, and they are the ones the Results use. Stepping `MR` to `VR` flips switch 3, and stepping `VR` to `IR` restores the boundary terms in both the variance and the gain while leaving the variance form where it already was; neither step is confined to what it flips, for the reason given in Appendix 1. The two together are the `R`-to-`IR` gap.

Least squares, which is off the lattice

Classical nonlinear least squares has no setting of the three switches. It propagates the mean occupancy alone, $\boldsymbol{\mu}_\Delta = \mathbf{P}(\Delta)^\top \boldsymbol{\mu}_0$, predicts $\bar{y}^{\text{pred}}_{0\to\Delta}$ from Eq. A16, and assigns to every residual one scale common to the whole record, marginalised rather than fitted (Methods). There is no gain, so no occupancy object is ever conditioned on the data.

The diagnostics, derived

The body states the three checks and the two matrices they produce. This appendix carries what is behind them: the expansion that turns a score into a parameter displacement, the exact sense in which the two parts of the distortion compose, and the two scalar summaries with the floor below which neither can be read.

The sandwich and the bias are one first-order expansion

Both corrections come from the same step. Expanding the total score around the parameters $\boldsymbol{\theta}_0$ the recordings were generated from,

$$S(\hat{\boldsymbol{\theta}}) \;\approx\; S(\boldsymbol{\theta}_0) + \nabla_{\boldsymbol{\theta}} S(\boldsymbol{\theta}_0)\,(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0), \tag{A38}$$

and using $S(\hat{\boldsymbol{\theta}}) = 0$ at a maximum-likelihood estimate, with the empirical Hessian replaced by its expectation $-\mathbf{H}$,

$$\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0 \;\approx\; \mathbf{H}^{-1} S(\boldsymbol{\theta}_0). \tag{A39}$$

Taking the covariance of Eq. A39 over independent recordings gives the sandwich,

$$\Sigma \;=\; \mathbf{H}^{-1}\mathbf{J}\,\mathbf{H}^{-1} \;=\; \mathbf{H}^{-1/2}\,\mathbf{C}\,\mathbf{H}^{-1/2}, \tag{A40}$$

and taking its expectation gives the bias, $b \,=\, \mathbf{H}^{-1}\mathbb{E}[S]$, which is nonzero exactly when the mean score is. The two are therefore the same approximation applied to the two moments of one displacement, which is why a member can fail them separately: $\mathbb{E}[S]$ can vanish while $\mathrm{Cov}(S)$ departs from $\mathbf{H}$, and at $\boldsymbol{\theta}_{\text{pool}}$ it does so by construction. When $\mathbf{C} = \mathbf{I}$, Eq. A40 returns $\mathbf{H}^{-1}$ and the correction is inert. Where $\mathbf{H}$ is singular, every inverse is taken on its retained eigenspace (Methods).

Both corrections are first order, and the sandwich is verified directly against the empirical covariance of the estimates over the simulated recordings and against the coverage of the nominal region (Figure 2).

How the two parts of the distortion compose

The sample distortion $\mathbf{C}_s$, built from the within-interval score covariance $\mathbf{J}_s$, and the correlation distortion $\mathbf{R}$ compose exactly. The exact identity is

$$\mathbf{C} = \mathbf{K}\,\mathbf{R}\,\mathbf{K}^{\top}, \qquad \mathbf{K} = \mathbf{H}^{-1/2}\,\mathbf{J}_s^{1/2},$$

which holds as algebra for any $\mathbf{H}$ and $\mathbf{J}_s$, since substituting the definitions collapses the chain. The factor $\mathbf{K}$ satisfies $\mathbf{K}\mathbf{K}^{\top} = \mathbf{C}_s$, so it is a legitimate square-root factor of the sample distortion, but it is not the symmetric principal square root, and the form $\mathbf{C} = \mathbf{C}_s^{1/2}\,\mathbf{R}\,\mathbf{C}_s^{1/2}$ holds only where $\mathbf{H}$ and $\mathbf{J}_s$ commute. What composes with no assumption at all is the information volume,

$$\log\det\mathbf{C} = \log\det\mathbf{C}_s + \log\det\mathbf{R},$$

an exact identity because the determinant is multiplicative. The per-parameter diagonal, the trace, the eigenvalues and the Frobenius norm do not compose across the split; only the volume does, and only the volume is decomposed in the maps.

The two scalars, and the distance they are the components of

With $\lambda_1, \ldots, \lambda_d$ the eigenvalues of $\mathbf{C}$ on the retained subspace, the magnitude $m = e^{\bar{e}}$ and the anisotropy $a = e^s$ of Eq. 4 are the exponentials of the mean and the standard deviation of $\log \lambda_i$. They are not two arbitrary summaries: they are the two orthogonal components of the affine-invariant Riemannian distance from the identity,

$$D_{\mathrm{AI}} = \left(\sum_i \log^2 \lambda_i\right)^{1/2} = \sqrt{d\,(\bar{e}^{\,2} + s^2)}, \tag{A41}$$

so that a member sits at $m = a = 1$ if and only if $D_{\mathrm{AI}} = 0$, and the two coordinates say whether a departure is a common factor on every direction or a spread between them. The pair is reported throughout and D_{AI} is reported nowhere in its place, because the decision the Results support is exactly the one the pair separates and the single number collapses: whether a scalar correction can work at all ($a = 1$) and what factor it should use (m).

Neither reads below its own sampling floor. The anisotropy of an estimated distortion is positive by construction even when the truth is isotropic, since the eigenvalues of a matrix estimated from a finite cloud spread on their own. Drawing a thousand estimates in two dimensions from a perfectly calibrated covariance and forming the same statistic gives a median of 1.038 and a 95th percentile of 1.081, so a measured spread of that size is not evidence of shape; and that is a lower bound on the floor, because the reported covariance the statistic is formed against is itself estimated.

Appendix 4

Repairing an interval without changing the likelihood

A reader who accepts the miscalibration can still ask what it would cost to repair it without changing the likelihood, and the alternatives answer different questions. The standard error across independent fits to separate patches is free, but it describes the uncertainty of a population mean, mixes statistical with biological variability, and leaves any bias in place. Resampling whole sweeps is valid and keeps the within-sweep correlation, but it brackets whatever the estimator converges to without saying whether that estimator is right. Resampling residuals is not valid here, because independent resampling destroys the correlation that is the problem; a parametric bootstrap assumes the model one was checking; a block bootstrap needs a block length well above a correlation time that has to be measured first. Simulating from the fitted model and forming the sandwich answers the same question the diagnostic does, at a cost set by the precision needed on a covariance, and still simulates from the model one wanted to verify. None of these, the sandwich included, reaches the Bayesian evidence: every repair here corrects the covariance a likelihood reports, while the evidence is an integral of the likelihood itself, which no covariance correction touches, so a comparison built on evidences is repaired only by changing the likelihood.

Appendix 5

The construction and its prior art

This appendix sets the construction against the work around it: the linear-Gaussian filter it coincides with and what that correspondence leaves to be supplied, what exists already, where, and the one thing none of it does, the derivatives the diagnostic runs on, an earlier calibration of a filter of this kind, the rung the ladder is missing, and a recent argument against filtering on accuracy and on cost.

The same conditioning has a name in the linear-Gaussian world, the integrated-measurement filter on an augmented state. Carrying the interval integral alongside the state there is what conditioning the interval average on the states at both ends of the window is here, which is why the two constructions agree to about 10^{-8}.

Neither framework hands the construction over. The augmentation says what form the filter takes and not what goes into it. The form needs the probability of crossing the window from one state to another, and the mean and variance of the interval-averaged conductance given that pair; those are classical for a single chain, and carrying them to an ensemble of channels at a cost that does not grow with the channel count is what this family does (Appendix 1). The exact Bayesian update is not available either, since the occupancy and the interval average are not jointly Gaussian, so the emission is closed by a Gaussian and the posterior is carried forward as a second closure, and what those two closures cost is what the body of this paper measures.

What exists, and where. The augmented-integral device is standard in time-series econometrics (***Zadrozny, 1988***). The exact treatment of an integrated observation over a Markov chain is available for a single chain (***Bäuerle et al., 2015***; ***Blackwell, 2018***), and for a single molecule under photon counting with trajectories sampled by Monte Carlo (***Kilic et al., 2021***); the recording filter has been inside single-channel likelihoods since the late 1990s (***Michalek et al., 1999***; ***Venkataramanan et al., 2000***; ***Qin et al., 2000b***). None of them is the ensemble: the same treatment for a patch of channels, at a cost that does not grow with the channel count, which is what makes it usable on a macroscopic recording.

The derivatives are what the diagnostic could not be run without, and analytical ones are not new. ***Milescu et al.*** (***2005***) computes them for a likelihood over independent intervals, and ***Qin et al.*** (***2000a***) for a single-channel recursion that runs the length of the record, through the forward and backward variables and through the derivative of the matrix exponential. What differentiating this recursion adds is what this recursion carries: the derivative of the spectral reconstruction inside it (Appendix 1), and the filter's own update, where the occupancy is held on the probability simplex by a smooth damping, because an operation not differentiable where it binds pumps variance into the score (Methods).

Münch et al. (***2022***) asked this paper's question of a filter of this family and answered it by counting coverage over repeated simulated data sets. A coverage rate costs one posterior fit per replicate at every design point, which keeps such a study to a line of conditions, while a score and an information at known parameters cost one pass each, which is what puts the 210 cells of the design plane within reach; and what comes back is a matrix rather than a rate, saying by how much and in which parameter the reported uncertainty departs. Their own checks were the whiteness of the residual and a rule of thumb in the channel count, and the axis this study finds moves their rung hardest is neither: its channel-number bias runs from four fifths of the parameter at the coarsest sampling to seven per cent at the finest. What their filter generalizes is the emission, through state-dependent open-channel noise and a fluorescence observable, and that generalization costs them the analytical update and obliges a two-moment closure of their own, the same obstruction the update of Eq. 2

meets. Their emission carries a term this study does not model, and their design points are not the cells measured here.

The ladder itself has a rung missing. The family carries no open-loop member with the full cross-interval covariance, the maximum-likelihood route of ***Celentano and Hawkes*** (***2004***) and ***Stepanyuk et al.*** (***2011***), which carries the correlation between time points as a prior and never conditions on the trace, so what the ladder measures is what conditioning buys within this family, and it does not separate recursion from correlation modelling as such; that rung would test the distinction and is not run here.

Del Core and Mirams (***2025***) argue against filtering on both counts, accuracy and cost, and measure both: their moment-equation method returns a lower relative error than the extended Kalman filter they compare it with, and reaches the optimum in about half an hour against about thirty-six, a difference they attribute to integrating the differential moment equations between every pair of samples and correcting their initial conditions and sensitivities at each step, which for p parameters is $(4+8p)$ matrix sums, $(8+17p)$ matrix products and one inversion per sample. The attribution is what decides how the figure transfers. The member here takes each acquisition interval in one spectral decomposition of the generator, reused across every interval that shares a condition, and a fixed number of $k \times k$ products, in a single pass over the record, so the work per interval is fixed and cubic in the state count, the growth is polynomial rather than combinatorial, and the numerical integration their comparison prices is not part of it. What this family costs in practice is a different number for a different task: a nine-scheme Bayesian comparison for a ligand-gated receptor took twelve to fifty-two days per scheme on sixteen to thirty-two cores (***Moffatt and Pierdominici-Sottile, 2025***), an evidence computation over nine schemes on recorded data rather than an optimisation on synthetic traces.

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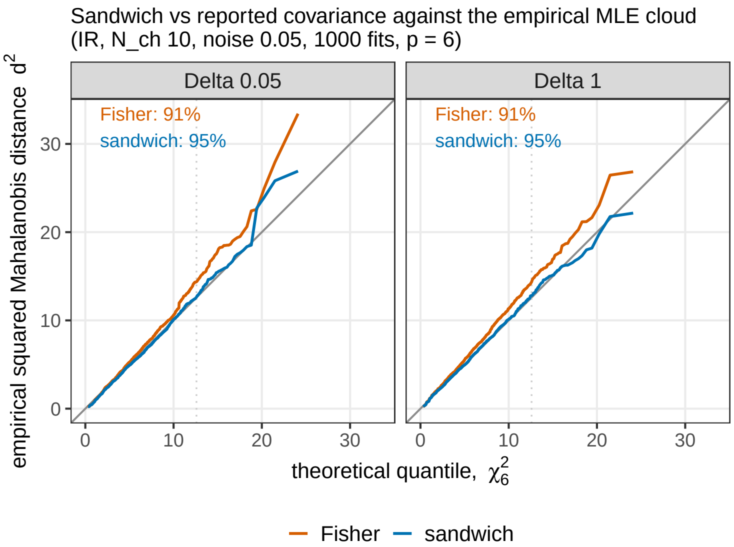

Figure 2—figure supplement 1. The reported covariance against the empirical one, in all six directions at once. The squared Mahalanobis distance of each estimate from the cloud mean, sorted and drawn against the quantiles of a χ^2 with six degrees of freedom, under the covariance the member reports and under the distortion-corrected sandwich. A calibrated covariance lies on the identity; above it the reported region is too small and under-covers. The reported Fisher covariance under-covers the nominal 95% region and the sandwich covers it, 0.914 against 0.947 to 0.949. Centring on the cloud mean tests the second moment alone, which is what the correction claims to repair. Two things differ from the body figure, both forced by the sample: the member is `IR` alone, and the cell is $N_{\text{ch}} = 10$ at $\widetilde{S} = 0.005$ with group size 100, the one cloud in the set carrying a thousand fits of that size.

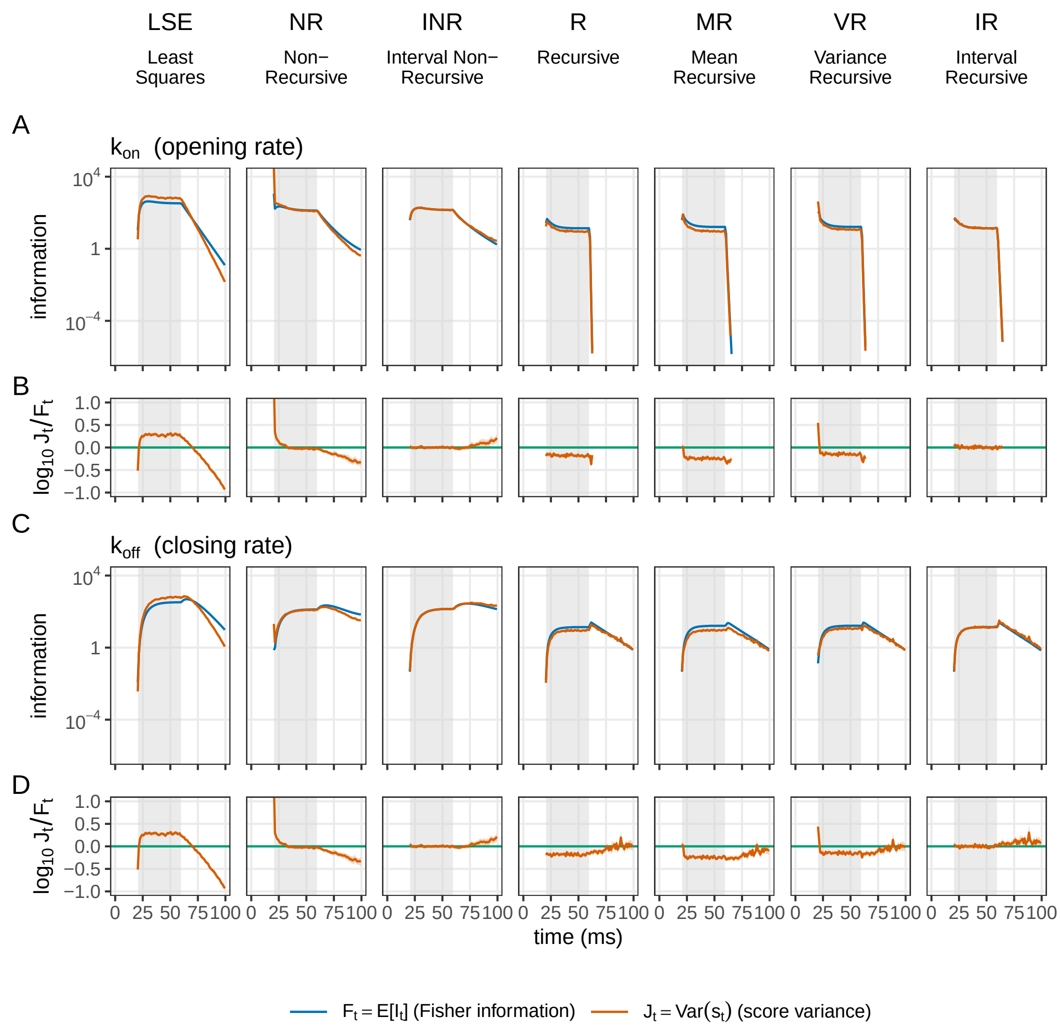

Figure 3—figure supplement 1. Per-step information against score variance, the opening rate and the closing rate. Seven members across the columns, the least-squares arm being the interval-averaged ILSE, and two parameters down the page as bands of two rows: k_{on} (A, B) and k_{off} (C, D). The upper row of each band carries the per-interval Fisher information and the score variance as magnitudes, which lie on each other where the member is calibrated at that interval; the lower row is $\log_{10}(J_t/F_t)$ with its 95% bootstrap interval, zero when they do. Per-interval calibration is not monotone along the cost ladder: on k_{off} the median reads $+0.184$ for ILSE, -0.038 for NR, $+0.007$ for INR, -0.145 for R, -0.229 for MR, -0.135 for VR and $+0.033$ for IR. Curves stop at the information mask, which is also where the upper axis ends. The least-squares ratio row is the same curve in every band of both halves, median 0.184: with a single constant variance the parameter cancels out of the ratio, leaving the true residual variance over the assumed one.

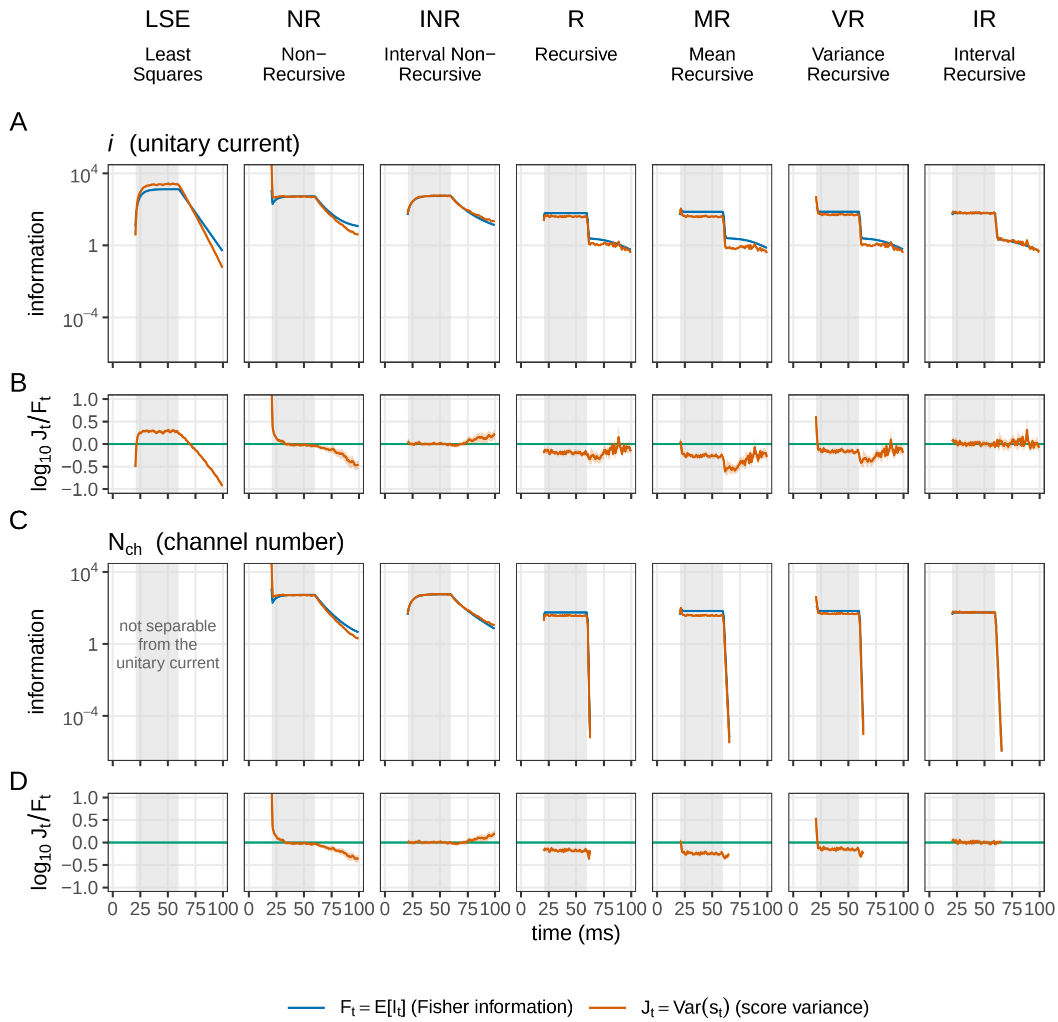

Figure 3—figure supplement 2. Per-step information against score variance, the unitary current and the channel number. The other half of the preceding supplement, on the same seven members and read the same way: the unitary current (A, B) and N_{ch} (C, D), the upper row of each band the per-interval Fisher information against the score variance and the lower row $\log_{10}(J_t/F_t)$ with its 95% bootstrap interval. In the two recursive columns the N_{ch} band reaches the information floor within six intervals of washout while the unitary-current band does not. The channel-number band carries no least-squares panel: N_{ch} is not separable from i there.

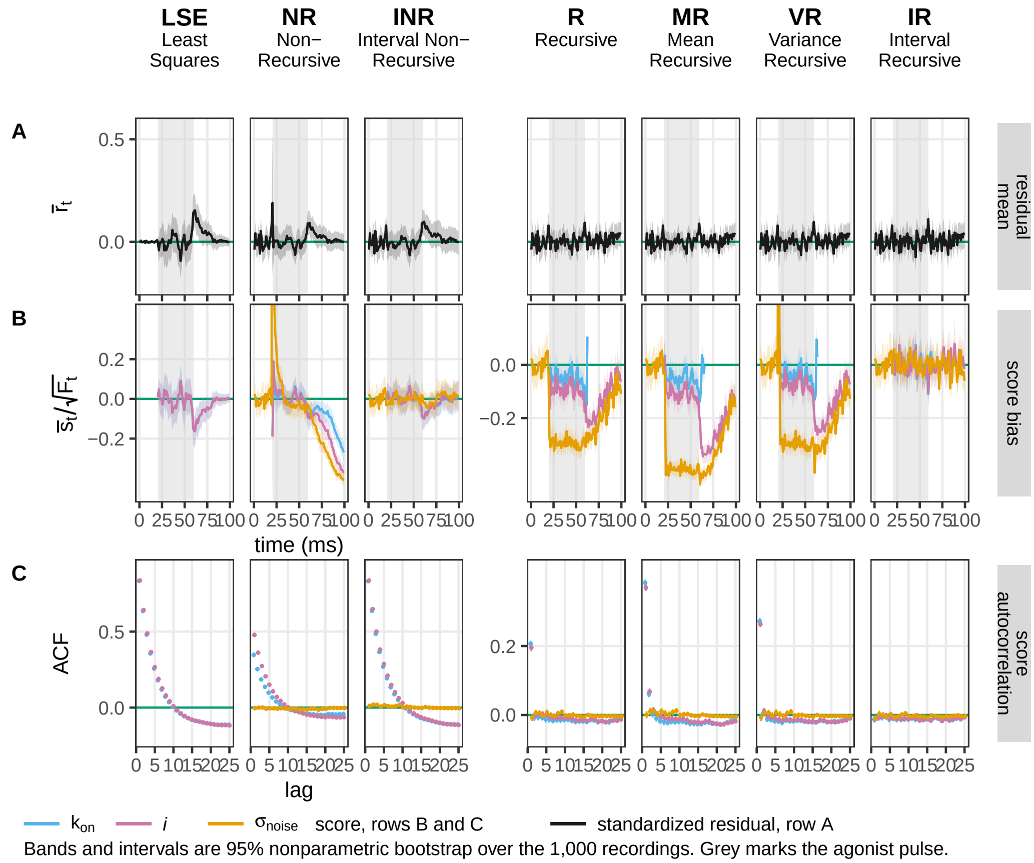

Figure 3—figure supplement 3. The checks this figure has no room for. The seven members of the preceding supplements at the same cell, carrying (A) the mean standardized residual, the one of the three properties of r_t not drawn above; (B) the score bias for the three parameters not coloured above, k_{on}, the unitary current and the instrumental noise level; and (C) their score autocorrelation. The first residual moment separates nobody, which is what makes the other two properties the informative ones. In (B) the ordering repeats the one in row (D) of the body figure: counting informative intervals whose interval excludes zero against a chance level of 0.05, IR reads 0.067, 0.050 and 0.050, INR 0.100, 0.100 and 0.000, and the three recursive members between 0.409 and 0.975. The noise level, measured before the agonist arrives, keeps a white score even where the kinetic directions do not. Least squares carries no noise curve: its noise level is a plug-in, not a free direction.

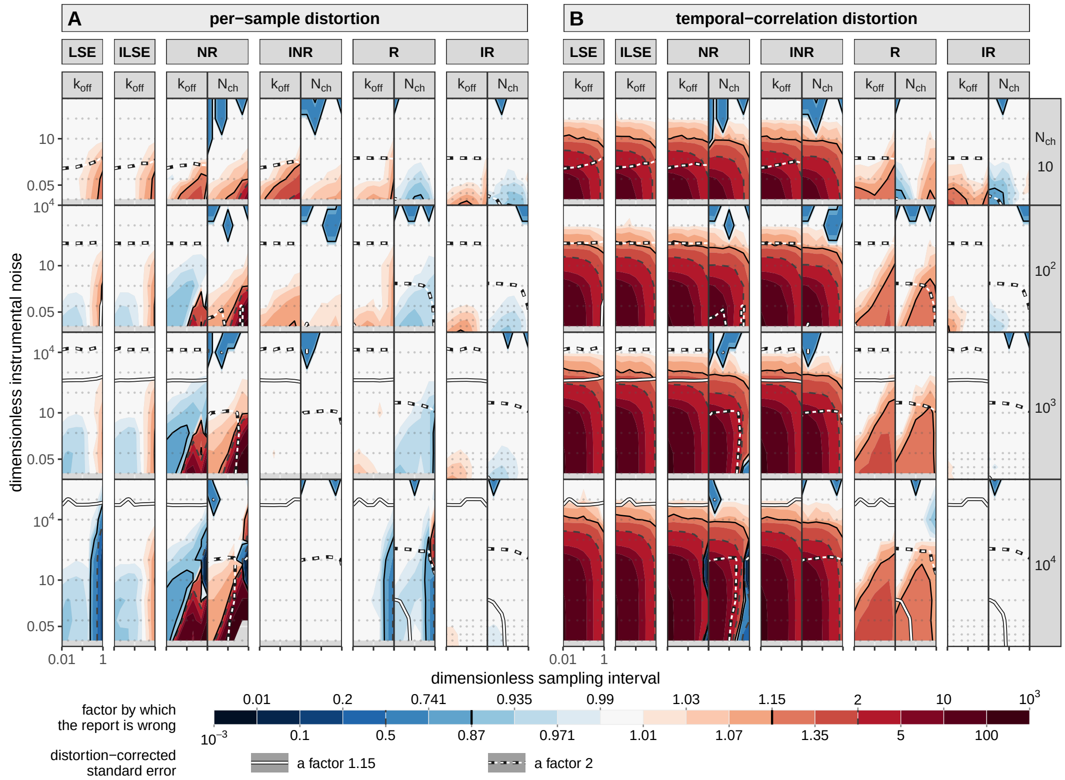

Figure 4—figure supplement 1. The distortion factored: what is a one-interval failure and what is a failure across intervals. Figure 4's second moment split into the two congruences the battery computes separately, the per-sample part $\mathbf{H}^{-1/2}\mathbf{J}_s\mathbf{H}^{-1/2}$, the departure of a single interval from the Gaussian the member assumes, and the correlation part $\mathbf{J}_s^{-1/2}\mathbf{J}\mathbf{J}_s^{-1/2}$, the memory across intervals that a likelihood diagonal in time cannot see (defined above). Same plane, same rows, same scale, same six members and the same two criteria as the body figure, nested component over member over parameter. Both factors are read at θ_{pool}, as in Figure 4B. It is the same split Figure 3 measures per parameter from the score, as rows (E) and (F), and read at the truth there rather than at the optimum; the two readings agree for the members that are centred and part company for the displaced ones, so a factor of this supplement is not to be matched against a factor of that figure without the anchor. `NR` and `INR` carry the *same* correlation part, 16.5, 19.5, 21.2, 21.2 against 16.5, 20.7, 21.8, 22.1 on the closing rate across the four decades of channel count, as they must, since they differ by a one-interval variance term and by nothing about time; what separates them is the per-sample part, where `NR` runs 1.24, 1.02, 5.68, 47.2 and diverges while `INR` runs 1.72, 1.15, 1.02, 1.00 and converges, and on the channel number the same pair ends at 569 against 1.00. The divergence the body figure shows for `NR` is therefore not a temporal failure at all: it is the one-interval variance missing a term that grows with the channel count. The recursion buys the other factor and only the other factor: `INR` to `IR` takes the correlation part from 22.1 to 1.000 and leaves the per-sample part where it was. The factorisation is exact in the log-determinant and not entry by entry, the reconstruction being $\mathbf{K}\,\mathbf{C}\,\mathbf{K}^{\mathsf{T}}$ with $\mathbf{K} = \mathbf{H}^{-1/2}\mathbf{J}_s^{1/2}$: $1.024 \times 1.433 = 1.468$ against a total of 1.481 for `R` at a thousand channels.

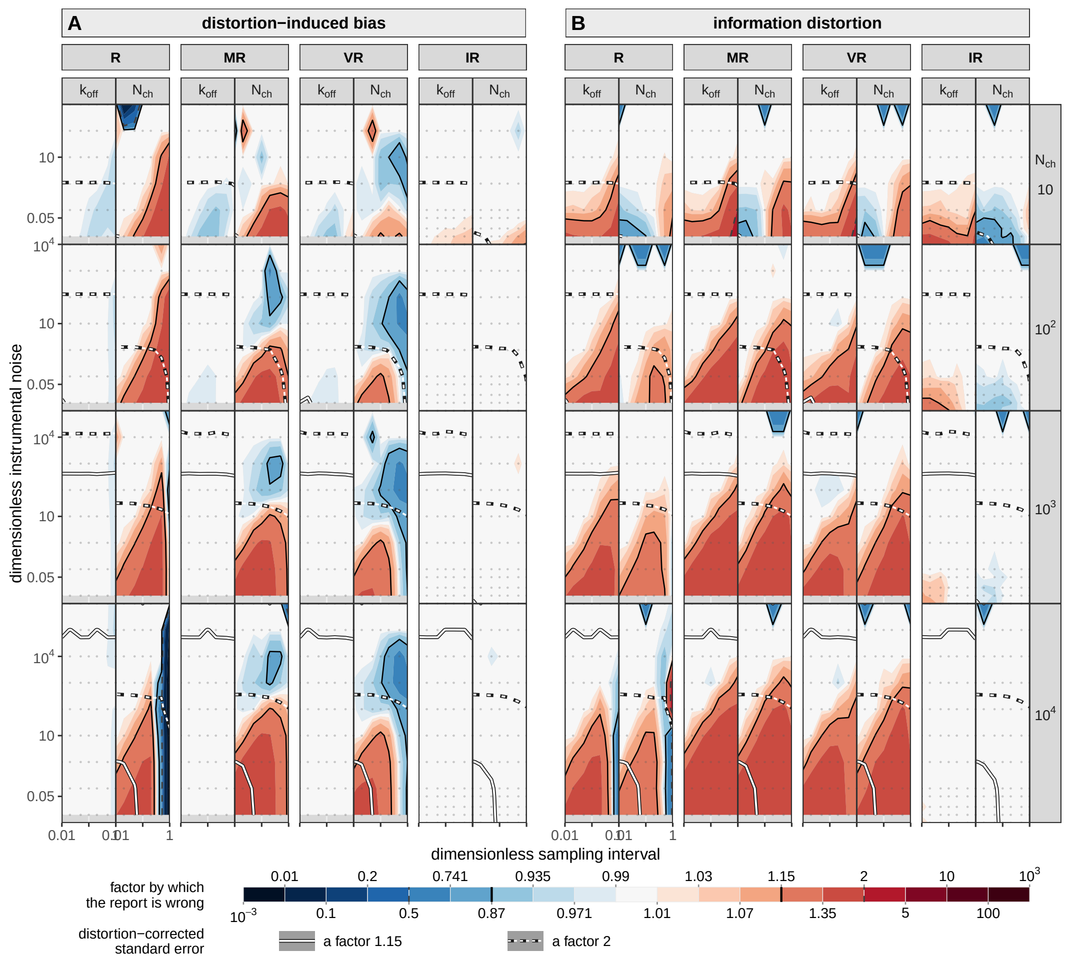

Figure 4—figure supplement 2. The same plane for the recursive family: R, MR, VR and IR. Figure 4 unchanged in every respect but the roster, the same two moments on the same shared factor scale with the same two line families and the same nesting, over the four rungs inside the recursive family, `R`, `MR`, `VR` and `IR` (Table 1). Sixteen columns rather than the body's twenty, since no least-squares arm appears and every member carries the channel number. The reading is in the text: each partial correction buys part of the first moment and gives it back in the second as channels are added, and neither approaches either interval member on either moment.

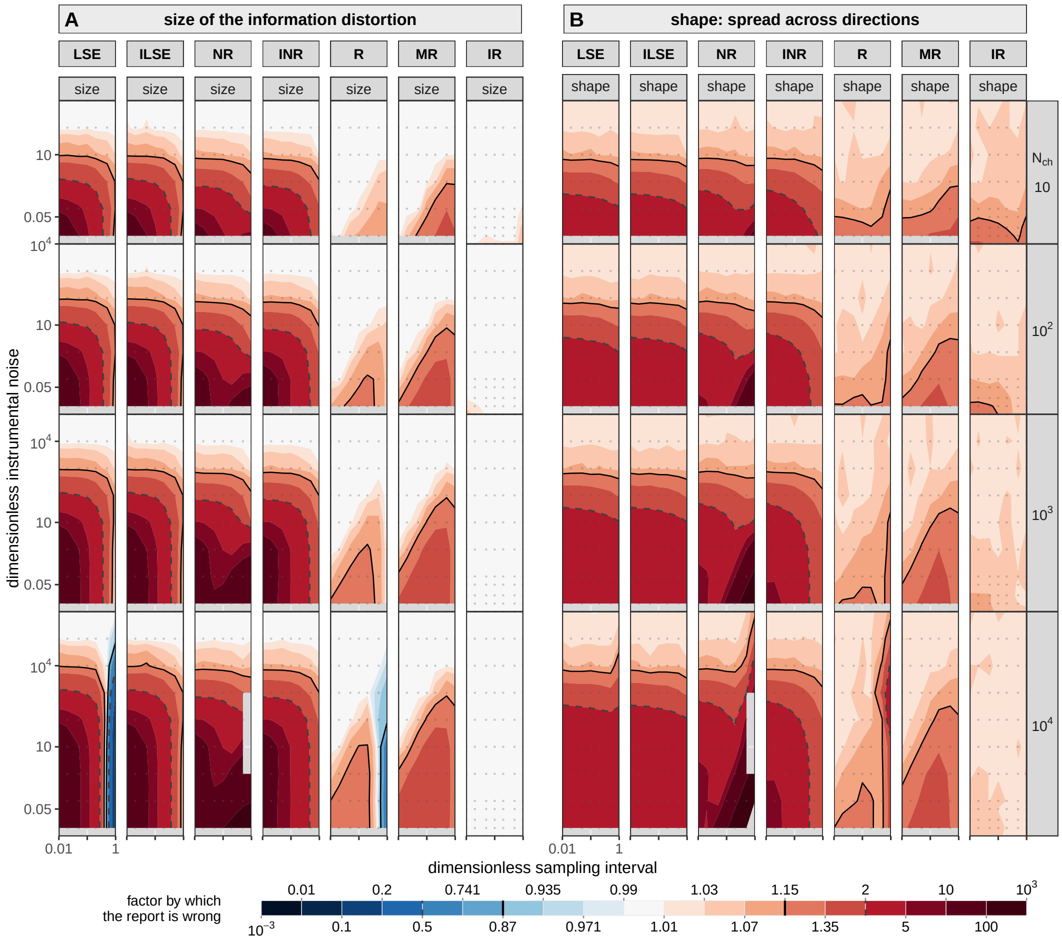

Figure 4—figure supplement 3. The magnitude of the information distortion against its anisotropy. Two invariants of the same distortion matrix over the same plane, for seven members: both least-squares arms, `NR`, `INR`, `R`, `MR` and `IR`. (A) its magnitude m and (B) its anisotropy a (Eq. 4), m signed about a null of one and a one-sided. The rule they give: a at one means a single effective-count rescaling is exact and m is the factor it should use; a above one means no such rescaling exists at any factor. The two least-squares arms are the same figure wherever the recording is finely sampled, and part at the coarsest interval: at $\widetilde{\Delta} = 1$ and 10^4 channels `LSE` reads 0.074 in magnitude, understating its own information thirteenfold, where `ILSE` reads 1.00. That is the mean model failing and not the variance model, and it is the deepest of the few cells anywhere on this page where a member is conservative rather than overconfident: 11 of `LSE`'s 210 cells and 4 of `R`'s, all of them at the two coarsest intervals and none for any other member.

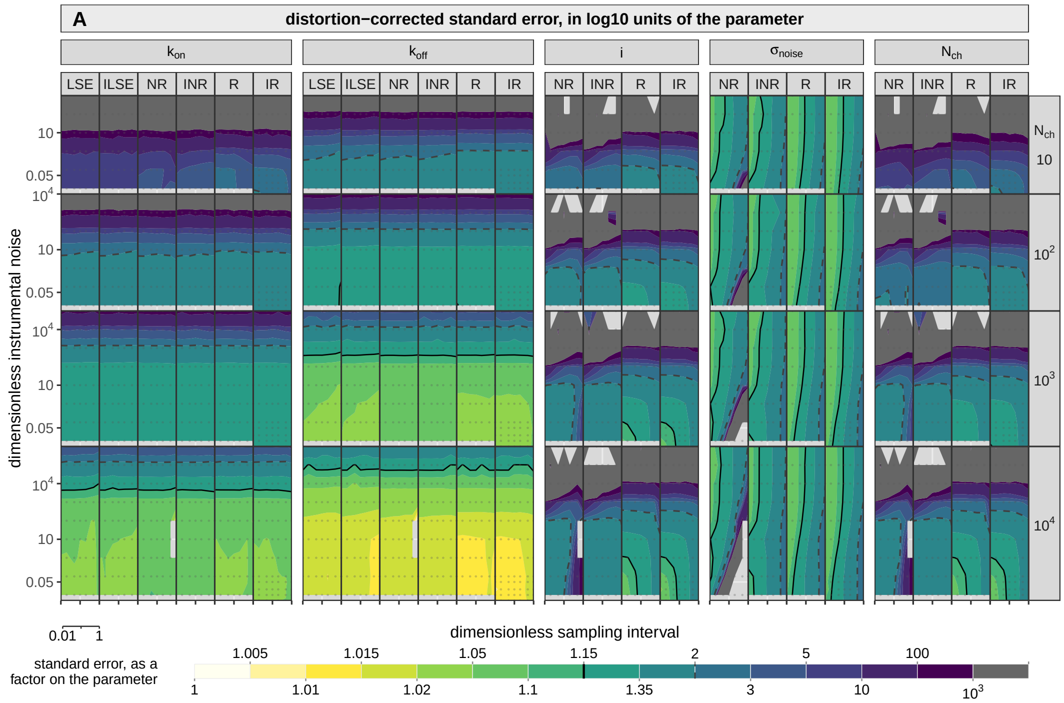

Figure 4—figure supplement 4. What each member achieves, as opposed to what it reports. The standard error each member delivers on each parameter: the square root of the diagonal of the distortion-corrected covariance, over the same plane as Figure 4 and in the same orientation. What the design axes buy is read directly off it, and the reading is in the text. It is also the field whose level set at a factor of two draws two of the boundaries of Figure 7. Blocks are per parameter, the members side by side inside each. The scale is sequential: pale is a parameter pinned, dark one barely determined. The two marks are an error bar of 15% of the parameter and an error bar of a factor two. The corrected error is drawn rather than the reported one, which Figure 4 shows wrong by up to a factor of 74, so only the corrected version compares members on the same footing. Both least-squares arms hold the unitary current and the noise level fixed and so appear on the rate blocks alone.

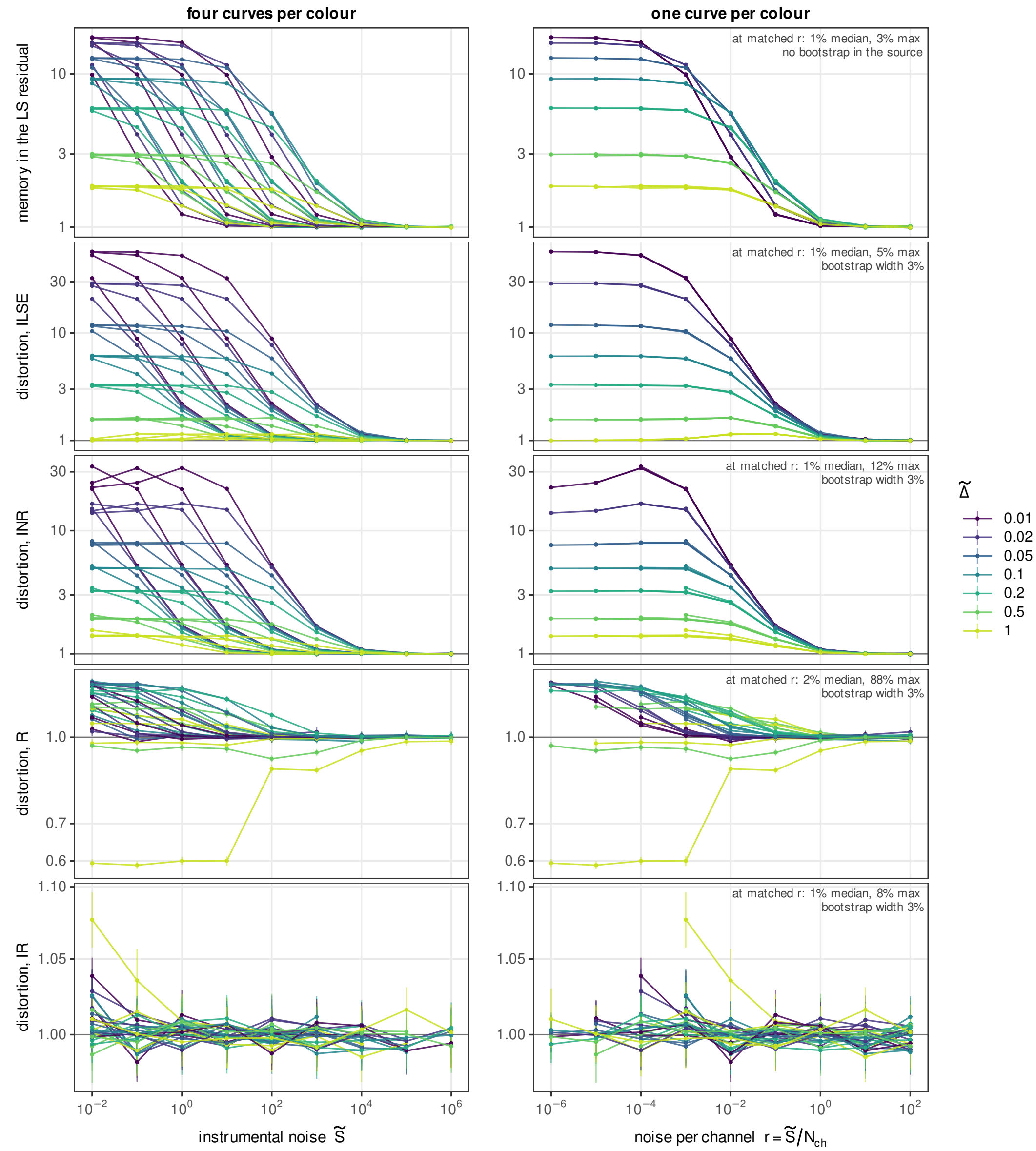

Figure 5—figure supplement 1. The instrumental noise and the channel count act only through their ratio. The same cells as the body figure, drawn twice. Each line is one channel count at one acquisition interval swept over the instrumental noise, twenty-eight per panel, and the two columns differ in nothing but the horizontal coordinate: on the left the instrumental noise $\widetilde{S}$, on the right the noise per channel $r = \widetilde{S}/N_{\mathrm{ch}}$. On the left the four channel counts are four separate curves per colour, a decade apart; on the right they merge, so seven of the twenty-eight are visible. Nothing is fitted, rescaled or normalized between the columns. The top row is the quantity the body figure's horizontal axis inverts, τ_9 of the least-squares residual, and it collapses with the rest: a reader who measures τ_9 on their own record has measured r without knowing it. The vertical bars are the bootstrap quantiles of each cell, absent on the top row alone: each point is known to about 3% and the four channel counts land within 1% of one another. The number in each right-hand panel is the residual disagreement between cells that share r exactly at the same interval, with no interpolation. The 5% of `ILSE` and the 8% of `IR` are within that scatter; the one large value, `R`'s, is the single ill-conditioned cell at $N_{\mathrm{ch}} = 10^4$ and $\widetilde{\Delta} = 1$ that Figure 4 paints at the deep end of its scale, without which `R` agrees to 23%. In the bottom row `IR`'s departures from unity are the width of its own error bar and not structure.

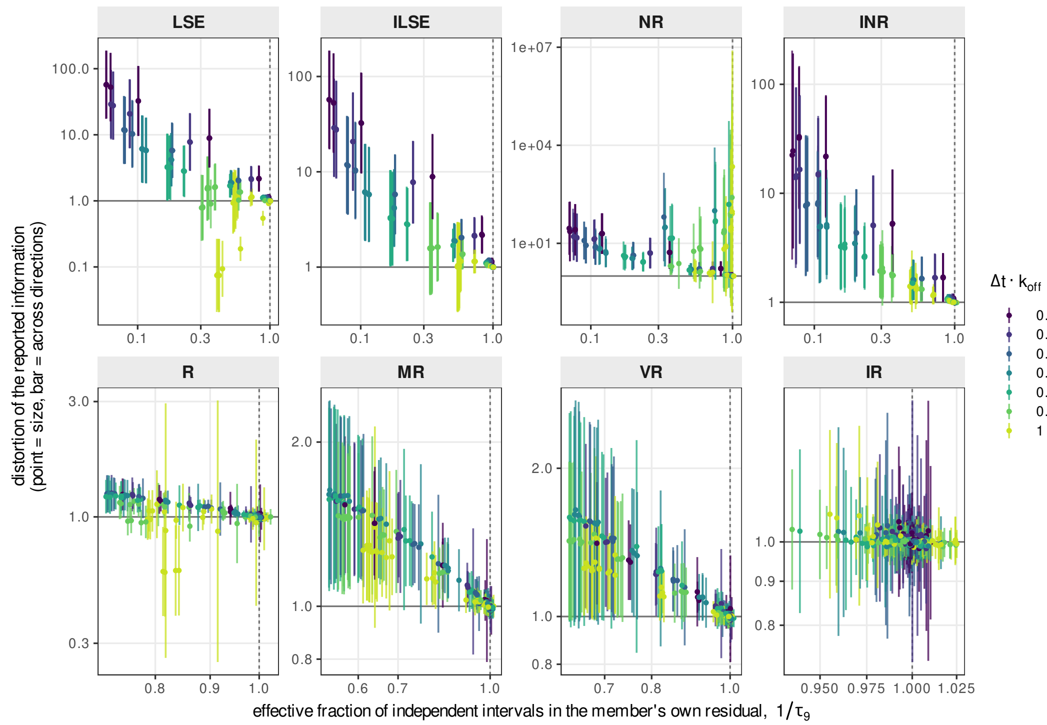

Figure 5—figure supplement 2. Each member against the memory left in its own residual. The whole ladder, with the horizontal axis of each panel now $1/\tau_9$ of that member's own standardized residual, which a reader cannot compute without running the member: a mechanism panel rather than a decision one, and it runs the same way as the body figure. It asks whether the memory a member fails to remove predicts the information it then misreports. The ladder is ordered by how much each fails to remove: `IR` occupies a point, its own τ_9 running 0.98 to 1.07 over the whole plane, `R` reaches 1.39, and the two least-squares arms span to 17.4. Within a member the prediction holds at a fixed sampling interval, the rank correlation between own memory and own distortion reading $+0.92$ for `ILSE`, $+0.96$ for `INR`, $+0.93$ for `MR` and $+0.91$ for `VR` over the whole plane and about $+0.99$ within one interval. It is necessary and not sufficient: `NR` carries 73 cells whose own τ_9 is below 1.05, an eighth of them distorted, and they sit at the two coarsest intervals, where consecutive samples are a relaxation time apart and there was no memory to remove. Each panel is on its own scales.